



INSTALLATION MANUAL

RTK GNSS Survey and Positioning System

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Version 2.1

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1 INTRODUCTION

This installation manual outlines the necessary steps required to install the Raspberry Pi OS onto the system SD cards and details the mandatory configuration tasks that must be completed prior to deploying the Farm Dog software. This workflow includes installing the required Wi-Fi drivers, setting up SSH access for remote terminal administration, and configuring static IP addresses for both the Base Station and the Rover.

Note: This document should be read in conjunction with the Farm Dog User Manual.

2 SOFTWARE REQUIREMENTS

Before initiating the installation process, make sure that the following requirements are met before proceeding:

- **OS Imaging Software:** To install Raspberry Pi OS onto an SD card, download Raspberry Pi Imager from the following website: www.raspberrypi.com/software
- **Remote Console Access:** For remotely accessing the Raspberry Pi for configuration, ensure you have access to Terminal / Command Prompt or software like Putty.
- **Remote Desktop Access (Optional):** If you want to use the Desktop version, ensure you have access to a VNC application (such as UltraVNC Viewer or RealVNC). Please note that VNC needs to be enabled on the Raspberry Pi first. For information on how to enable this function, please read the configuration of VNC section.

- **Network Environment:** During the initial installation and configuration process, ensure that the user's host computer and both Raspberry Pi units are connected on the same local network.

3 HARDWARE REQUIREMENTS

Before starting the physical installation process, ensure that you have active internet connection to download and install the necessary software packages and system updates for the Raspberry Pi modules.

Also make sure that you have access to the following:

- Micro SD cards
- Micro SD card reader
- A computer or laptop with USB ports to connect the card reader
- The USB Wi-Fi module and its external antenna
- The Raspberry Pi's

For hardware installation of the Base Station and Rover enclosures, please have the following tools and components ready:

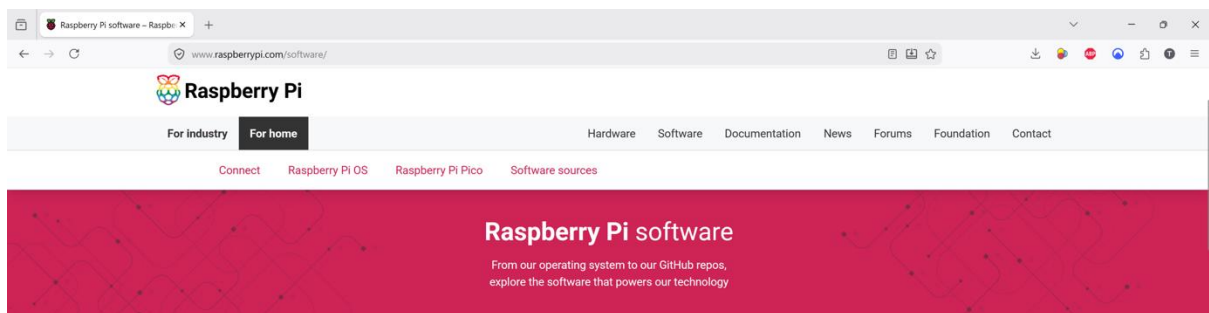
- Philips head screws and screwdriver
- Metal Standoffs for the Raspberry Pi boards
- Adhesive Velcro tape
- The IP66 ABS plastic enclosures (for both Base Station and Rover)
- A power drill with appropriate drill bits

4 SOFTWARE INSTALLATION

4.1 INSTALLING RASPBERRY PI OS

Before beginning the software installation, ensure you have a reliable internet connection and that the host computer is on the exact same local network as the two Raspberry Pi units.

- **Step 1:** Navigate to www.raspberrypi.com/software.
- **Step 2:** Download the “Raspberry Pi Imager” application.



Raspberry Pi Imager

Raspberry Pi Imager is the quick and easy way to install **Raspberry Pi OS** and other operating systems to a microSD card, ready to use with your Raspberry Pi.

Download and install Raspberry Pi Imager on a computer with an SD card reader. Insert the microSD card you'll use with your Raspberry Pi into the reader and run Raspberry Pi Imager.

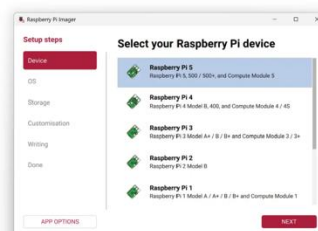
[Download for Windows](#)

[Read the documentation](#)

[Download for macOS](#)

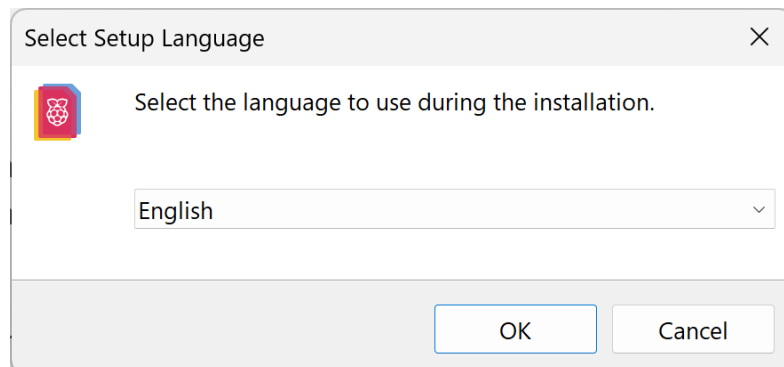
[Download for Linux \(x86_64\)](#)

To install on Raspberry Pi OS, type
`sudo apt install rpi-imager`
into a terminal window

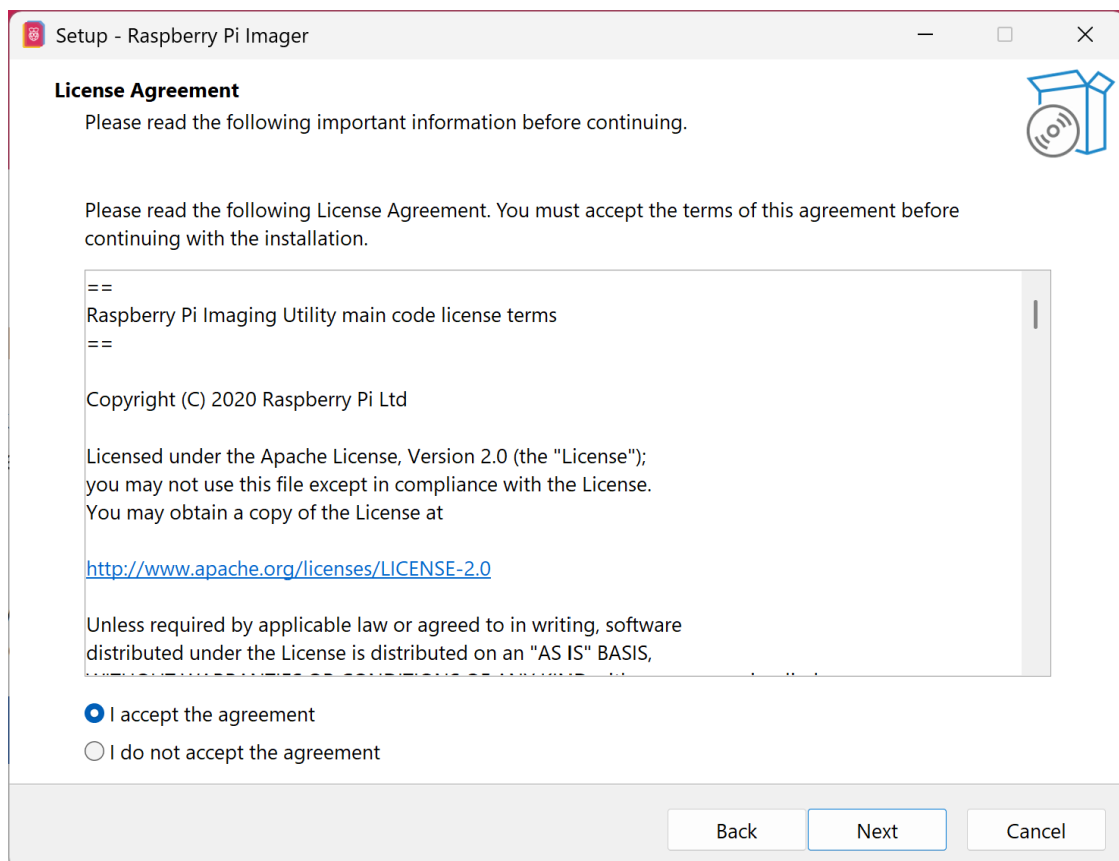


4.2 RASPBERRY PI IMAGER

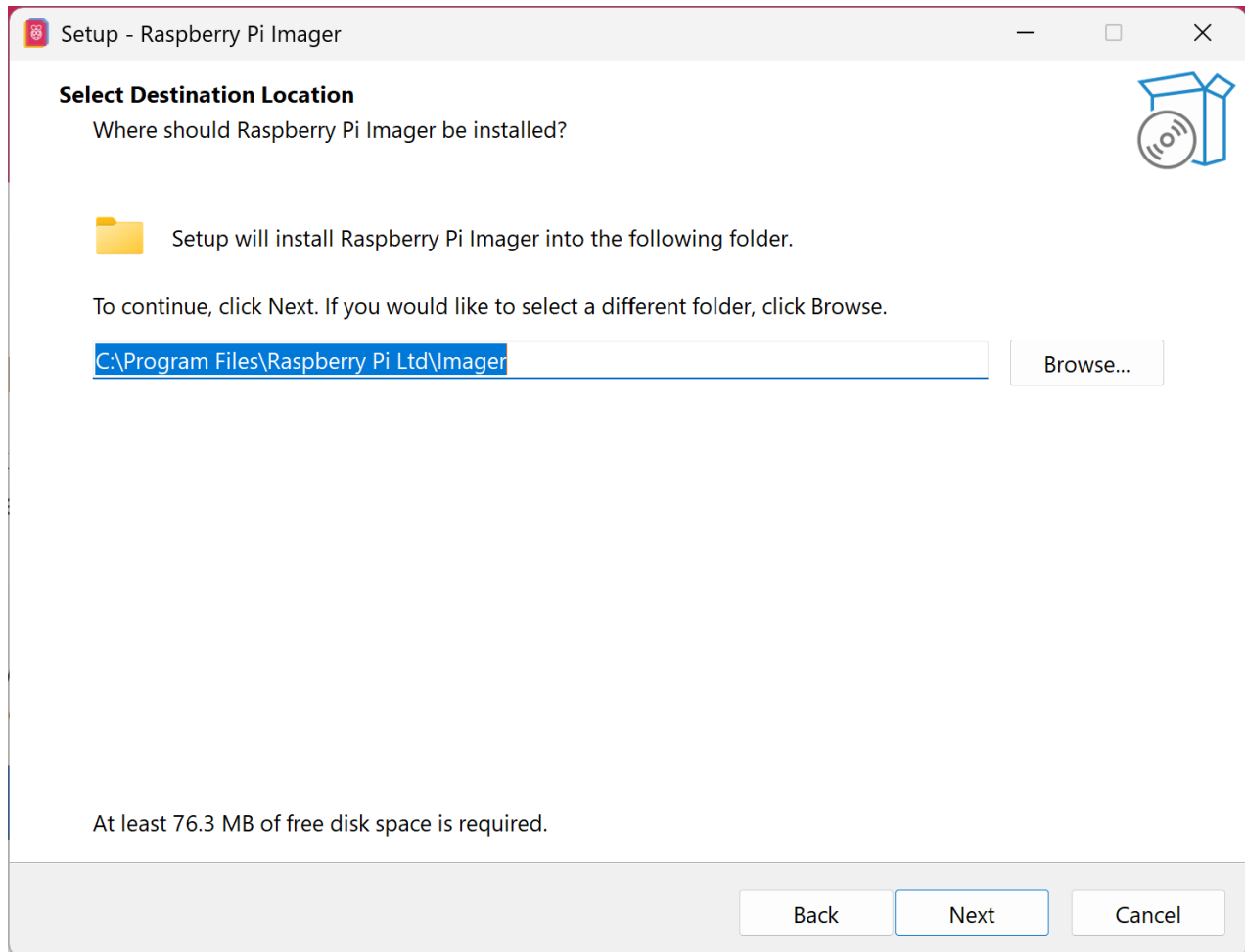
To install the Raspberry Pi Imager on a Windows operating system, execute the downloaded installer, select language and follow the setup wizard prompts:



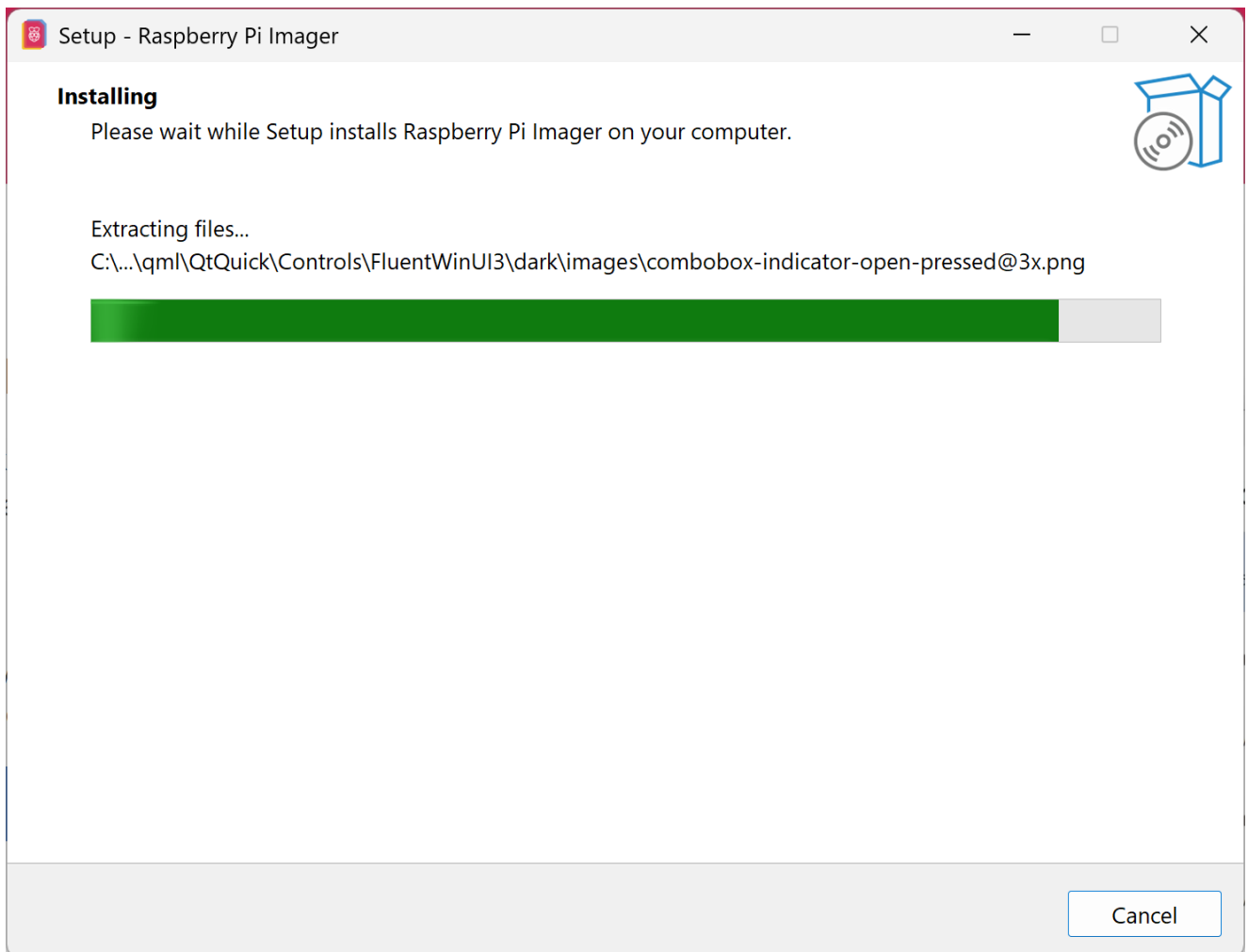
- **Step 1: Read and agree to the License Agreement.**



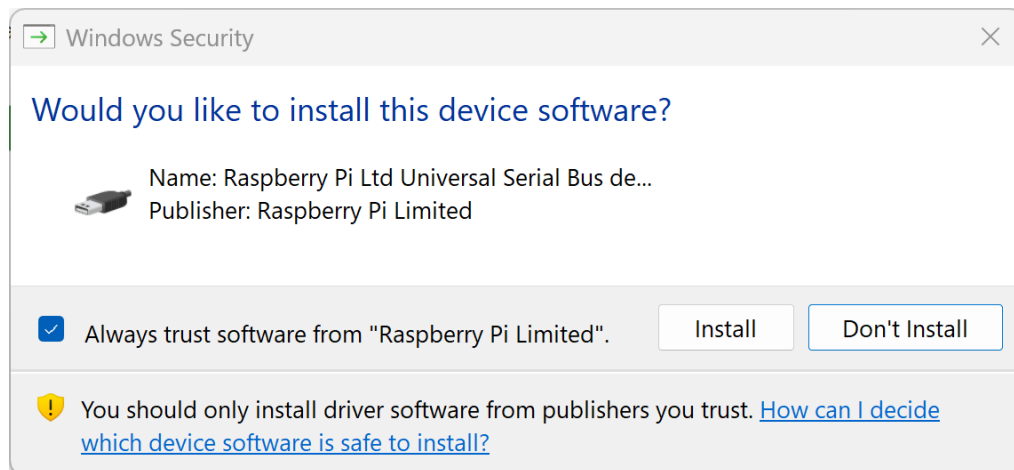
- **Step 2:** Select the destination folder where you would like the Raspberry Pi Imager to be installed.



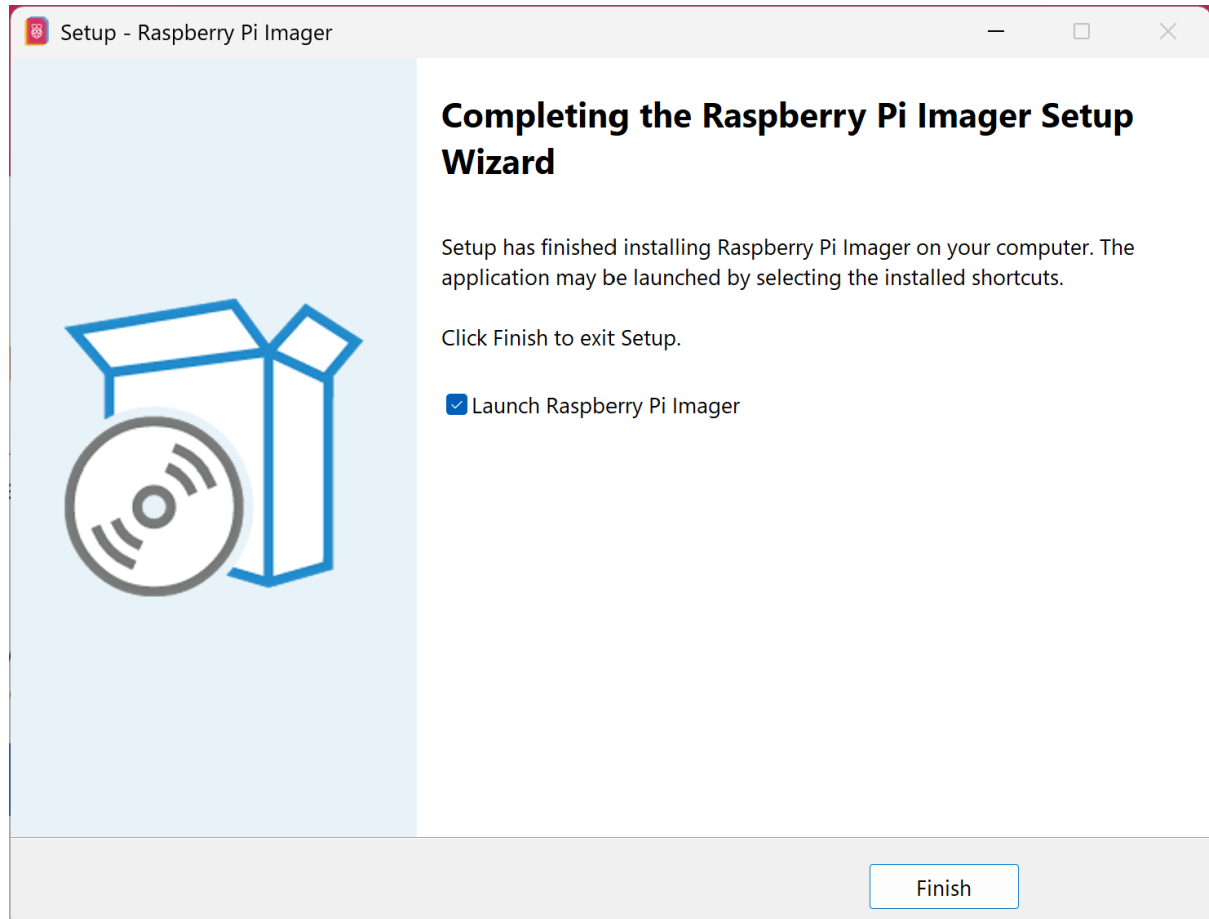
- **Step 3:** Wait for the extraction and installation process to complete (go grab a coffee whilst the program is being installed).



- **Step 4:** If prompted by Windows Security, grant permission to allow the Raspberry Pi Imager device software to be installed.



- **Step 5:** Click "**Finish**" once the setup wizard indicates the installation is complete.



FARM DOG

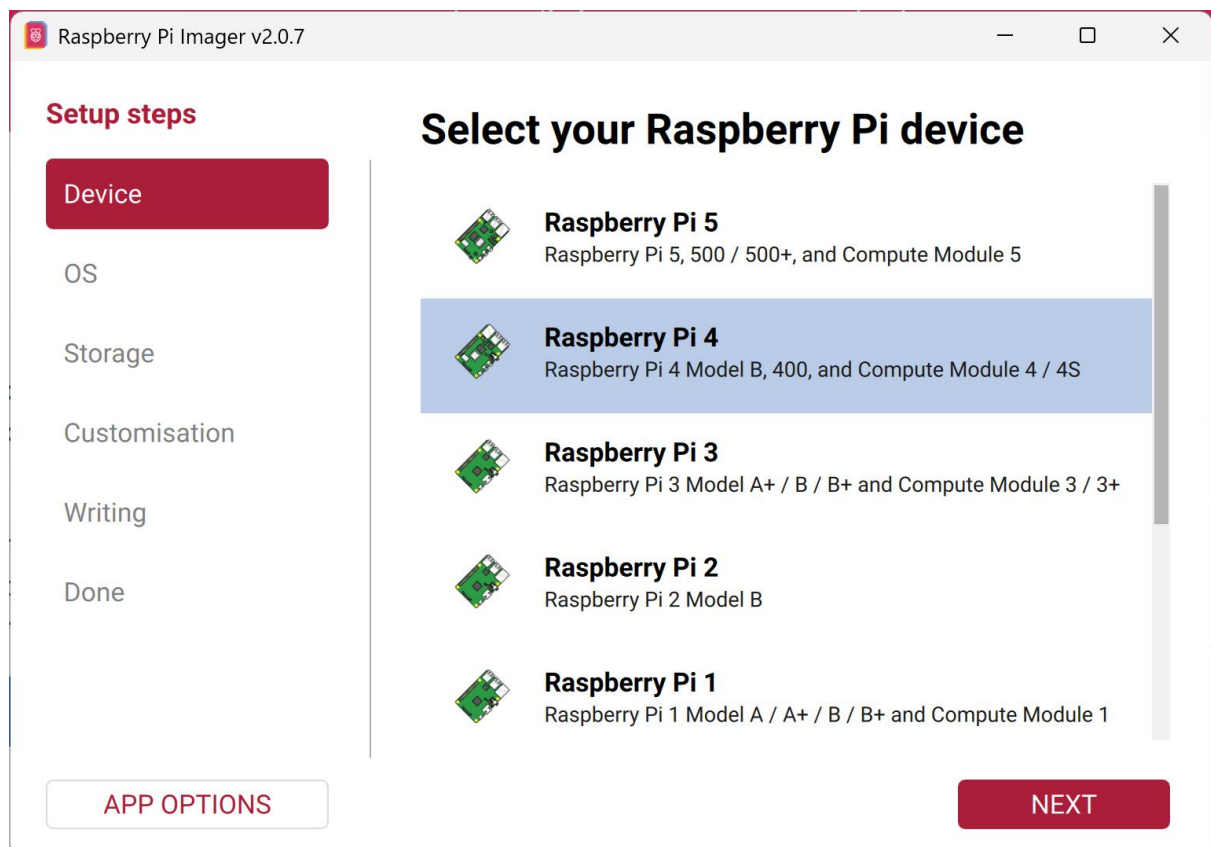
- **Step 6:** With the application successfully installed, prepare your Micro SD card and insert it into your computer's Micro SD card reader.



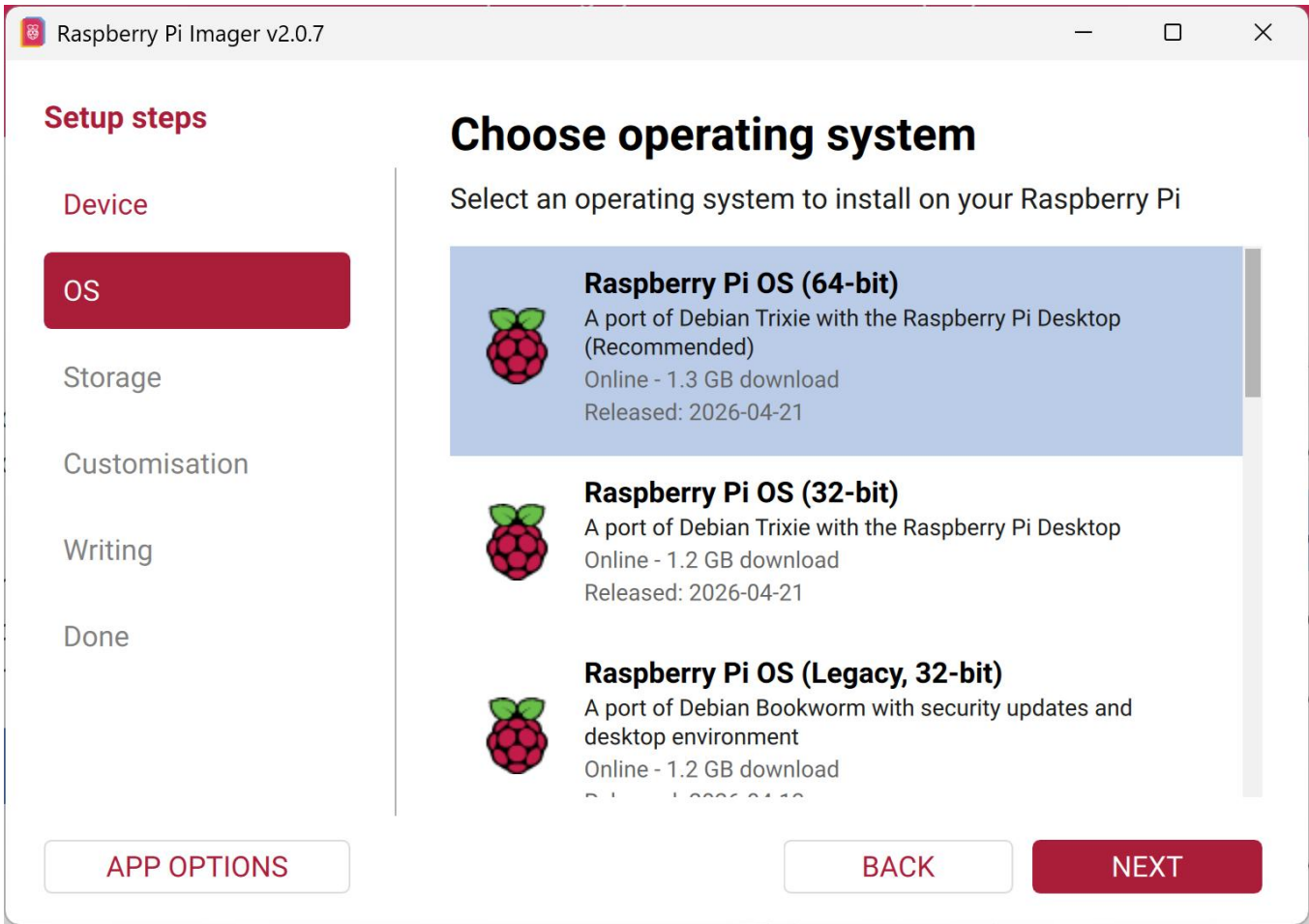
4.3 FLASHING AND CONFIGURING THE OPERATING SYSTEM

Once the Raspberry Pi Imager is installed, proceed with configuring and writing the OS to your Micro SD card.

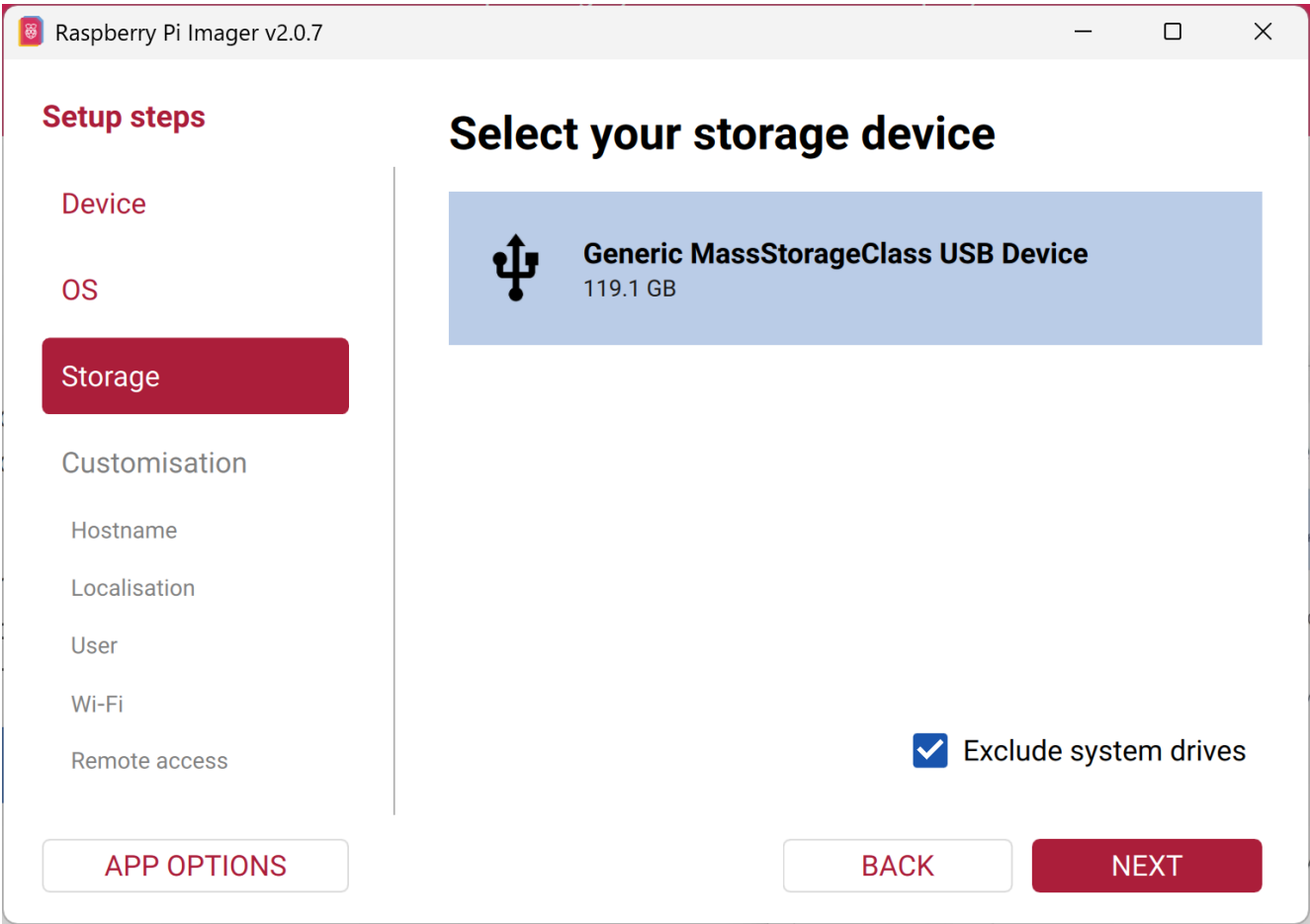
- **Step 1:** The Raspberry Pi Imager will launch automatically after clicking "Finish" during the installation process. Ensure your Micro SD card is inserted into the card reader.
- **Step 2:** Under the device selection menu, choose the appropriate hardware model, which in this case is the "**Raspberry Pi 4**".



- **Step 3:** For the operating system, select "**Raspberry Pi OS (64-bit)**" and click **Next**.



- **Step 4:** Select your inserted Micro SD card as the target storage device.



- **Step 5:** When prompted to customize settings, enter a designated **hostname** for the device and click **Next**.

Raspberry Pi Imager v2.0.7

Device

OS

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Customisation: Choose hostname

Enter your hostname

A hostname is a unique name that identifies your Raspberry Pi on the network. It should contain only letters, numbers, and hyphens.

APP OPTIONS

SKIP CUSTOMISATION

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NEXT

- **Step 6:** Configure the localization settings by selecting the appropriate capital city (e.g., Canberra for Australia), time zone, and keyboard layout.

Raspberry Pi Imager v2.0.7

OS

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Writing

Customisation: Localisation

Select your location for suggested time zone and keyboard layout

Capital city:

Canberra (Australia)

Time zone:

Australia/Perth

Keyboard layout:

au

APP OPTIONS

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NEXT

- **Step 7:** Establish a secure username and password for the device.

Raspberry Pi Imager v2.0.7

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Writing

Done

Customisation: Choose username

Create a user account for your Raspberry Pi

Username:

Enter your username

user-rover

Password:

Enter password

user-rover

Confirm password:

Re-enter password

user-rover

The username must be lowercase and contain only letters, numbers, underscores, and hyphens.

APP OPTIONS

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[SENSITIVE PERSONAL]

- **Step 8:** To connect the device to your secure Wi-Fi network automatically, enter the corresponding network SSID and password.

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Writing

Done

Customisation: Choose Wi-Fi

SECURE NETWORK

OPEN NETWORK

SSID:

Network name

Terry's Farm

Password:

Network password

Confirm password:

Re-enter password

☐ Hidden SSID

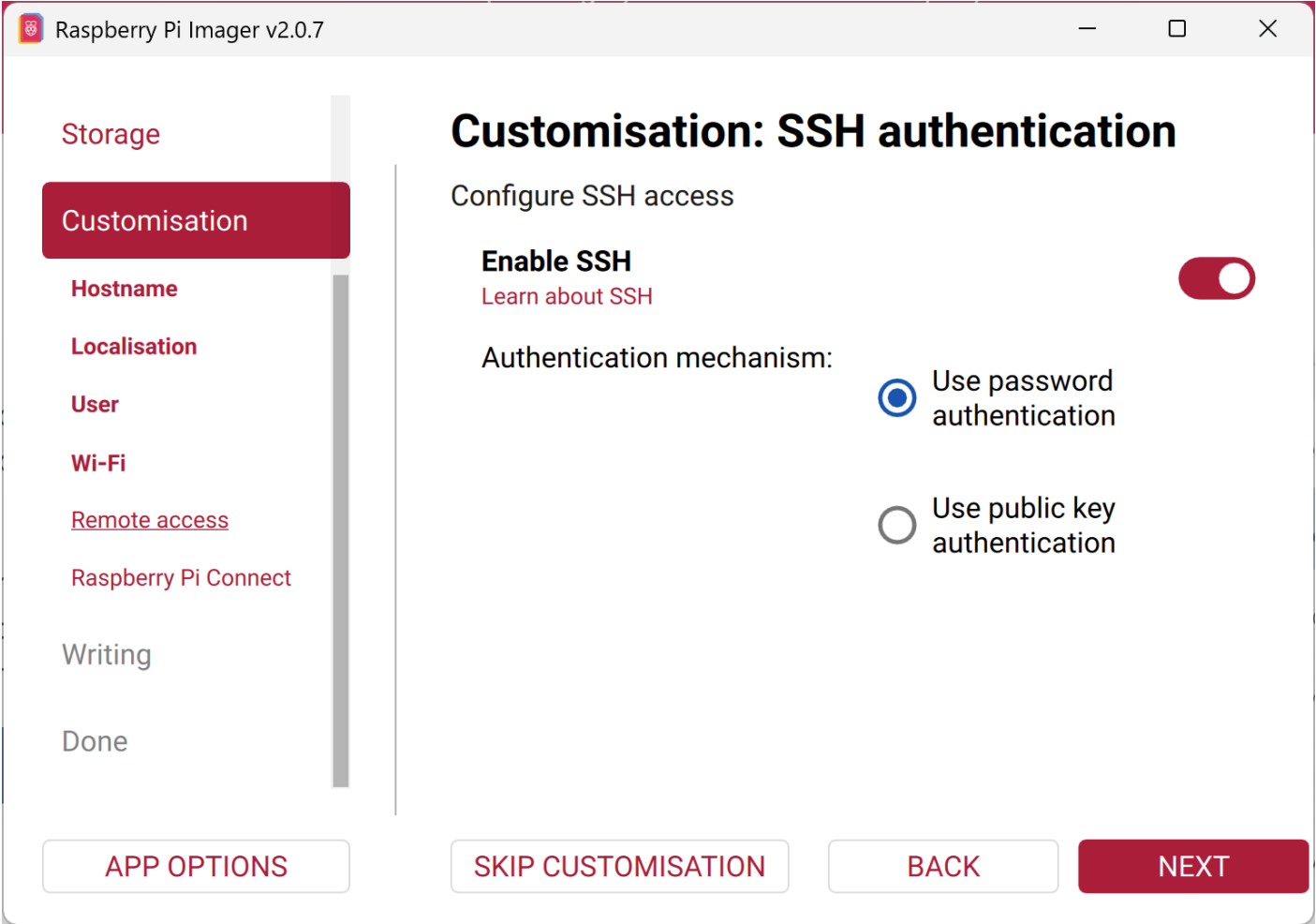
APP OPTIONS

SKIP CUSTOMISATION

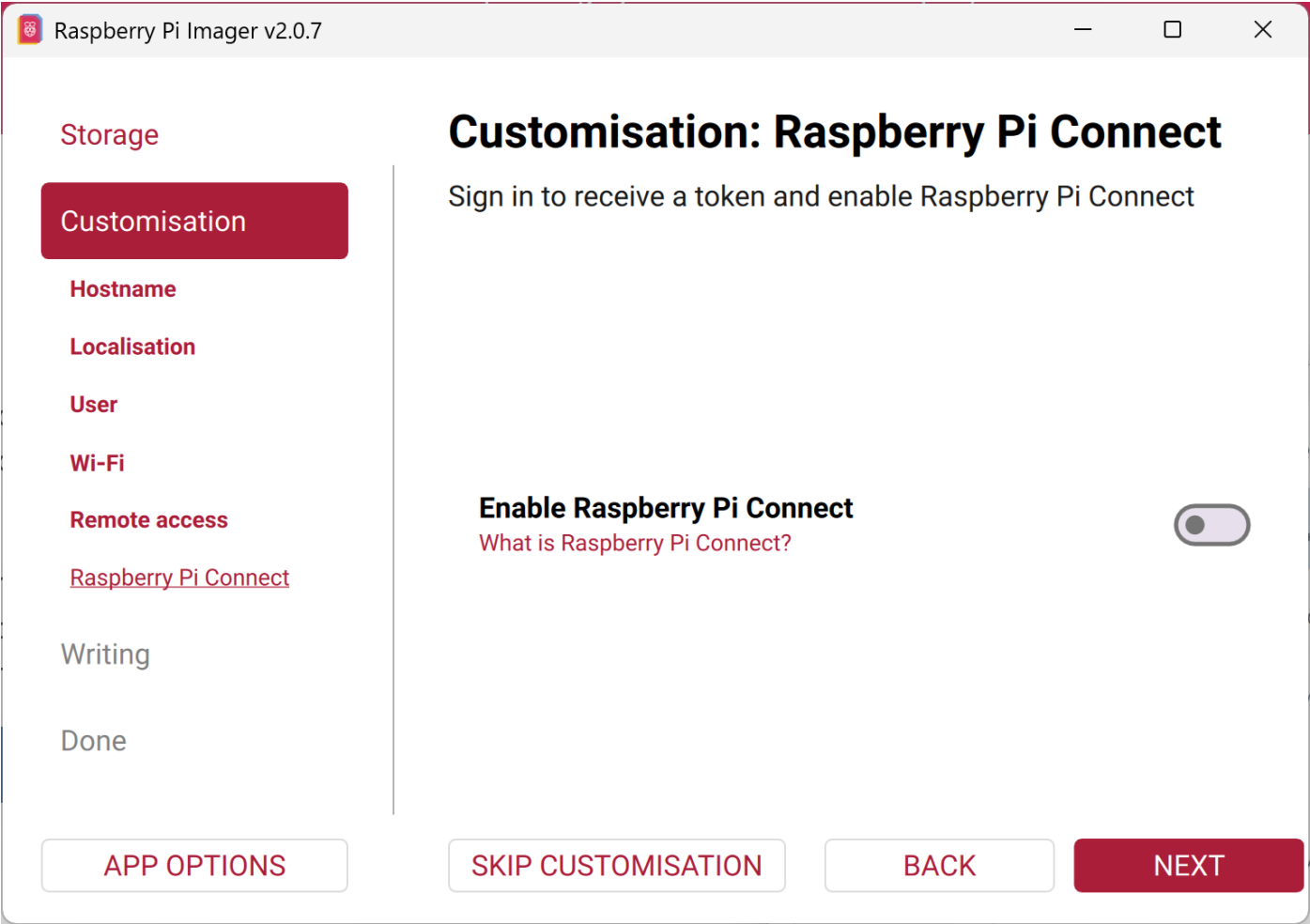
BACK

NEXT

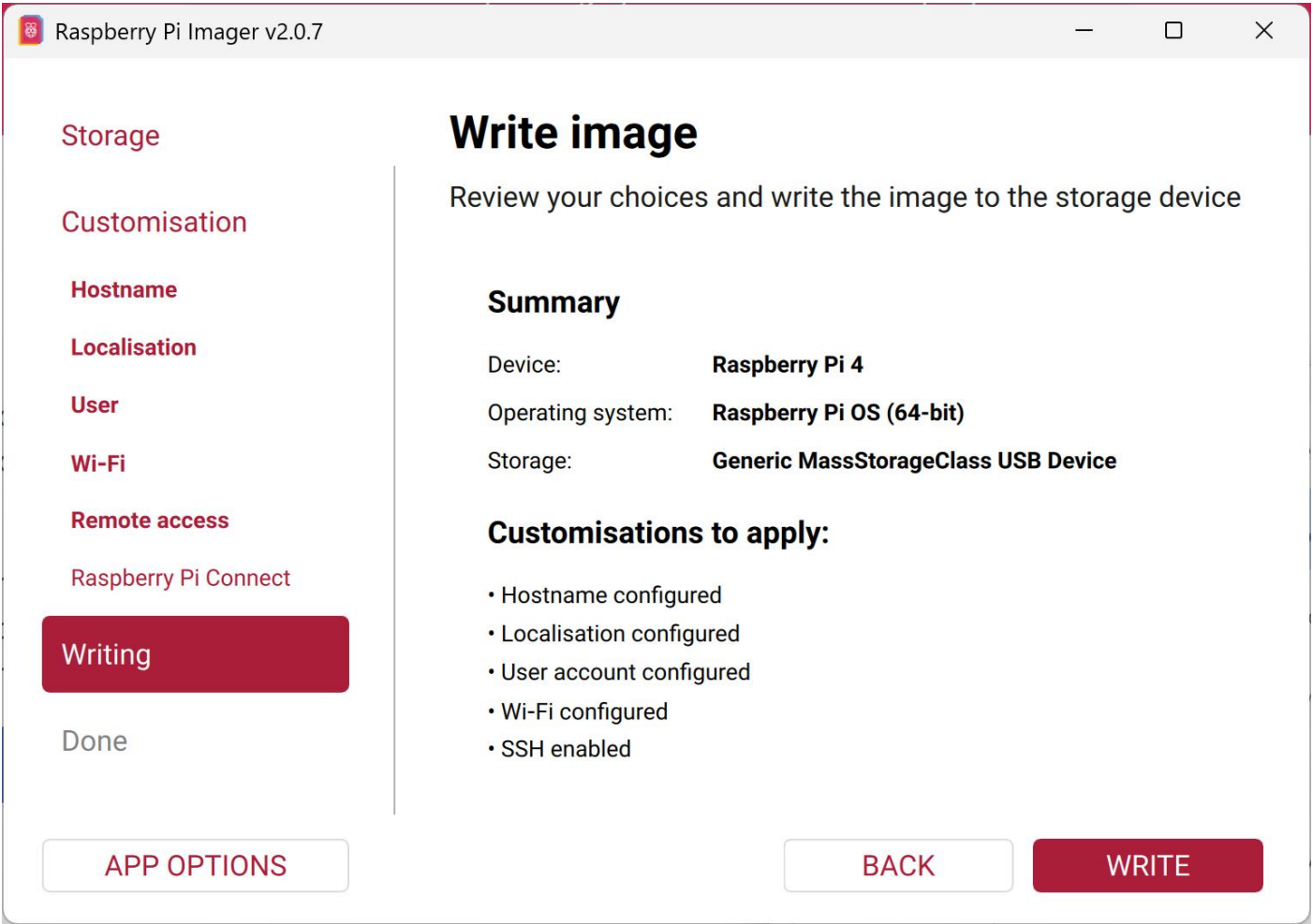
- **Step 9:** Enable SSH and choose password authentication to allow remote administrative access via the terminal.



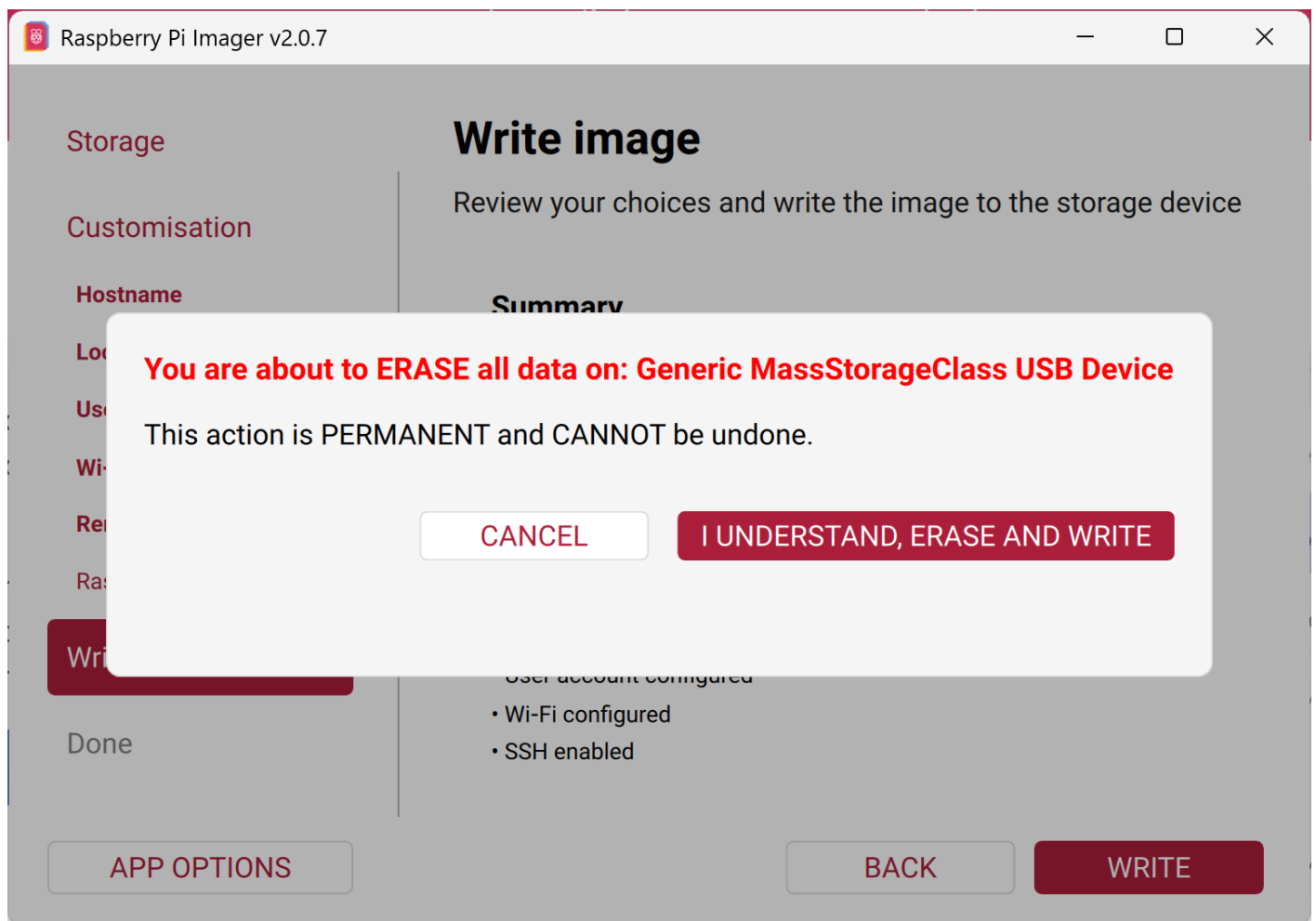
- **Step 10:** Leave the "Raspberry Pi Connect" option disabled and select **Next**.



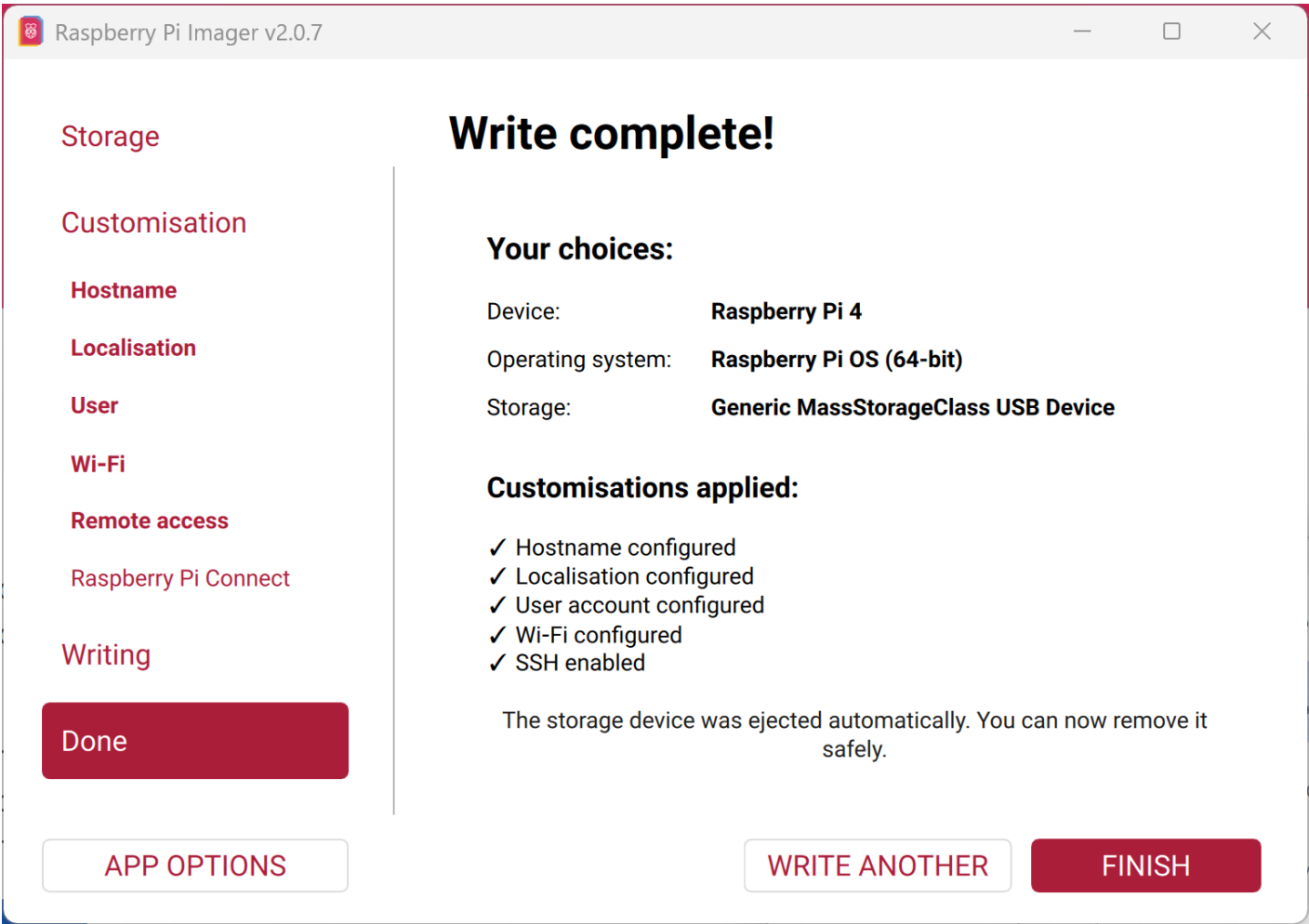
- **Step 11:** Review your configuration choices, then click **"Write"** to begin flashing the image to the SD card.



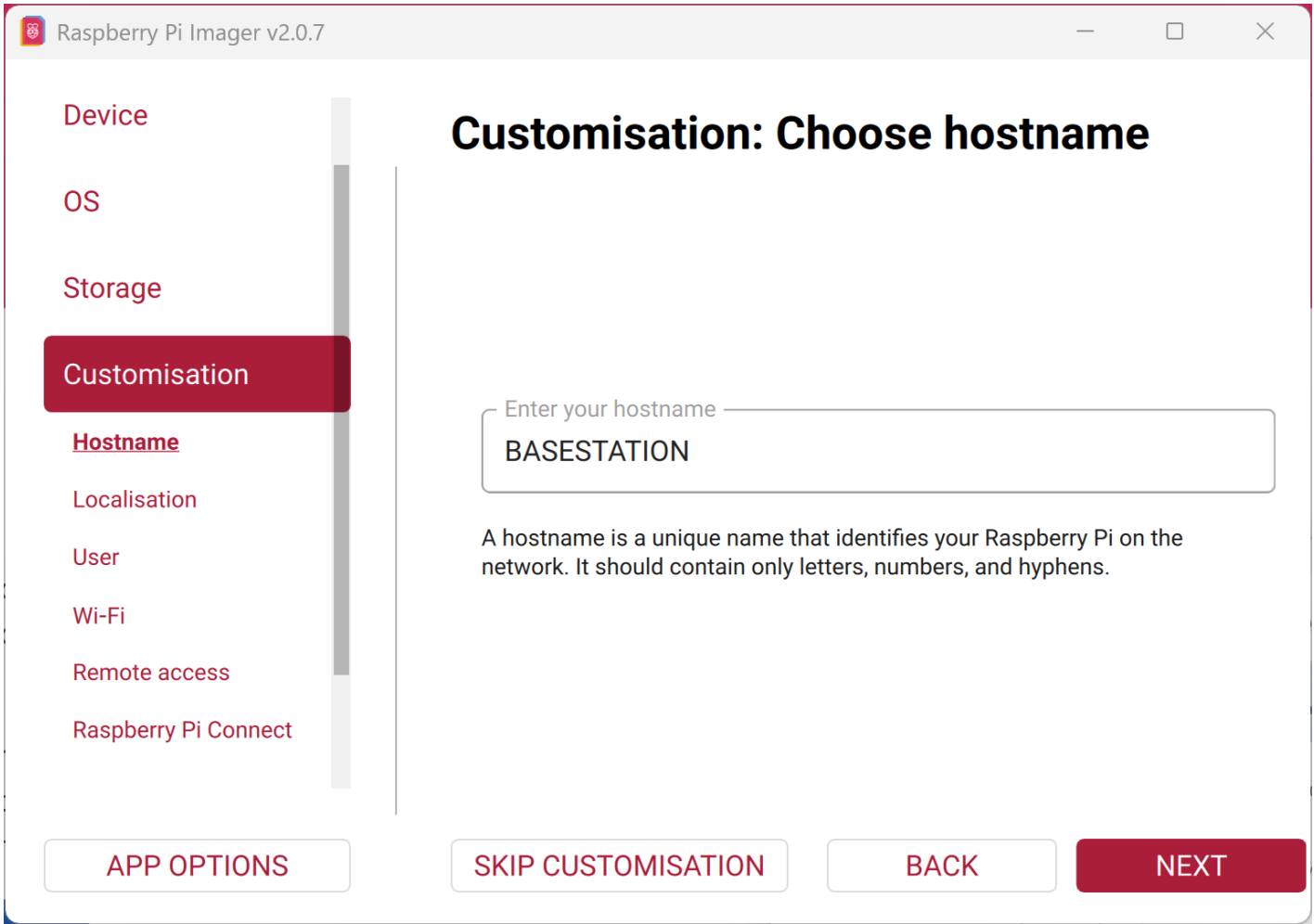
- **Step 12:** A system warning will appear advising that all existing data on the target drive will be erased; click **"I understand, erase and write"** to proceed.



- **Step 13:** Wait for the writing process to finalize. A success message will appear upon completion.



- **Step 14A:** Once the OS image has been successfully written for the Rover, repeat this exact process (Steps 2 through 13) for the Base Station. **Crucial Note:** Ensure that the hostname and username for the Base Station distinctly differ from the credentials established for the Rover.



- **Step 14B:** Once the OS image has been successfully written for the Rover, repeat this exact process (Steps 2 through 13) for the Base Station. **Crucial Note:** Ensure that the hostname and username for the Base Station distinctly differ from the credentials established for the Rover.

Raspberry Pi Imager v2.0.7

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Writing

Done

Customisation: Choose username

Create a user account for your Raspberry Pi

Username:

Enter your username

user-base

Password:

Saved (hidden) — leave blank t...

user-base

Confirm password:

Re-enter to change password

user-base

The username must be lowercase and contain only letters, numbers, underscores, and hyphens.

APP OPTIONS

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[SENSITIVE PERSONAL]

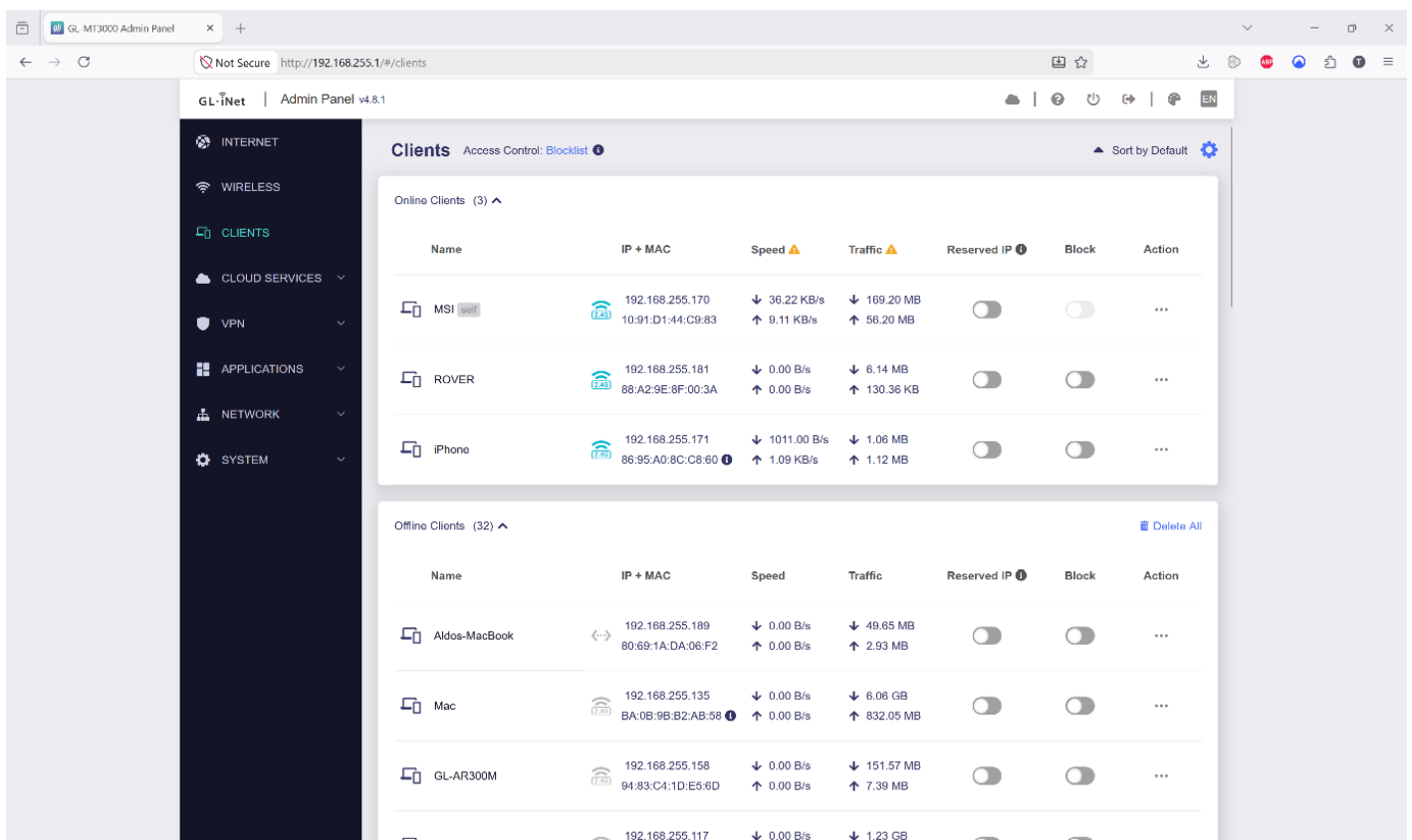
- **Step 15:** Safely unplug the Micro SD card reader from your computer or laptop and remove the Micro SD card.
- **Step 16:** Insert the newly flashed Micro SD card into the Micro SD slot on the Raspberry Pi 4B.
- **Step 17:** To power on the Rover or Base Station, plug the USB Type-C cable into the USB-C port on the side of the Raspberry Pi and connect the USB Type-A connector into the power bank or another appropriate power source.

5 RASPBERRY PI INITIAL SETUP

Once the device is powered on, proceed with the initial network configuration and system updates.

Note: It is highly recommended to use a wired Ethernet connection for this initial SSH session. Making changes to the wireless settings over a Wi-Fi connection may cause the network to drop out during configuration.

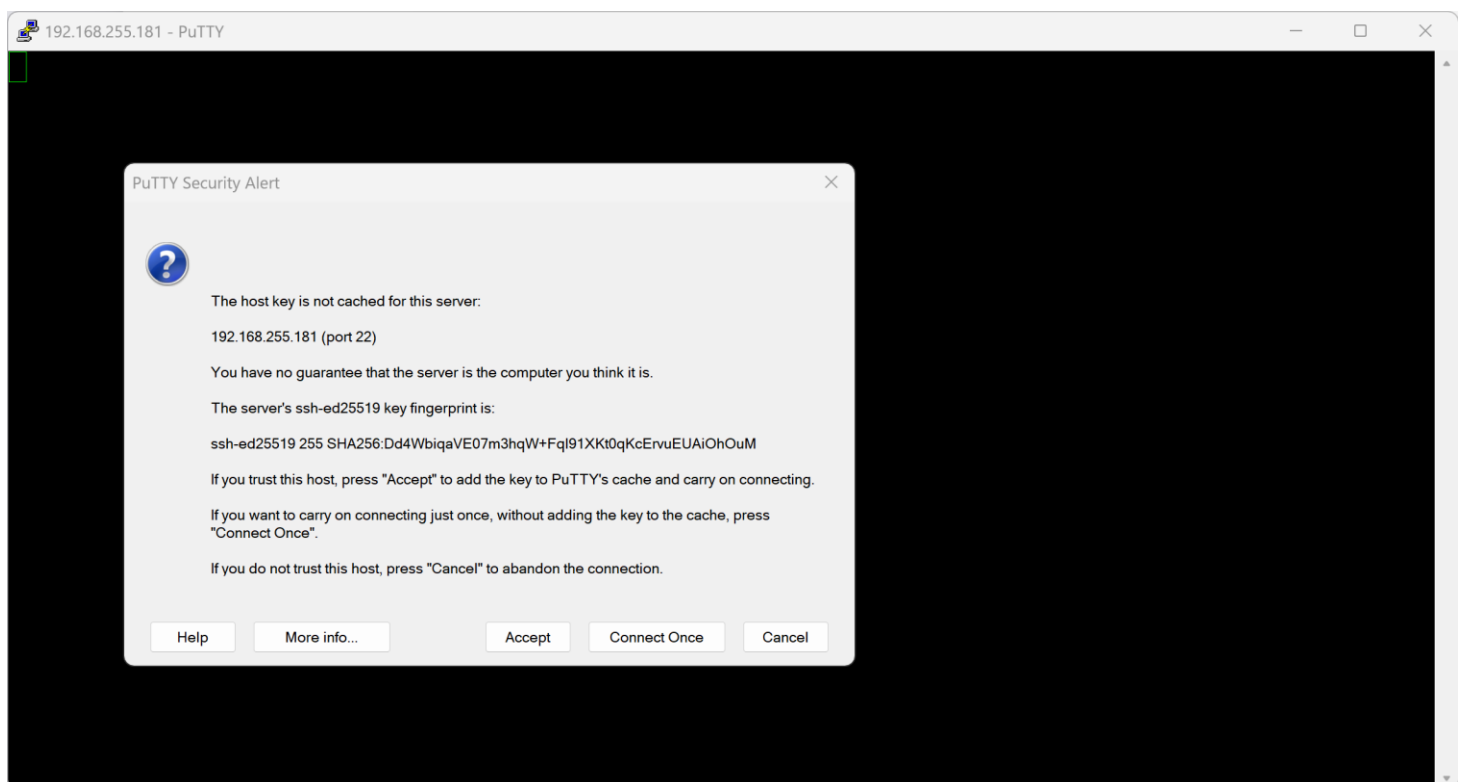
- **Step 1:** Access your Wireless Access Point (AP) administration portal to retrieve the dynamically assigned IP address of the Rover.



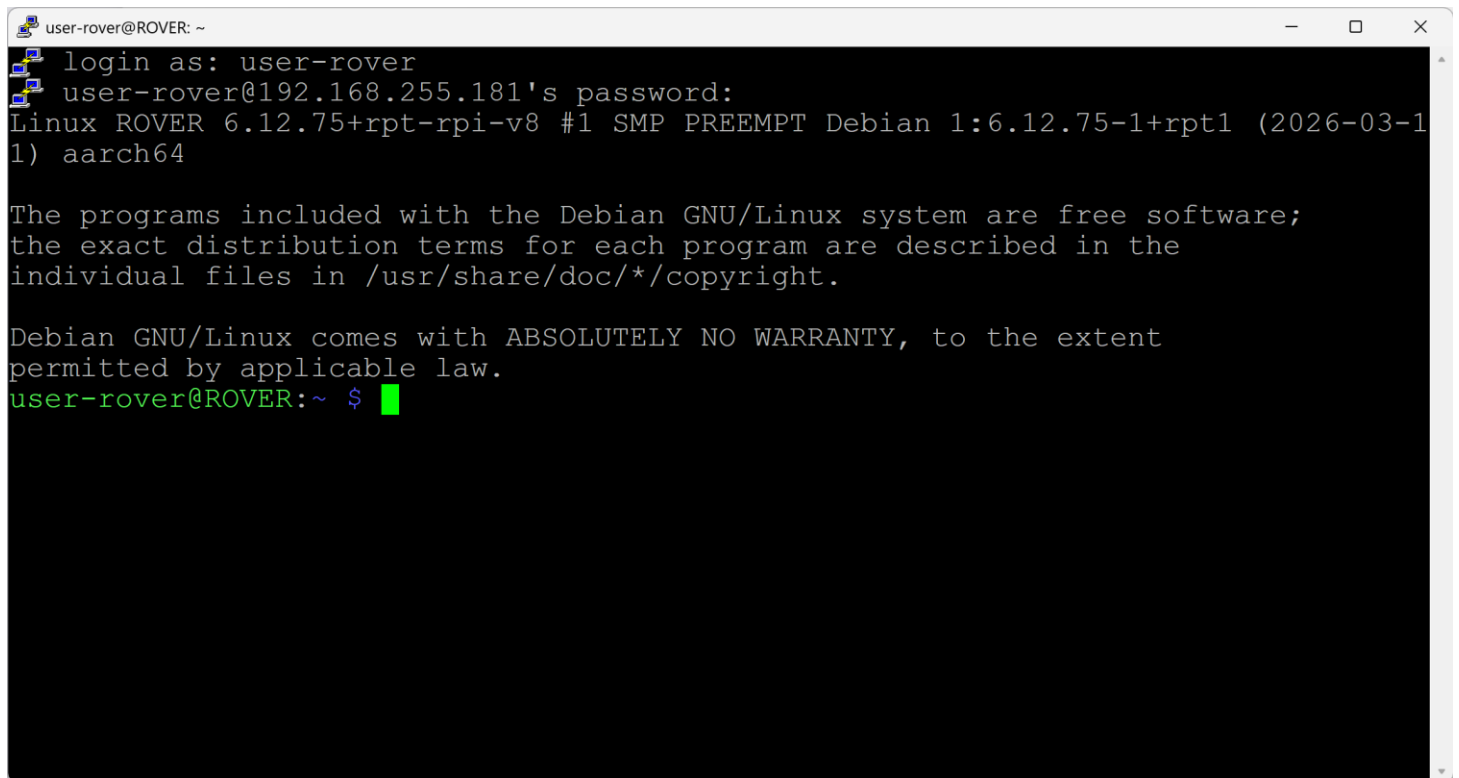
- **Step 2:** Open a Terminal (macOS/Linux) or an SSH client like PuTTY (Windows) to establish a remote connection to the Rover by executing the following command:

```
ssh -l user-rover [IP Address of rover]
```

- **Step 3:** If prompted with a security warning regarding an unrecognized host key, accept the key to establish the connection.



- **Step 4:** Log in using the specific username and password that you configured for the Rover earlier during the OS imaging process.

A terminal window titled 'user-rover@ROVER: ~' with standard window controls. The terminal shows a login sequence: 'login as: user-rover', 'user-rover@192.168.255.181's password:', and a system boot message 'Linux ROVER 6.12.75+rpt-rpi-v8 #1 SMP PREEMPT Debian 1:6.12.75-1+rpt1 (2026-03-11) aarch64'. It then displays the Debian GNU/Linux copyright notice and warranty disclaimer. The prompt 'user-rover@ROVER:~ \$' is shown with a green cursor.

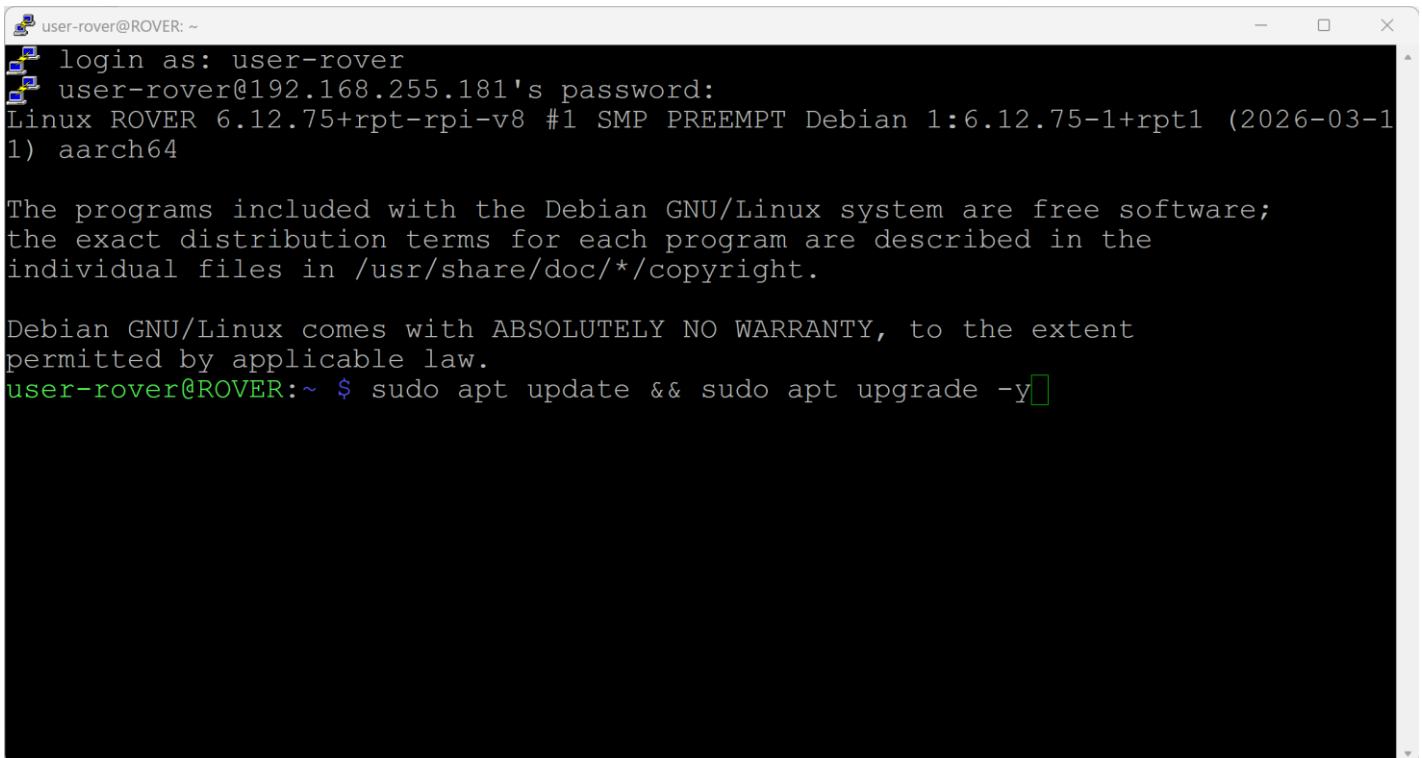
```
user-rover@ROVER: ~
login as: user-rover
user-rover@192.168.255.181's password:
Linux ROVER 6.12.75+rpt-rpi-v8 #1 SMP PREEMPT Debian 1:6.12.75-1+rpt1 (2026-03-11) aarch64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
user-rover@ROVER:~ $
```

- **Step 5:** Once successfully authenticated, execute the following command to ensure all system packages and dependencies are fully updated:

```
sudo apt update && sudo apt upgrade -y
```

A terminal window titled 'user-rover@ROVER: ~' with standard window controls. The terminal shows a login session for 'user-rover' at IP '192.168.255.181'. It displays the system information: 'Linux ROVER 6.12.75+rpt-rpi-v8 #1 SMP PREEMPT Debian 1:6.12.75-1+rpt1 (2026-03-11) aarch64'. It then shows the Debian GNU/Linux copyright notice. Finally, the user enters the command 'sudo apt update && sudo apt upgrade -y' and a green cursor is visible at the end of the line.

```
user-rover@ROVER: ~  
login as: user-rover  
user-rover@192.168.255.181's password:  
Linux ROVER 6.12.75+rpt-rpi-v8 #1 SMP PREEMPT Debian 1:6.12.75-1+rpt1 (2026-03-11) aarch64  
  
The programs included with the Debian GNU/Linux system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent  
permitted by applicable law.  
user-rover@ROVER:~ $ sudo apt update && sudo apt upgrade -y
```


FARM DOG

6 INSTALL FARM DOG PROGRAM

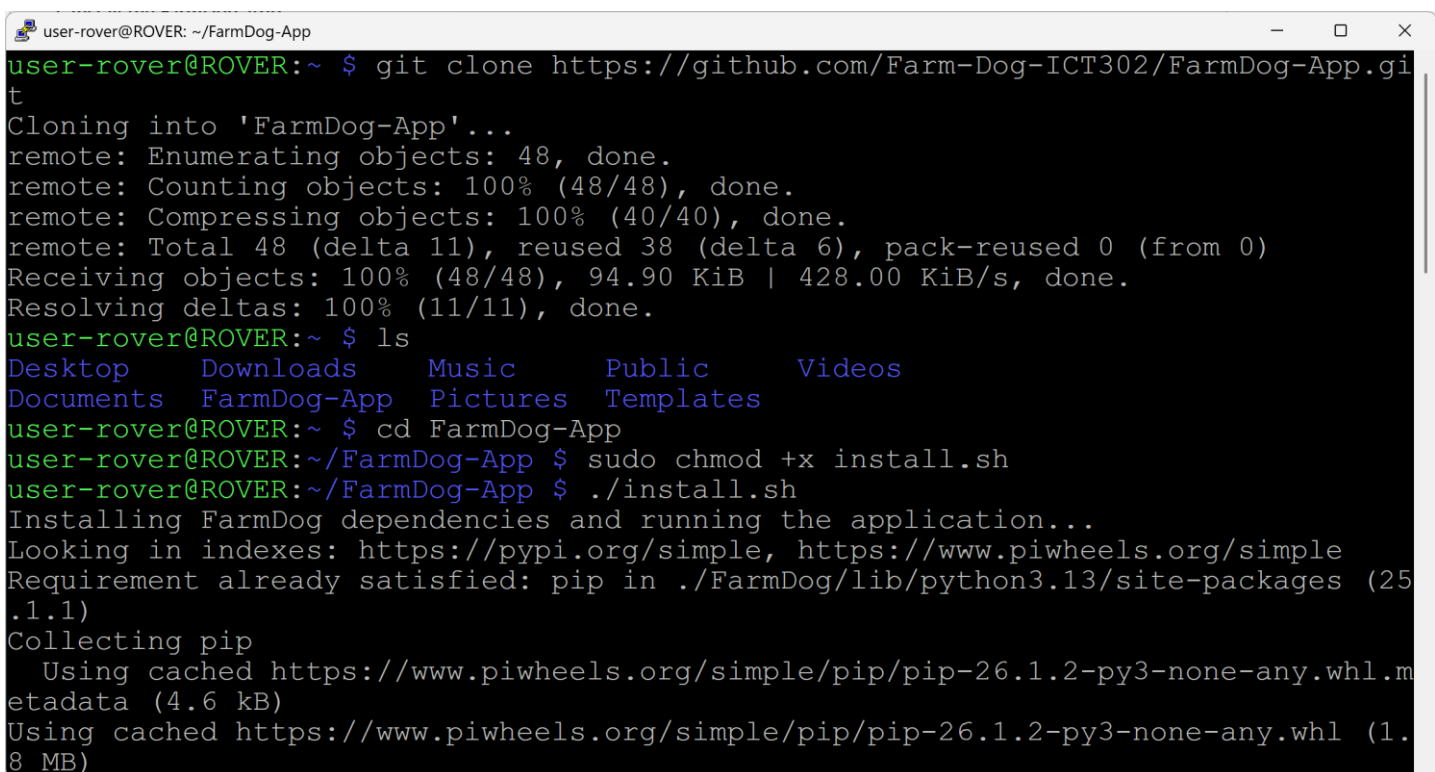
Once the operating system is fully updated, you can proceed with installing the proprietary software on the Rover unit.

- **Step 1:** On the Rover terminal, utilize the `git clone` command to pull the Farm Dog application repository directly from GitHub:

```
git clone https://github.com/Farm-Dog-ICT302/FarmDog-App.git
```

- **Step 2:** Change into the clone Farm-Dog-RTK-GNSS-System directory:

```
cd FarmDog-App
```

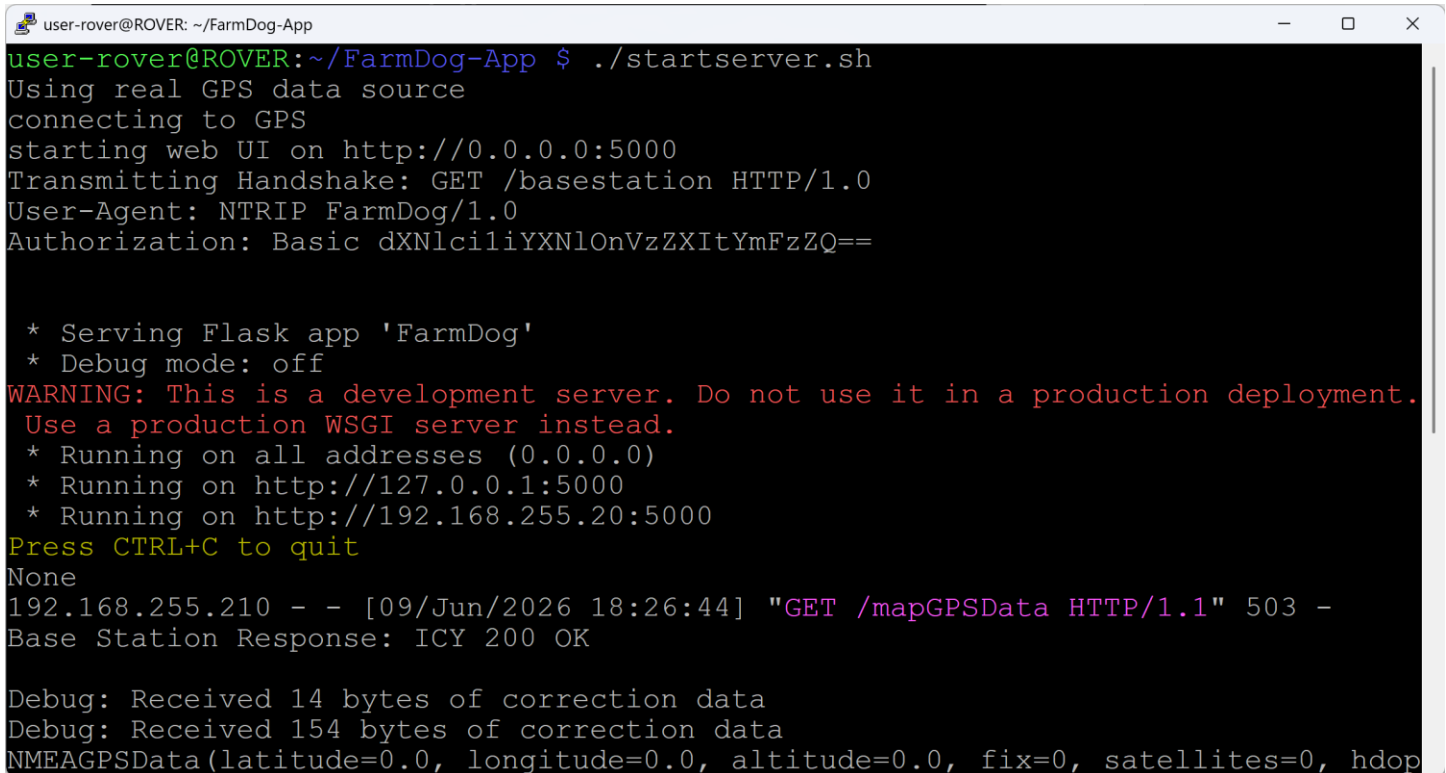


```
user-rover@ROVER: ~/FarmDog-App
user-rover@ROVER:~$ git clone https://github.com/Farm-Dog-ICT302/FarmDog-App.git
Cloning into 'FarmDog-App'...
remote: Enumerating objects: 48, done.
remote: Counting objects: 100% (48/48), done.
remote: Compressing objects: 100% (40/40), done.
remote: Total 48 (delta 11), reused 38 (delta 6), pack-reused 0 (from 0)
Receiving objects: 100% (48/48), 94.90 KiB | 428.00 KiB/s, done.
Resolving deltas: 100% (11/11), done.
user-rover@ROVER:~$ ls
Desktop  Downloads  Music      Public     Videos
Documents  FarmDog-App  Pictures  Templates
user-rover@ROVER:~$ cd FarmDog-App
user-rover@ROVER:~/FarmDog-App$ sudo chmod +x install.sh
user-rover@ROVER:~/FarmDog-App$ ./install.sh
Installing FarmDog dependencies and running the application...
Looking in indexes: https://pypi.org/simple, https://www.piwheels.org/simple
Requirement already satisfied: pip in ./FarmDog/lib/python3.13/site-packages (25.1.1)
Collecting pip
  Using cached https://www.piwheels.org/simple/pip/pip-26.1.2-py3-none-any.whl.metadata (4.6 kB)
Using cached https://www.piwheels.org/simple/pip/pip-26.1.2-py3-none-any.whl (1.8 MB)
```

FARM DOG

- **Step 3:** Execute the initialization script to start the local server and run the application:

```
./startserver.s
```

A terminal window titled 'user-rover@ROVER: ~/FarmDog-App' showing the execution of the './startserver.sh' script. The output includes messages about using real GPS data, connecting to GPS, starting a web UI on http://0.0.0.0:5000, and transmitting a handshake. It also shows a warning about using a development server and the Flask app 'FarmDog' running on three addresses. The terminal ends with a debug message showing received correction data and a JSON object for NMEAGPSData.

```
user-rover@ROVER: ~/FarmDog-App
user-rover@ROVER:~/FarmDog-App $ ./startserver.sh
Using real GPS data source
connecting to GPS
starting web UI on http://0.0.0.0:5000
Transmitting Handshake: GET /basestation HTTP/1.0
User-Agent: NTRIP FarmDog/1.0
Authorization: Basic dXNlciliYXNlOnVzZXItYmFzZQ==

* Serving Flask app 'FarmDog'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment.
Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://192.168.255.20:5000
Press CTRL+C to quit
None
192.168.255.210 - - [09/Jun/2026 18:26:44] "GET /mapGPSData HTTP/1.1" 503 -
Base Station Response: ICY 200 OK

Debug: Received 14 bytes of correction data
Debug: Received 154 bytes of correction data
NMEAGPSData(latitude=0.0, longitude=0.0, altitude=0.0, fix=0, satellites=0, hdop=
```

- **Step 4:** (Optional Diagnostic) You can optionally install a graphical user interface tool called PyGPSClient to monitor the raw GNSS receiver data directly. To install it, run the following command:

```
pip install pygpsclient
```

7 INSTALLING RTKBASE FOR BASE STATION

With the operating system installed and updated, proceed with installing the RTKBase software to handle the GNSS correction data.

- **Step 1:** Access your Wireless Access Point (AP) administration portal to retrieve the dynamically assigned IP address of the Base Station.
- **Step 2:** Ensure your GNSS receiver (u-blox ZED-F9P) is physically connected via USB to the Base Station Raspberry Pi.
- **Step 3:** Open an SSH terminal session to the Base Station and execute the following commands sequentially to download, grant permissions, and run the RTKBase installation script:

```
cd ~

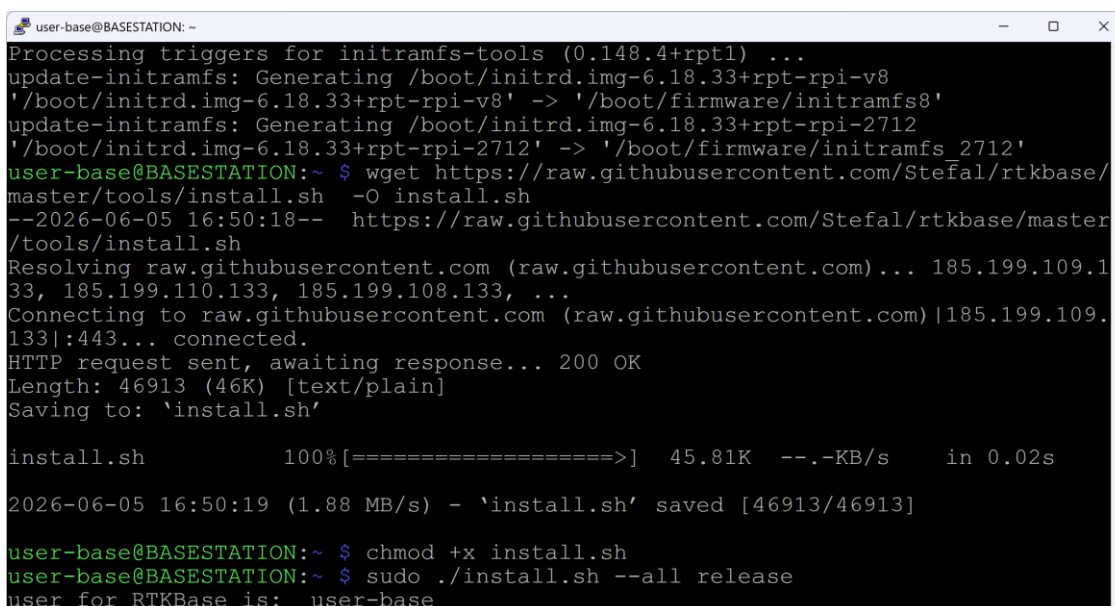
wget

https://raw.githubusercontent.com/Stefal/rtkbase/

master/tools/install.sh -O install.sh

chmod +x install.sh

sudo ./install.sh --all release
```



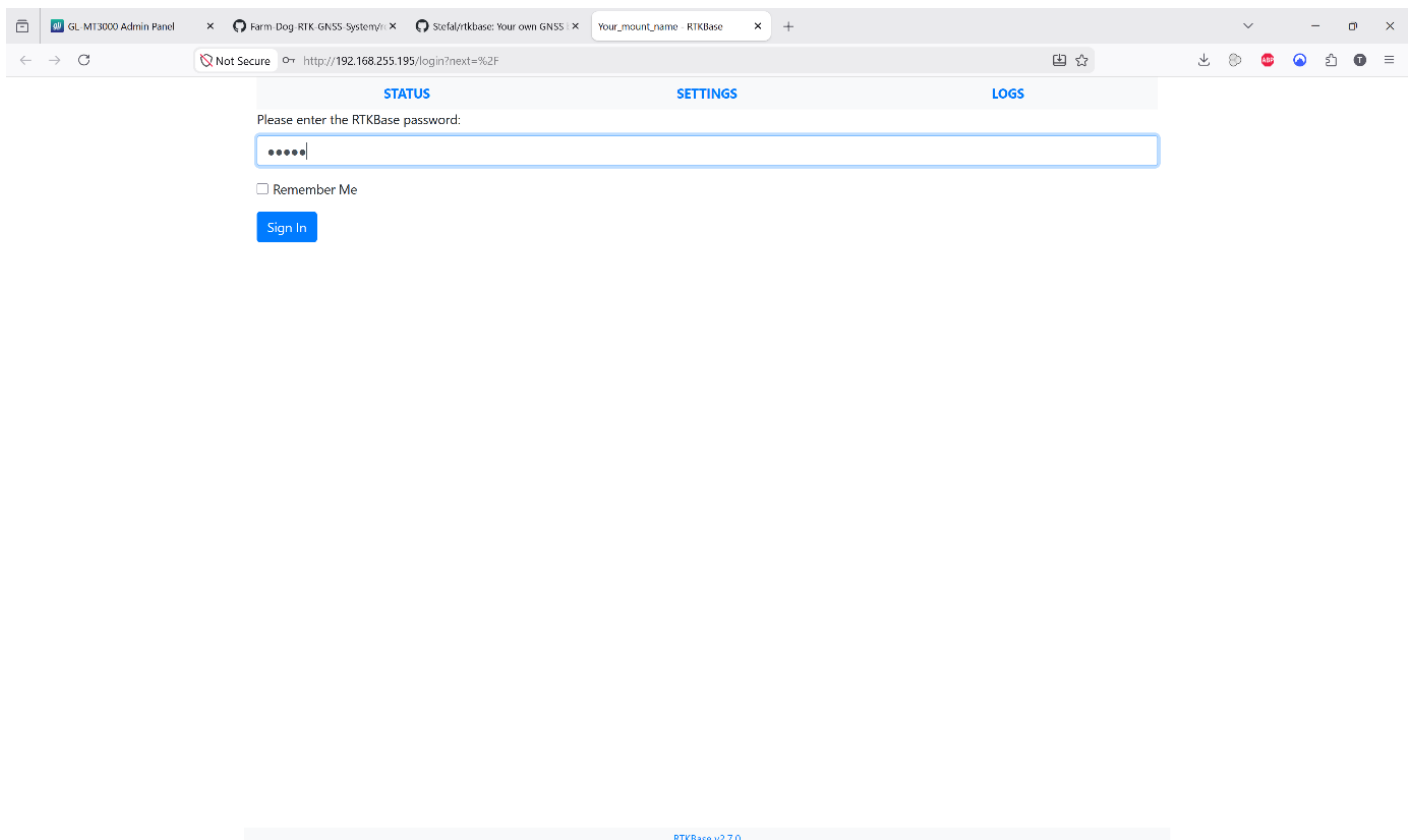
```
user-base@BASESTATION: ~
Processing triggers for initramfs-tools (0.148.4+rpt1) ...
update-initramfs: Generating /boot/initrd.img-6.18.33+rpt-rpi-v8
'/boot/initrd.img-6.18.33+rpt-rpi-v8' -> '/boot/firmware/initramfs8'
update-initramfs: Generating /boot/initrd.img-6.18.33+rpt-rpi-2712
'/boot/initrd.img-6.18.33+rpt-rpi-2712' -> '/boot/firmware/initramfs_2712'
user-base@BASESTATION:~ $ wget https://raw.githubusercontent.com/Stefal/rtkbase/
master/tools/install.sh -O install.sh
--2026-06-05 16:50:18-- https://raw.githubusercontent.com/Stefal/rtkbase/master
/tools/install.sh
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.109.1
33, 185.199.110.133, 185.199.108.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.109.
133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 46913 (46K) [text/plain]
Saving to: 'install.sh'

install.sh          100%[=====] 45.81K  --.-KB/s   in 0.02s

2026-06-05 16:50:19 (1.88 MB/s) - 'install.sh' saved [46913/46913]

user-base@BASESTATION:~ $ chmod +x install.sh
user-base@BASESTATION:~ $ sudo ./install.sh --all release
user for RTKBase is: user-base
```

- **Step 4:** Once the terminal script finishes processing, open a web browser on your host computer and navigate to `http://ip\_of\_your\_sbc` (the installation script will output the exact IP address you need to use). Log in to the web interface using the default password: `admin`.



- **Step 5:** Navigate to the “Settings” tab, click on "Options" under the Main Service section, and then click "Detect/Configure" to allow the system to identify the connected GNSS Receiver.

GL MT3000 Admin PanelFarm Dog RTK GNSS SystemStefal/rtkbase: Your own GNSSYour mount name: RTKBase

Not Securehttp://192.168.255.195/settings

STATUSSETTINGSLOGS

Services:

Main service

OnOptions

Base coordinates

47.0983869 -1.2655108 36.40

Antenna's coordinates: latitude (decimal degree) longitude (decimal degree) elevation (meter)

Com port

ttyGNSS

Detect/Configure

Gnss receiver com port (without /dev/)

Com port settings

115200:8:n:1

Com port settings: baudrate: data bits: parity: stop bits: flow control

Receiver type

U-blox_ZED-F9P

Gnss receiver brand and model

Receiver format

ubx

Gnss receiver format (ubx,rtcm3,tcm3,tcm3,tcm3,...)

Antenna info

ADVNULANTENNA

ADVNULANTENNA or information (model,serial number) for IGS calibrated antenna

Tcp host addr

127.0.0.1

Empty value enable external access to the Gnss receiver from another computer. Default is 127.0.0.1 (no external access)

Tcp port

5015

Local tcp port

Save

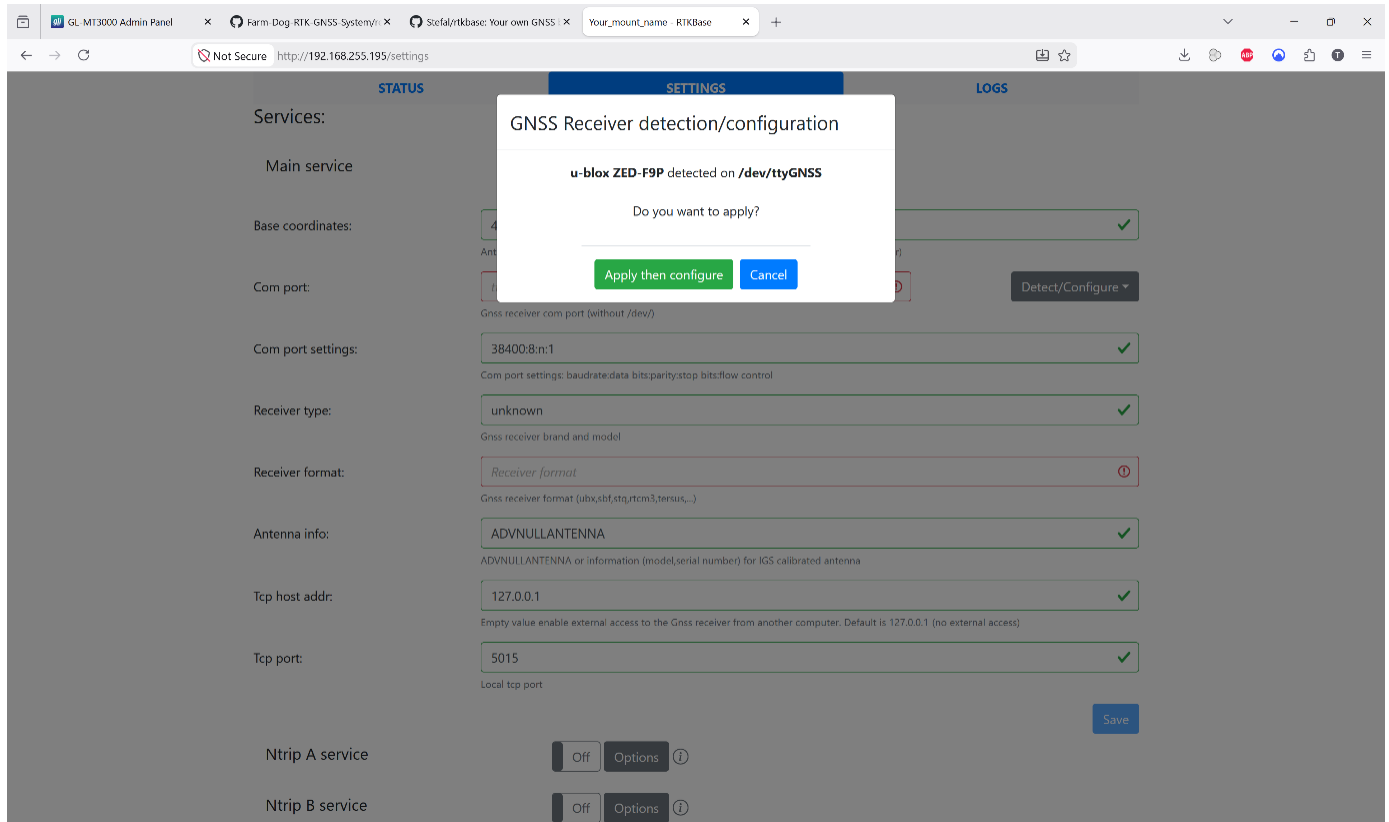
Ntrip A service

OffOptions

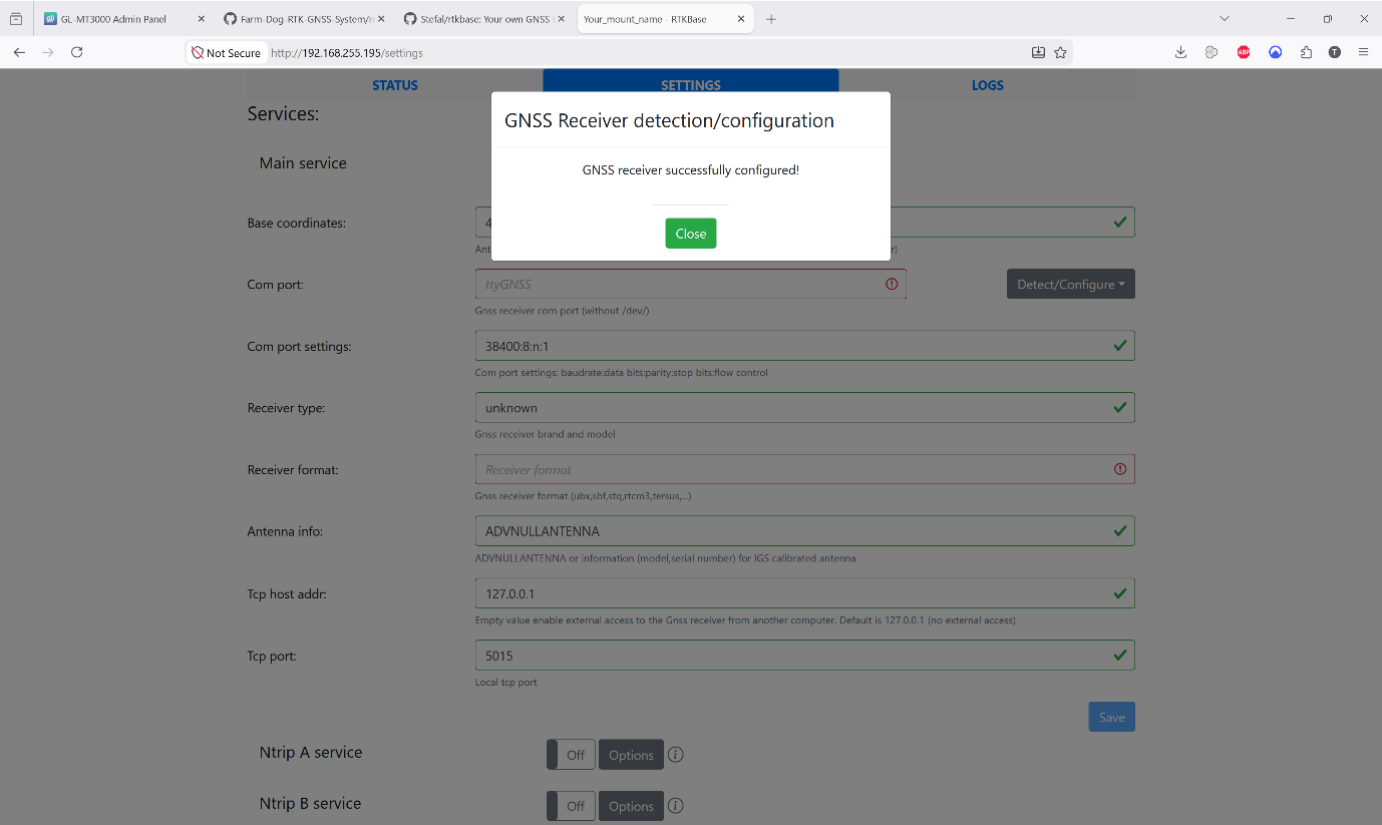
Ntrip B service

OffOptions

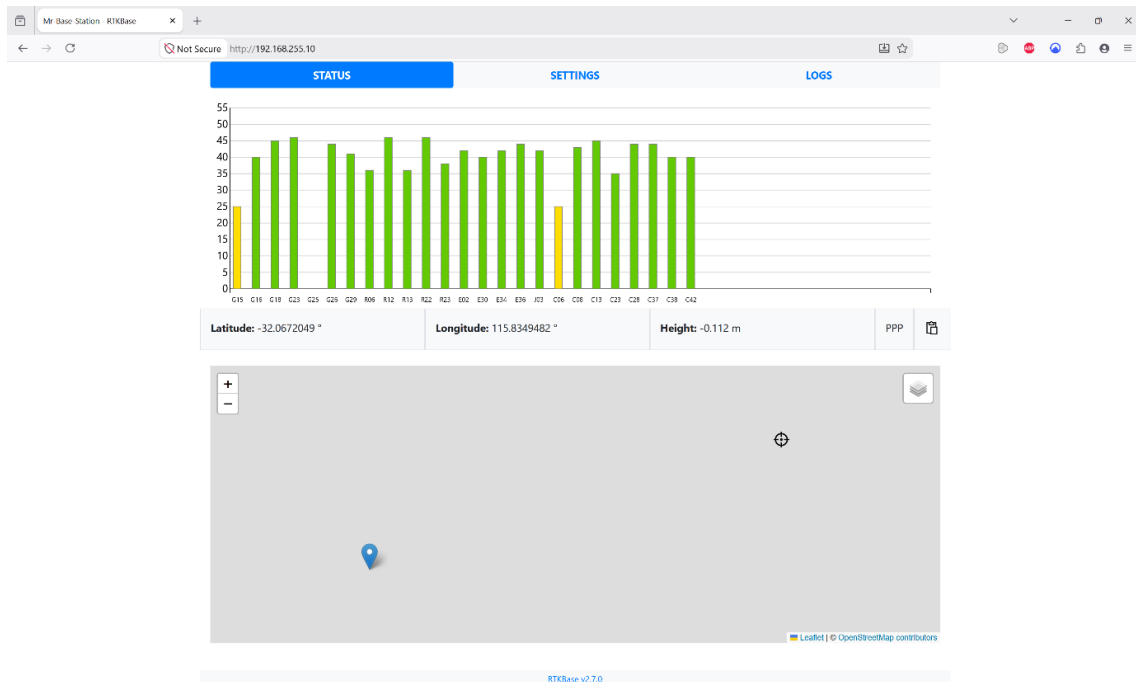
- **Step 6:** The system should automatically detect the ZED-F9P module and prompt a confirmation message. Click "Apply then configure".



- **Step 7:** A success message will appear confirming that the GNSS Receiver has been configured successfully. Click "Close".



- **Step 8:** Navigate to the “**Status**” page and copy the active coordinates of the Base Station. Paste these exact values into the “**Base Coordinates**” field in the settings to establish the fixed reference location, then click **Save**.



The screenshot shows the 'SETTINGS' page of the RTKBase web interface. At the top, there are three tabs: 'STATUS', 'SETTINGS' (selected), and 'LOGS'. Below the tabs, the 'Services' section is visible. The 'Main service' is set to 'On'. The 'Base coordinates' field is set to '47.0983869 -1.2655108 36.40'. The 'Com port' is set to 'ttyGNSS'. The 'Com port settings' are set to '115200:8n:1'. The 'Receiver type' is set to 'U-blox ZED-F9P'. The 'Receiver format' is set to 'ubx'. The 'Antenna info' is set to 'ADVNULANTENNA'. The 'Tcp host addr' is set to '127.0.0.1'. The 'Tcp port' is set to '5015'. There are 'Save' and 'Options' buttons for each setting. The 'Ntrip A service' and 'Ntrip B service' are both set to 'Off'.

- **Step 9:** To enable the correction broadcast, toggle the “Ntrip Caster Service” to "On" and click "Options". Enter the designated username and password for the Base Station user, set the local caster mount name to "basestation", and click **Save**.

The screenshot shows the 'Ntrip Caster Service' configuration page in the Farm Dog RTK GNSS System Admin Panel. The browser address bar shows the URL 'http://192.168.255.195/settings'. The page contains several configuration fields and service toggles.

Tcp port: 5015 (checked)

Local tcp port: (empty)

Save button

Ntrip A service: Off Options ⓘ

Ntrip B service: Off Options ⓘ

Ntrip Caster service: On Options ⓘ

Local caster username: user-base (checked)
Local Ntrip caster username you will use in your rover to connect

Local caster password: user-base (checked) ⓘ
Local Ntrip caster password you will use in your rover to connect

Local caster port: 2101 (checked)
Local Ntrip Caster port number, e.g. 2101

Local caster mount name: basestation (checked)
Local Ntrip caster mount name

Local caster rtcm messages: 1004,1005(10),1006,1008(10),1012,1019,1020,1033(10),1042,1045,1046,1077,1087,1097,1107,1127,1230 (checked)
Local caster Rtc messages list: msg(interval in seconds),msg(interval in seconds),...

Local caster receiver options: -TADJ=1 (checked)
Receiver dependent options, e.g. -TADJ=1 for U-Blox F9P

Save button

Rtcm tcp service: Off Options ⓘ

Rtcm serial service: Off Options ⓘ

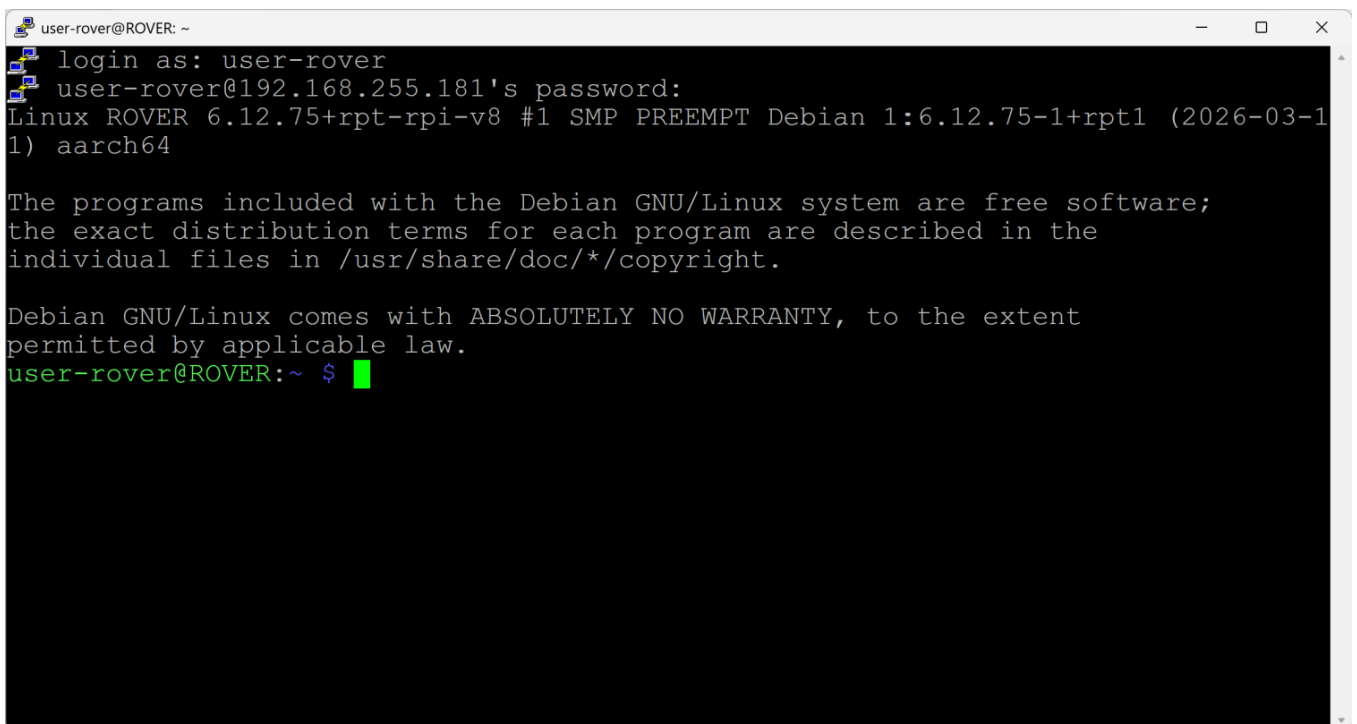
8 INSTALL DRIVERS FOR SIMPLECOM NW621 USB WI-FI ADAPTER

This section walks you through the installation of the USB Wi-Fi adapter that is used for the Base Station and the Rover. It also includes steps to modify the configuration file to ensure a more stable Wi-Fi connection.

- **Step 1:** Plug the USB Wi-Fi module into the USB 3.0 port on the Raspberry Pi.
- **Step 2:** Navigate to the following GitHub page to download a third-party open-source driver:

<https://github.com/morrownr/88x2bu-20210702>

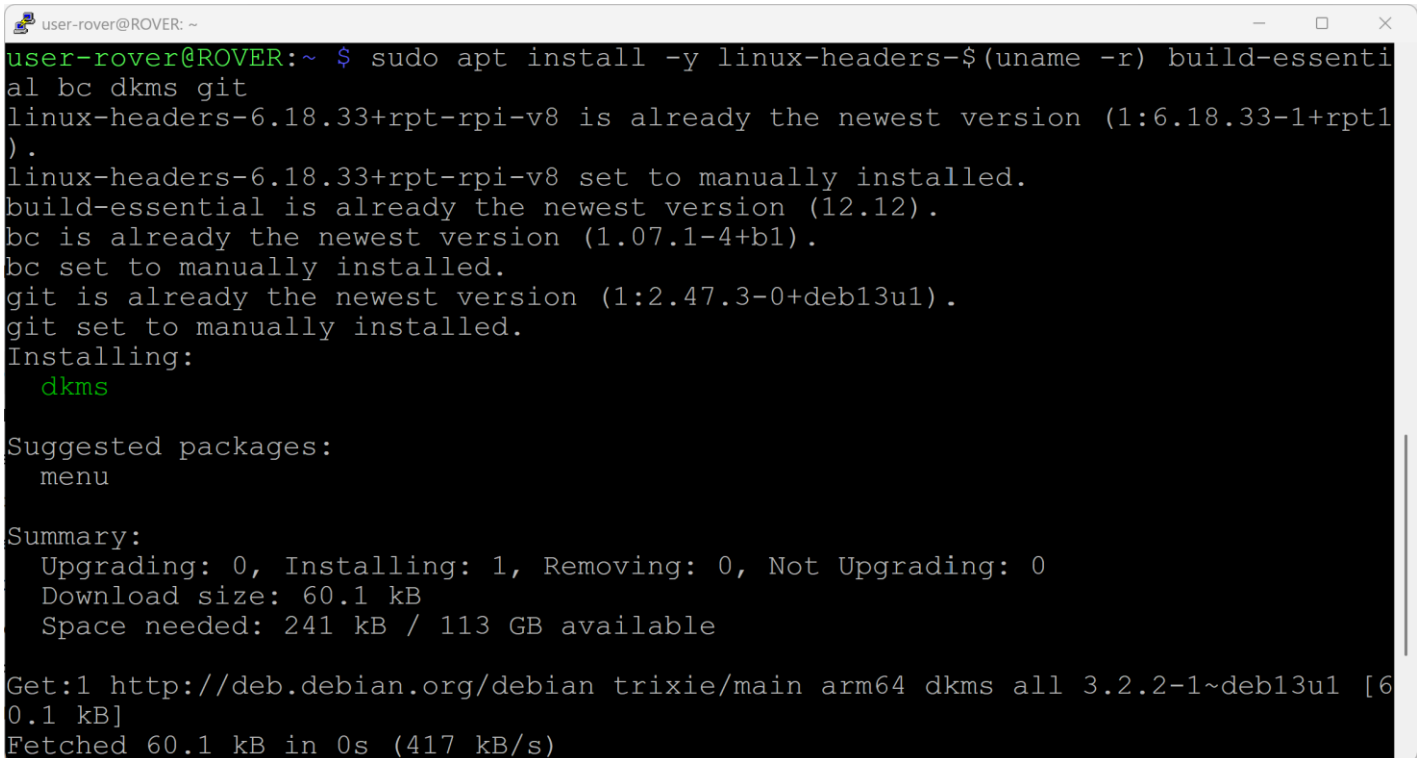
- **Step 3:** Connect to the Raspberry Pi using SSH through PuTTY.



```
user-rover@ROVER: ~  
login as: user-rover  
user-rover@192.168.255.181's password:  
Linux ROVER 6.12.75+rpt-rpi-v8 #1 SMP PREEMPT Debian 1:6.12.75-1+rpt1 (2026-03-11) aarch64  
  
The programs included with the Debian GNU/Linux system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent  
permitted by applicable law.  
user-rover@ROVER:~ $
```

- **Step 4:** Install the required packages with the following command:

```
sudo apt install -y linux-headers-$(uname -r)
build-essential bc dkms git
```

A terminal window titled 'user-rover@ROVER: ~' showing the execution of 'sudo apt install -y linux-headers-\$(uname -r) build-essential bc dkms git'. The output indicates that linux-headers-6.18.33-rpt-rpi-v8, build-essential, bc, and git are already the newest versions and are set to manually installed. The package dkms is being installed. Suggested packages include menu. A summary shows 1 package being installed with a download size of 60.1 kB and a space needed of 241 kB. The dkms package is fetched from the trixie/main arm64 repository.

```
user-rover@ROVER:~ $ sudo apt install -y linux-headers-$(uname -r) build-essential bc dkms git
linux-headers-6.18.33+rpt-rpi-v8 is already the newest version (1:6.18.33-1+rpt1).
linux-headers-6.18.33+rpt-rpi-v8 set to manually installed.
build-essential is already the newest version (12.12).
bc is already the newest version (1.07.1-4+b1).
bc set to manually installed.
git is already the newest version (1:2.47.3-0+deb13u1).
git set to manually installed.
Installing:
  dkms

Suggested packages:
  menu

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 0
  Download size: 60.1 kB
  Space needed: 241 kB / 113 GB available

Get:1 http://deb.debian.org/debian trixie/main arm64 dkms all 3.2.2-1~deb13u1 [60.1 kB]
Fetched 60.1 kB in 0s (417 kB/s)
```

- **Step 5:** Create a directory to hold the downloaded driver using this

command:

```
o mkdir -p ~/src
```

- **Step 6:** Move to the newly created directory:

```
o cd ~/src
```

- **Step 7:** Download the driver from the GitHub repository using the `git`

clone command:

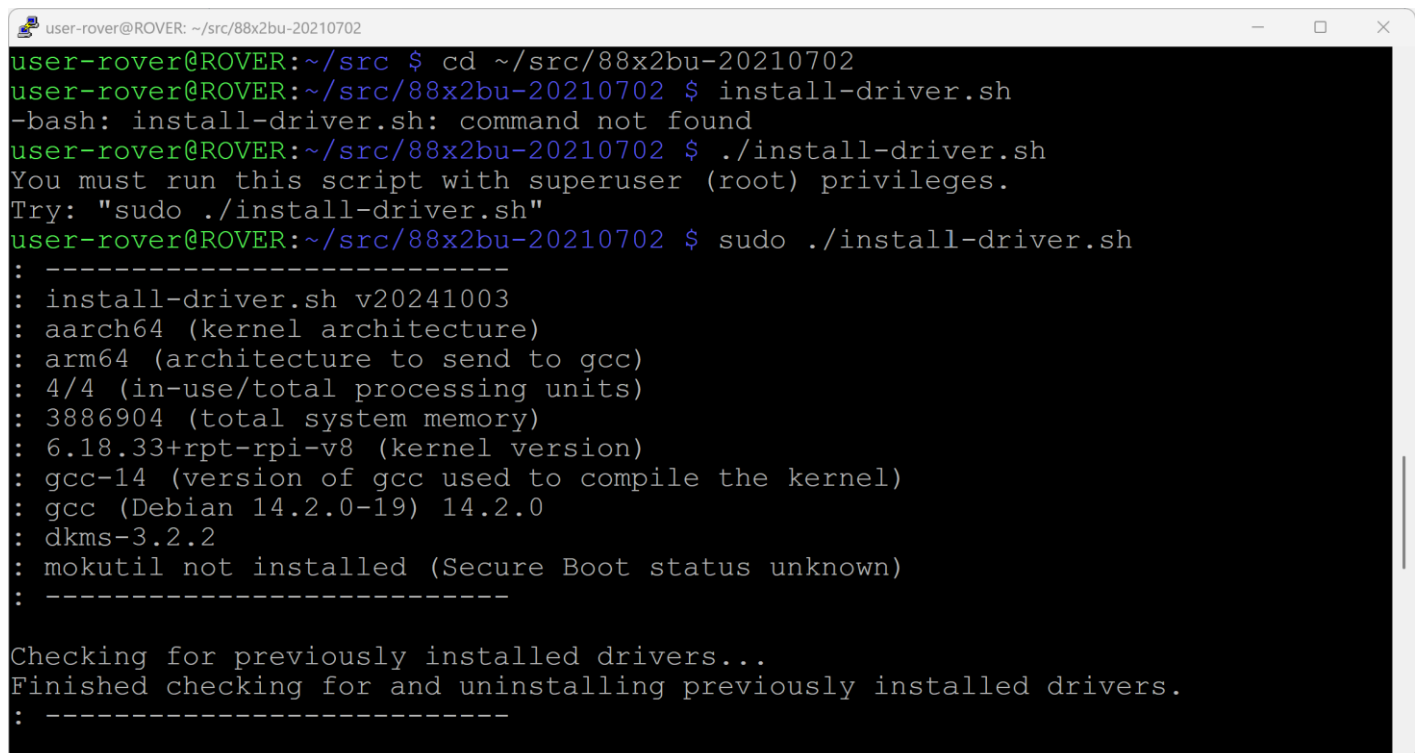
- o `git clone https://github.com/morrownr/88x2bu-20210702.git`

- **Step 8:** Move to the newly created driver directory:

- o `cd ~/src/88x2bu-20210702`

- **Step 9:** Run the installation script with this command:

- o `sudo ./install-driver`



```
user-rover@ROVER: ~/src/88x2bu-20210702
user-rover@ROVER:~/src $ cd ~/src/88x2bu-20210702
user-rover@ROVER:~/src/88x2bu-20210702 $ install-driver.sh
-bash: install-driver.sh: command not found
user-rover@ROVER:~/src/88x2bu-20210702 $ ./install-driver.sh
You must run this script with superuser (root) privileges.
Try: "sudo ./install-driver.sh"
user-rover@ROVER:~/src/88x2bu-20210702 $ sudo ./install-driver.sh
: -----
: install-driver.sh v20241003
: aarch64 (kernel architecture)
: arm64 (architecture to send to gcc)
: 4/4 (in-use/total processing units)
: 3886904 (total system memory)
: 6.18.33+rpt-rpi-v8 (kernel version)
: gcc-14 (version of gcc used to compile the kernel)
: gcc (Debian 14.2.0-19) 14.2.0
: dkms-3.2.2
: mokutil not installed (Secure Boot status unknown)
: -----
Checking for previously installed drivers...
Finished checking for and uninstalling previously installed drivers.
: -----
```

- **Step 10:** After the driver has been successfully installed, you will be asked if you want to edit the driver options file now. Type **"Yes"** and press Enter.

```
user-rover@ROVER: ~/src/88x2bu-20210702
..... done.
The driver was built by dkms successfully.
: -----
Installing /lib/modules/6.18.33+rpt-rpi-v8/updates/dkms/88x2bu.ko.xz
Running depmod.... done.
The driver was installed by dkms successfully.
: -----

Info: Update this driver with the following commands as needed:

$ git pull
$ sudo sh install-driver.sh

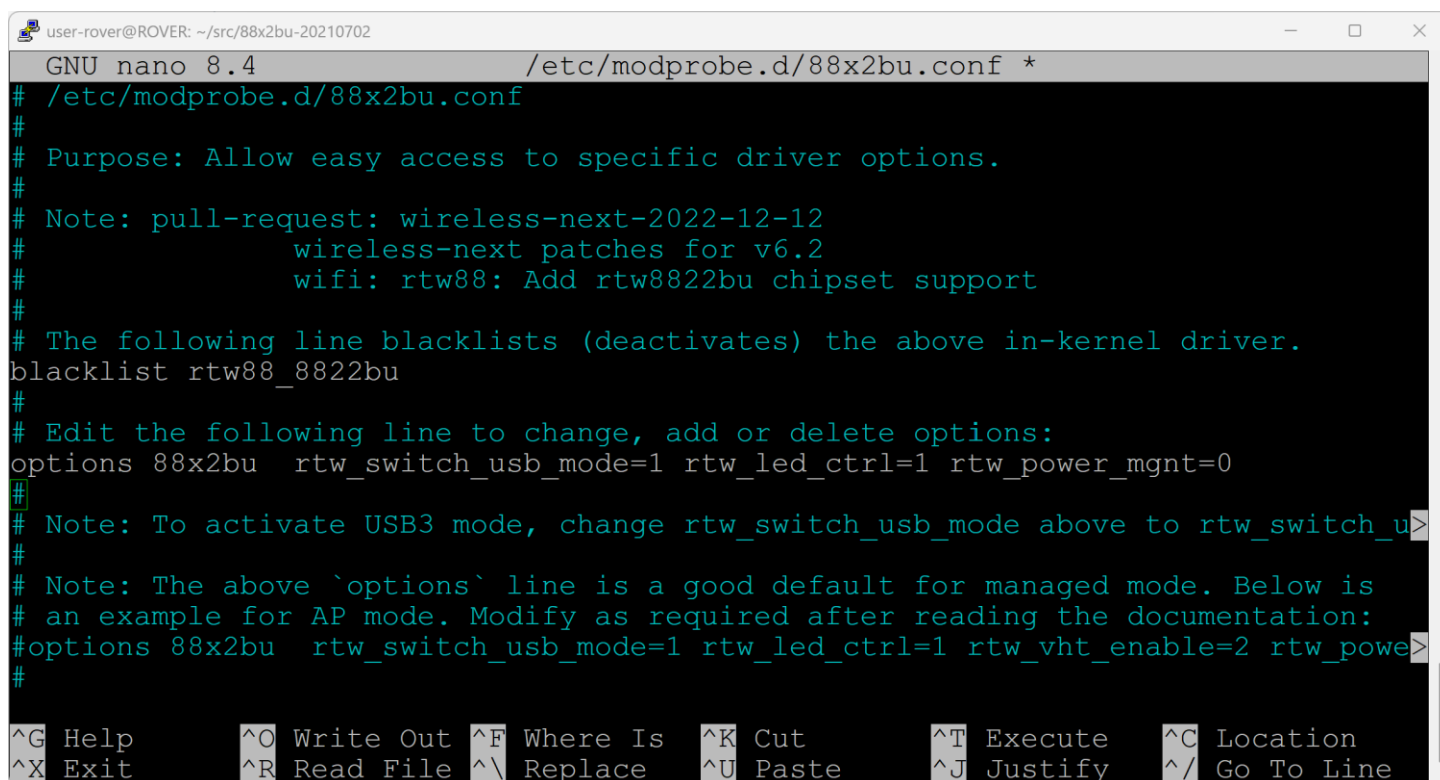
Note: Updates to this driver SHOULD be performed before distro
      upgrades such as Ubuntu 23.10 to 24.04.
Note: Updates to this driver SHOULD be performed before major
      upgrades such as kernel 6.5 to 6.6.
Note: Updates can be performed as often as you like. It is
      recommended to update at least every 3 months.
Note: Work on this driver, like the Linux kernel, is continuous.

Enjoy!

Do you want to edit the driver options file now? (recommended) [Y/n] ☐
```

- **Step 11:** Inside the driver options file, locate the commented line: *"Edit the following line to change, add or delete options:"*.
- **Step 12:** Change the option for the USB mode from `rtw_switch_usb_mode=0` to `rtw_switch_usb_mode=1` so the USB Wi-Fi module is forced to use the USB 3.0 mode instead of the slower USB 2.0 mode.

- **Step 13:** Add `rtw_power_mgnt=0` to the exact same line to turn off power saving.
- **Step 14:** Save the file by pressing **CTRL+O** and hit **Enter** to keep the same file name. After this, press **CTRL+X** to exit the file.



```
user-rover@ROVER: ~/src/88x2bu-20210702
GNU nano 8.4 /etc/modprobe.d/88x2bu.conf *
# /etc/modprobe.d/88x2bu.conf
#
# Purpose: Allow easy access to specific driver options.
#
# Note: pull-request: wireless-next-2022-12-12
#       wireless-next patches for v6.2
#       wifi: rtw88: Add rtw8822bu chipset support
#
# The following line blacklists (deactivates) the above in-kernel driver.
blacklist rtw88_8822bu
#
# Edit the following line to change, add or delete options:
options 88x2bu rtw_switch_usb_mode=1 rtw_led_ctrl=1 rtw_power_mgnt=0
#
# Note: To activate USB3 mode, change rtw_switch_usb_mode above to rtw_switch_u>
#
# Note: The above `options` line is a good default for managed mode. Below is
# an example for AP mode. Modify as required after reading the documentation:
#options 88x2bu rtw_switch_usb_mode=1 rtw_led_ctrl=1 rtw_vht_enable=2 rtw_powe>
#
^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line
```

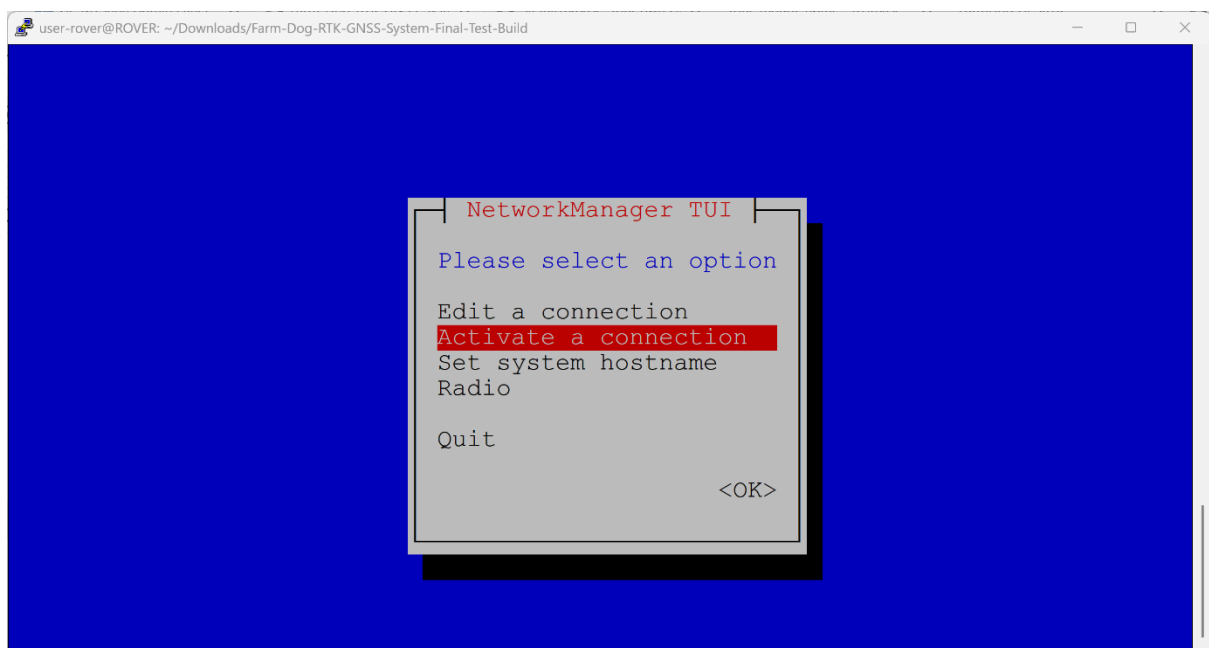
9 ACTIVATE THE USB WI-FI CONNECTION AND SET STATIC IP ADDRESS

After making the changes to the driver options, activate the Wi-Fi connection on the USB module and set a static IP address for the base station and the rover so both devices have a fixed IP address.

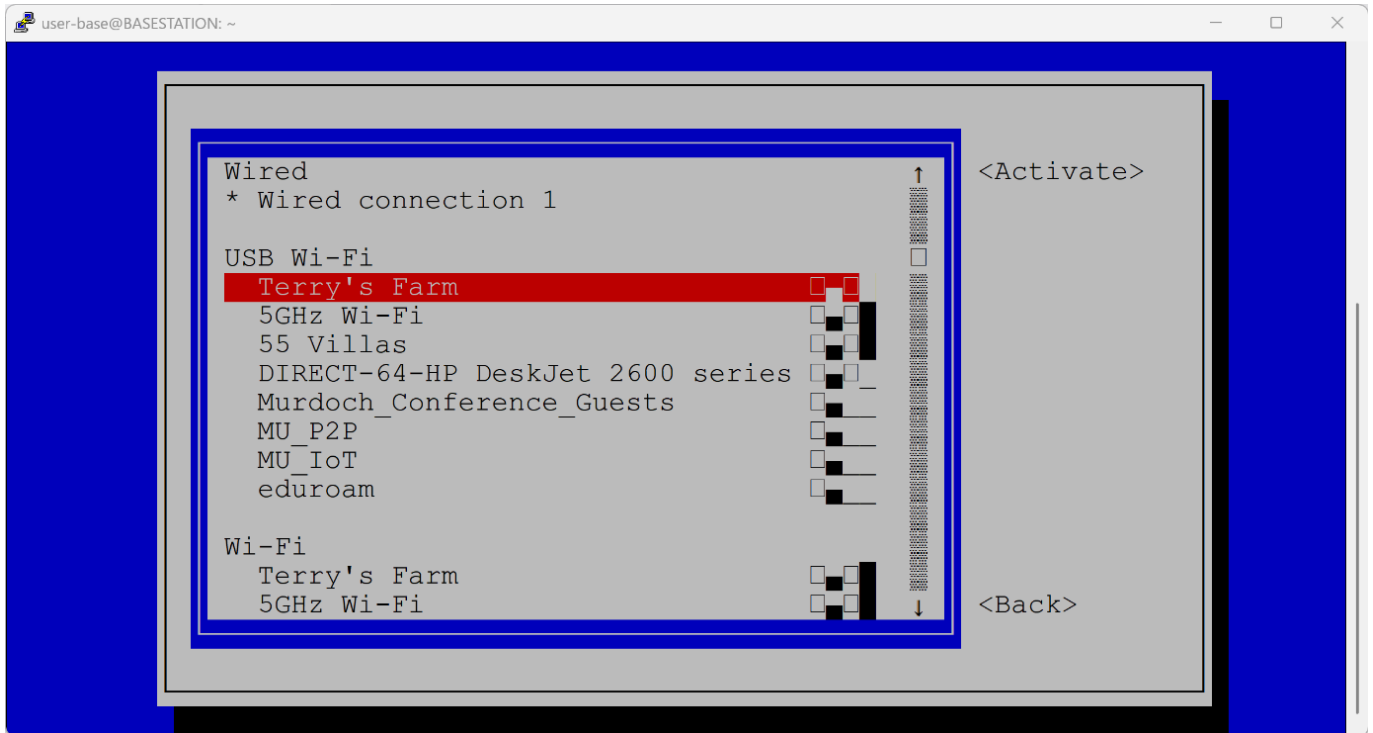
9.1 ACTIVATE THE WI-FI CONNECTION

Important Note: Before continuing, it is advised that the Raspberry Pi is connected via Ethernet since the onboard Wi-Fi of the Raspberry Pi will disconnect when activating the USB Wi-Fi connection and therefore, the connection will be lost.

- **Step 1:** In Putty, type the following command to get to the TUI network manager:
 - o `sudo nmtui`
- **Step 2:** Use your down arrow key to highlight "**Activate a connection**" and hit Enter.

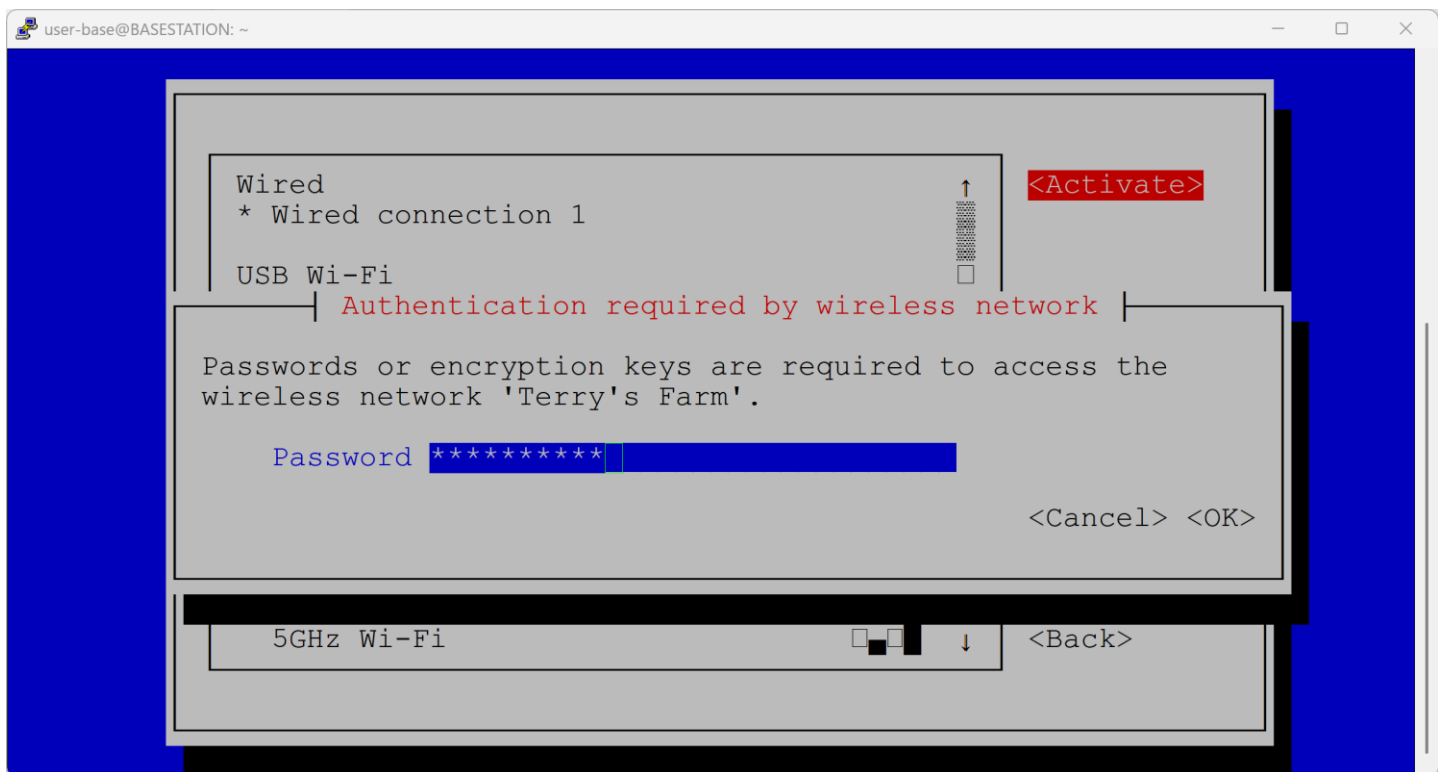


- **Step 3:** You are now in the Wi-Fi connection activation menu. Navigate to the USB Wi-Fi network that you want to activate under the USB Wi-Fi heading.



- **Step 4:** Use the right arrow key to navigate to "**Activate**" and press Enter.

- **Step 5:** When asked for the Wi-Fi password, type it in. After that, navigate to **"OK"** and press Enter.

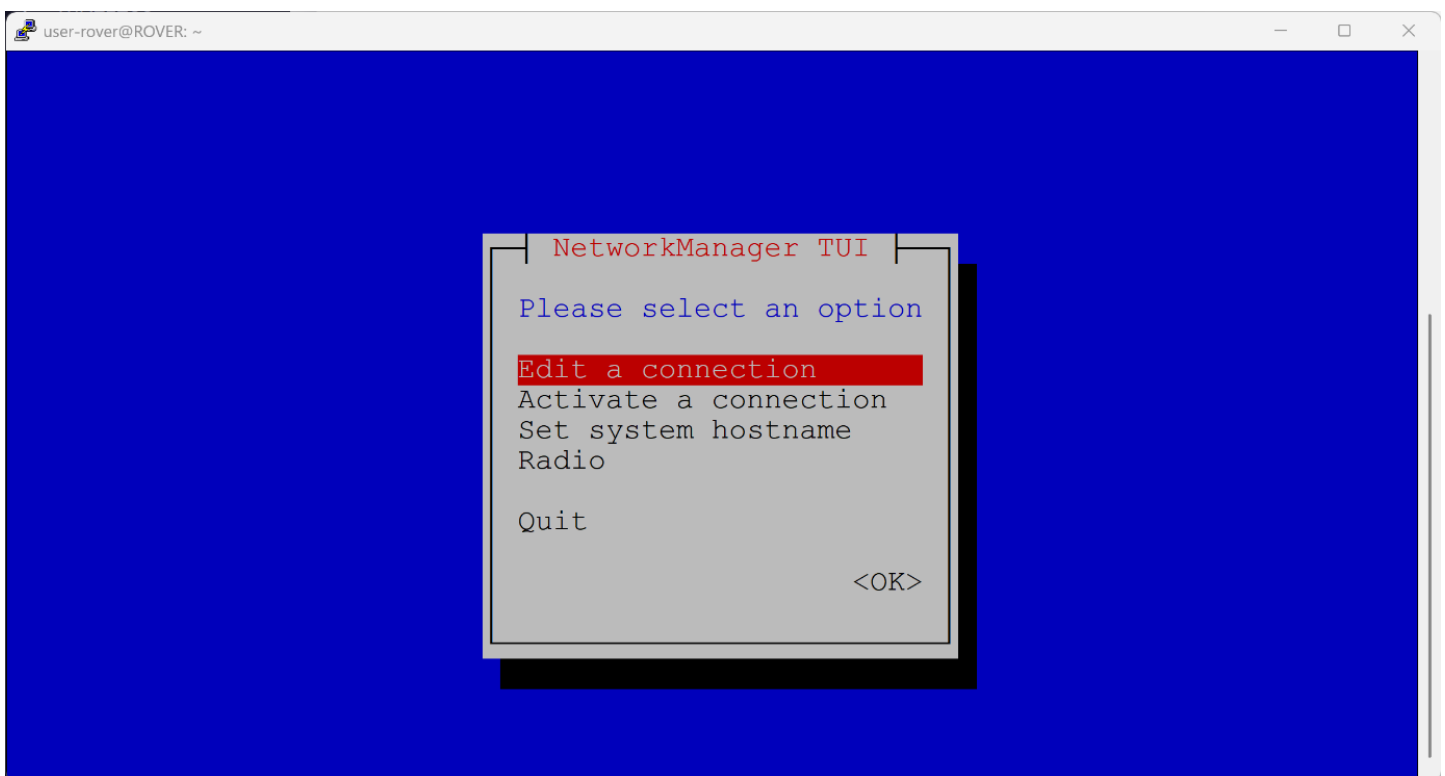


- **Step 6:** Once you are back in the previous window, navigate to **"Back"** and press Enter to get back to the main screen of the TUI network manager.

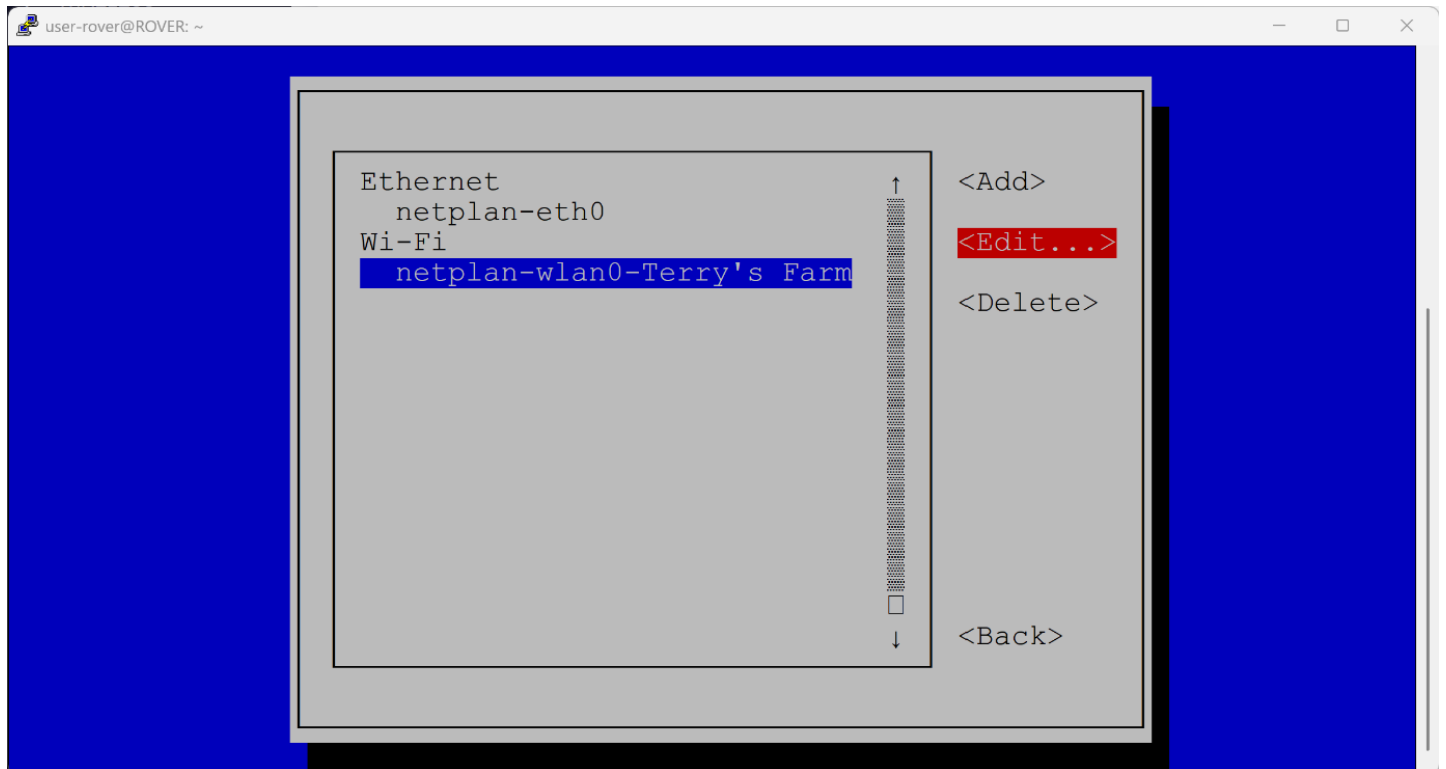
9.2 SET STATIC IP ADDRESS

From the main screen, select “Edit a Connection” and press enter.

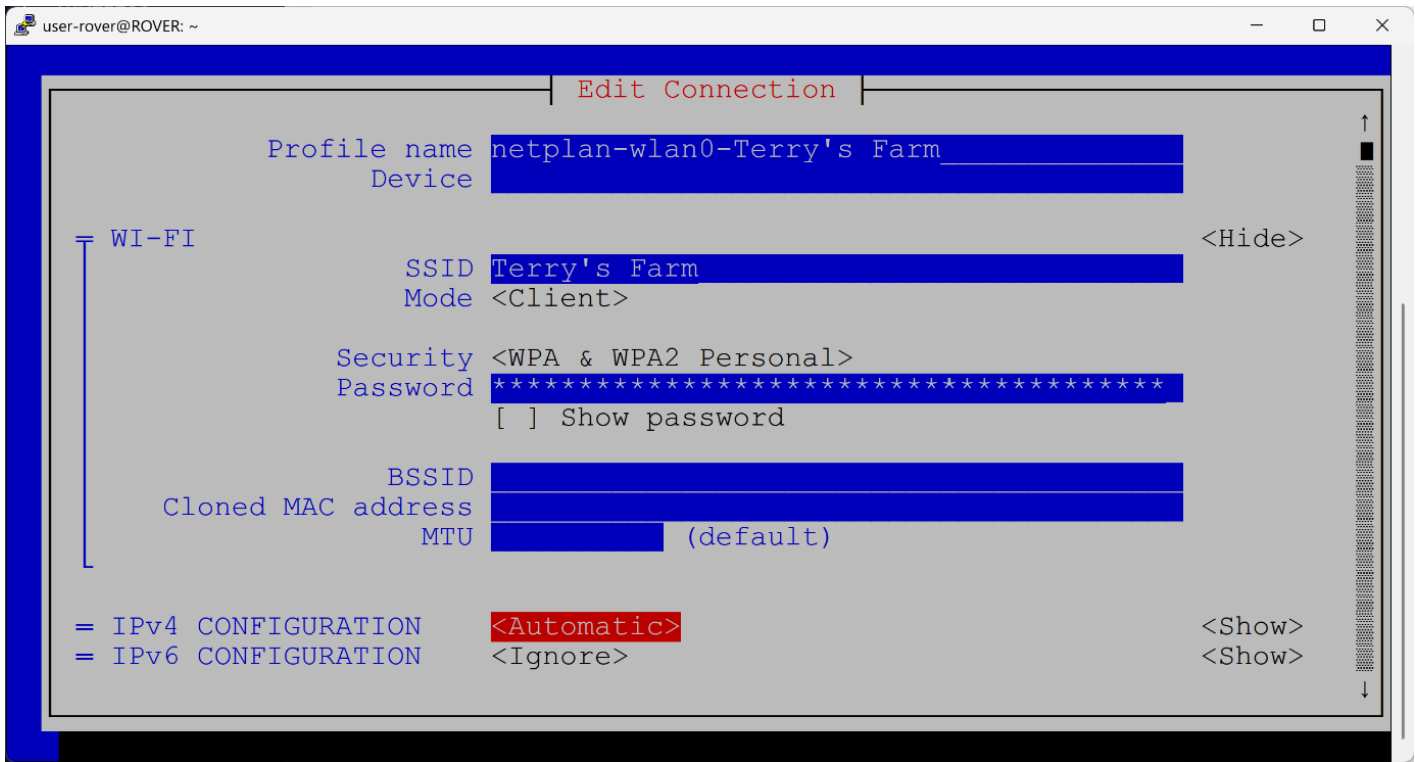
- **Step 1:** From the main TUI screen, select “**Edit a Connection**” and press Enter.



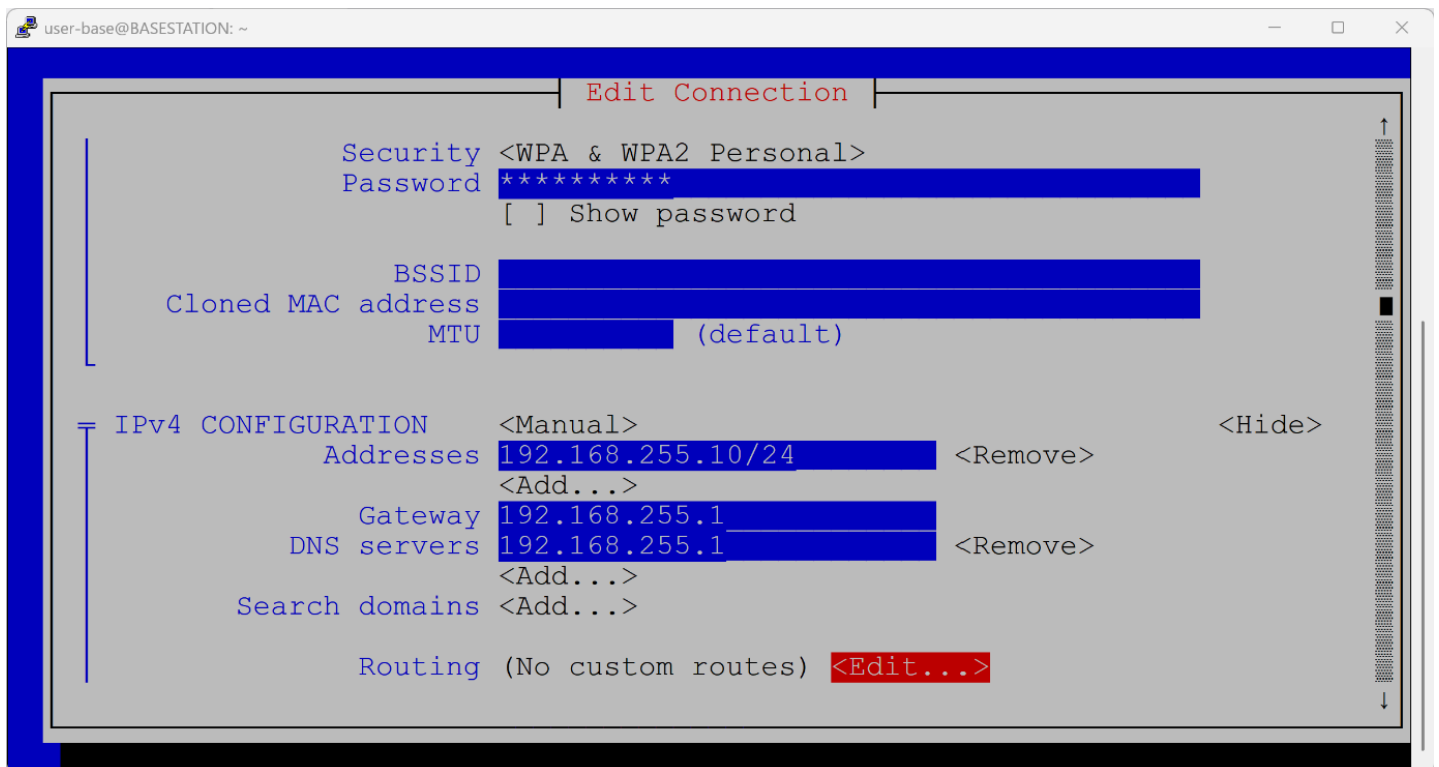
- **Step 2:** From the list of available networks, choose the Wi-Fi connection that you previously activated and select “**Edit...**”.



- **Step 3:** Navigate down to **IPv4 Configuration** and change the setting from *Automatic* to **Manual**.



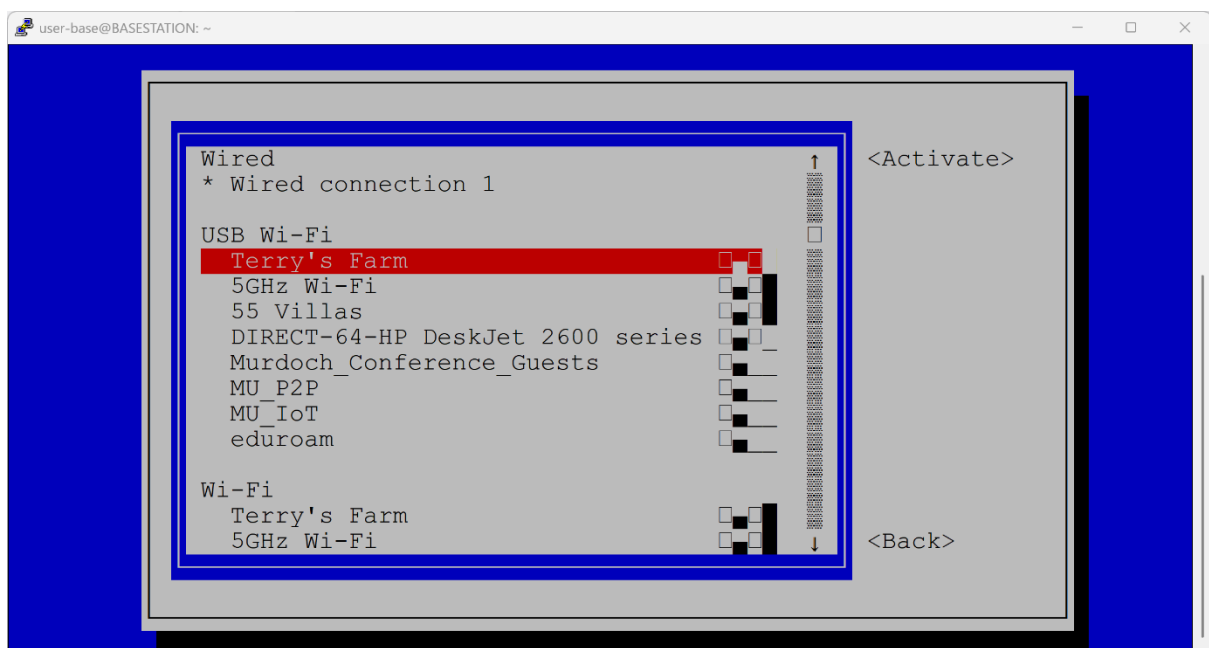
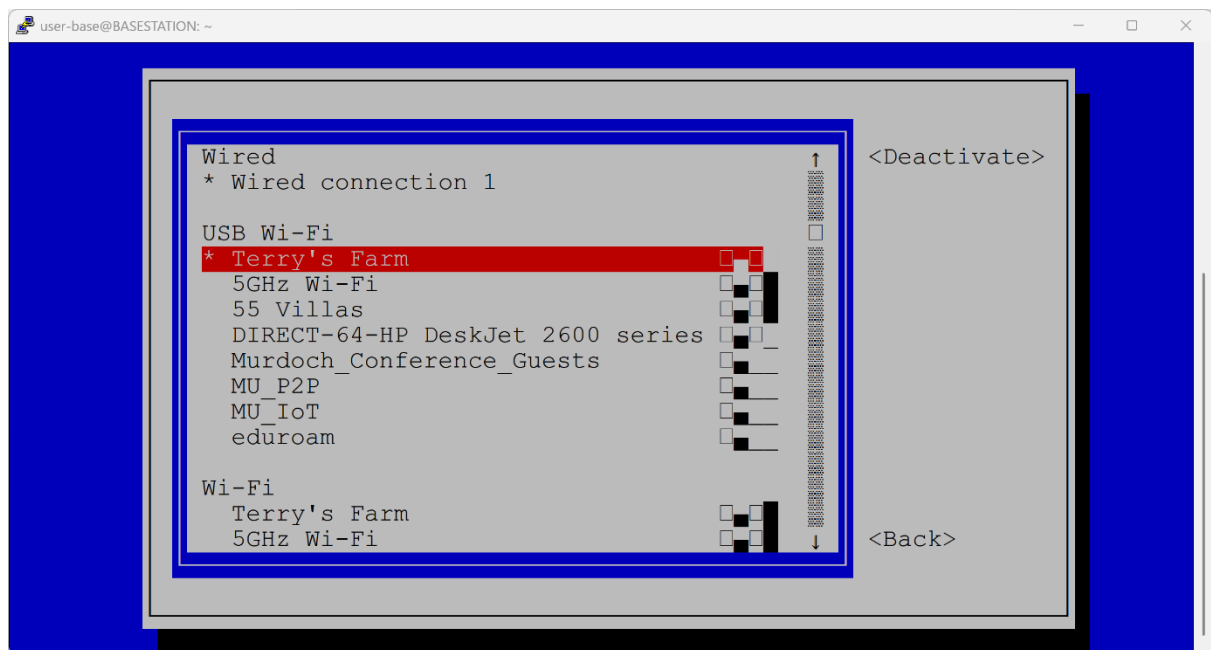
- **Step 4:** Input an unallocated static IP address for the device. This must include the subnet mask in CIDR format (e.g., /24), the IP address of the default gateway, and the IP address of the DNS server.



- **Step 5:** Once the network parameters are entered, exit the TUI network manager by selecting “**Back**” and then “**Quit**” on the main screen.

- **Step 6:** In order for the new static IP address to apply, you must either deactivate and reactivate the current connection, or simply enter the following command into the Terminal to restart the device:

- o `sudo reboot`



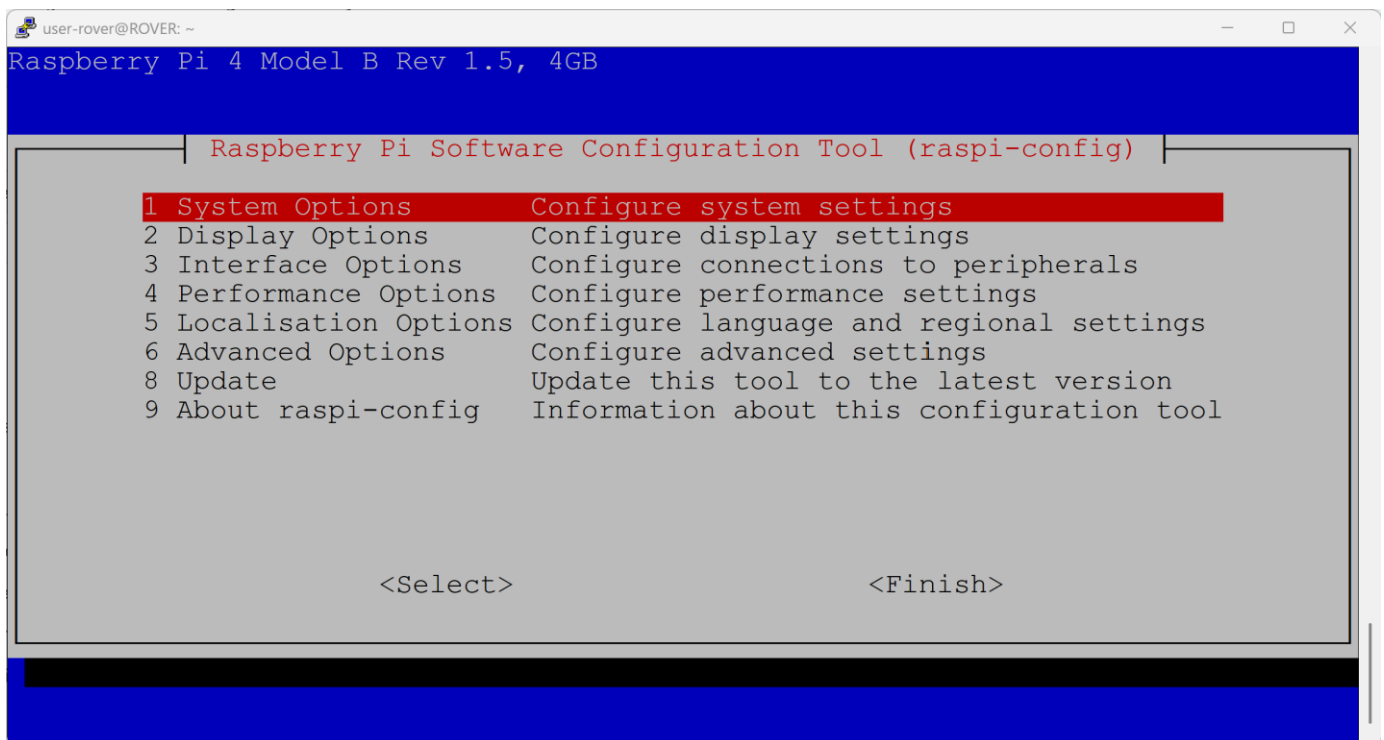
- **Step 7:** Once you have successfully completed all the steps in Section 9 for the Base Station, repeat the entire process for the Rover.

This wraps up the entire network and static IP configuration!

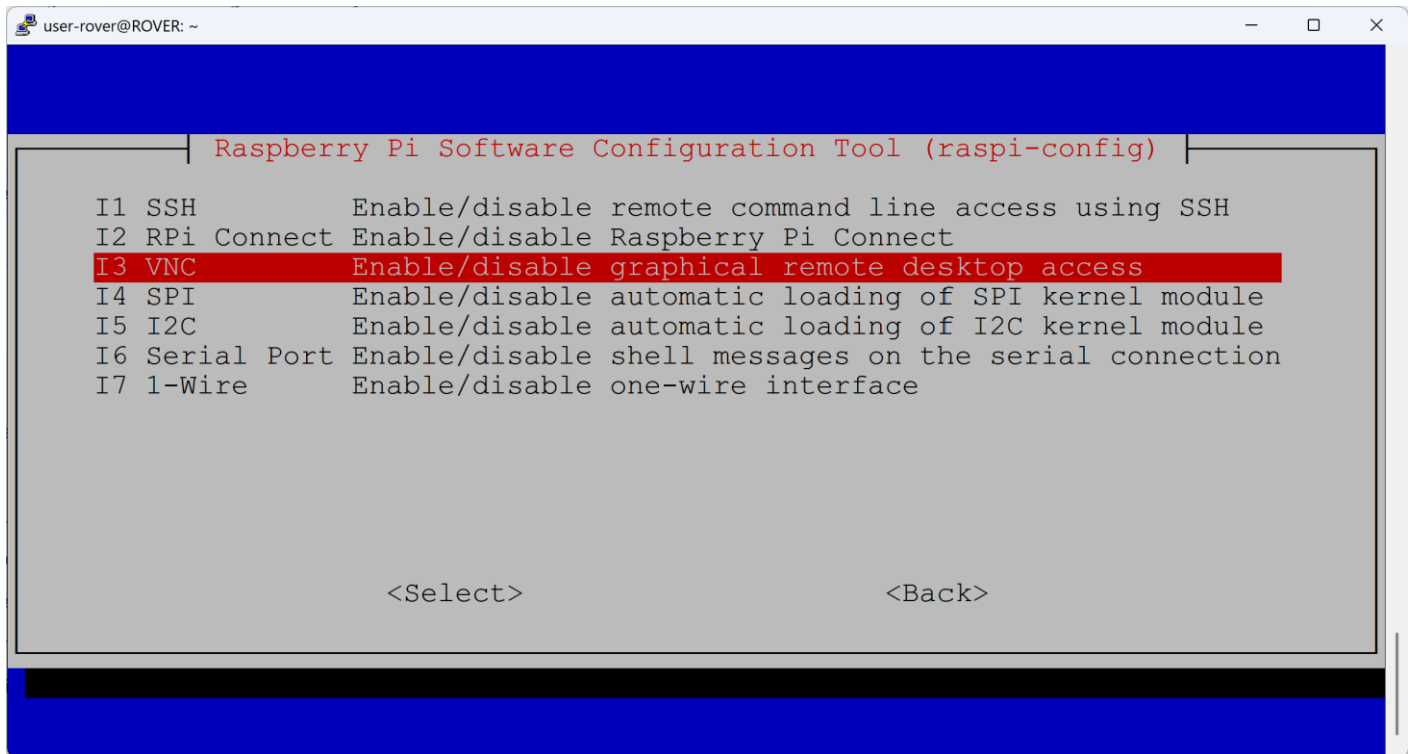
10 CONFIGURATION FOR VNC REMOTE DESKTOP

To enable VNC and provide the system with a graphical user interface for remote administration, you must enable the service through the built-in configuration tool.

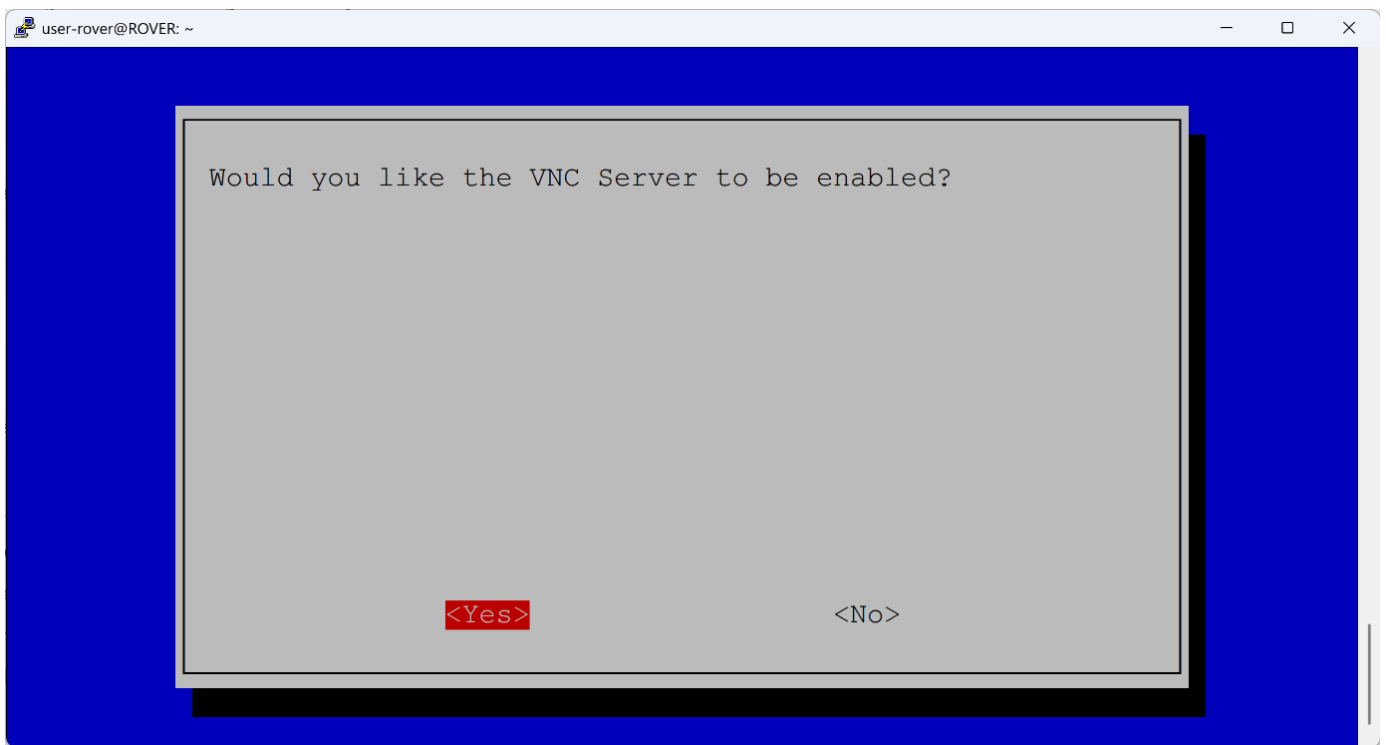
- **Step 1:** Enter the following command into your active SSH Terminal:
 - o `sudo raspi-config`
- **Step 2:** This command opens the Raspberry Pi Software Configuration Tool. Use your arrow keys to navigate down and select “**3 Interface Options**”.



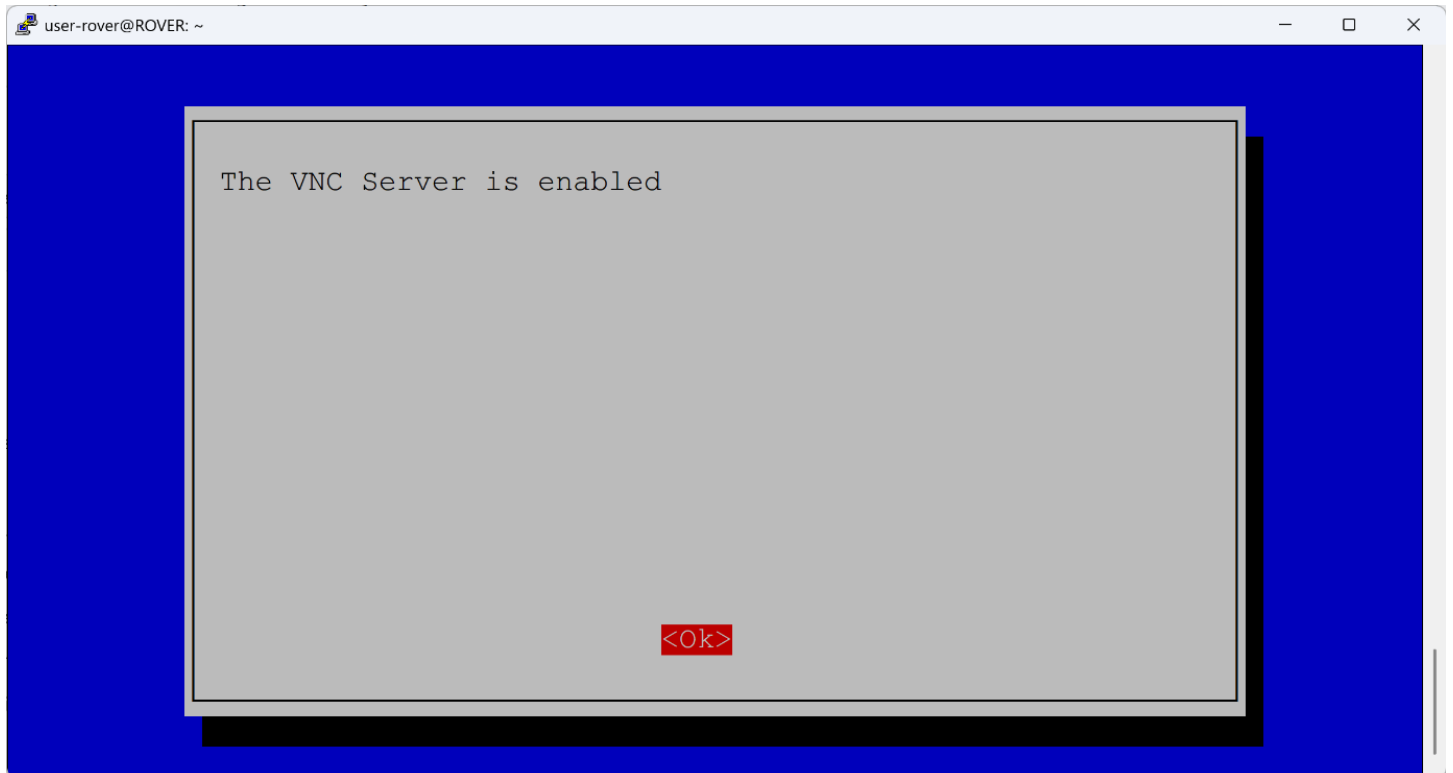
- **Step 3:** In the next window, navigate down and select “**I3 VNC**”.



- **Step 4:** When prompted to enable the service, select “**Yes**” to enable the VNC Server.



- **Step 5:** You should receive a confirmation screen stating that the VNC Server has been successfully enabled. Select **"OK"** to exit the prompt and then navigate to **"Finish"** to exit the configuration tool.



11 ADDITIONAL SOFTWARE

Additional software was installed to assist the team throughout the testing phase. LibreOffice was utilized to temporarily view generated CSV files directly on the mobile Rover, and a Samba NAS (Network Attached Storage) was set up to easily export those CSV files over the network to a laptop for temporary mapping.

11.1 ROVER

- **Step 1:** Update and refresh the system packages by executing the following command:

- `sudo apt update && sudo apt upgrade -y`

- **Step 2:** Install LibreOffice, Samba, and Btop by running:

- `sudo apt install libreoffice btop samba samba-common-bin -y`

- **Step 3:** Open the Samba configuration file using a text editor such as nano:

- `sudo nano /etc/samba/smb.conf`

- **Step 4:** Add or modify the settings inside the configuration file to match the following parameters:

- [global]

```
workgroup = WORKGROUP

server string = Rover RTK Station

server role = standalone server

log file = /var/log/samba/log.%m

max log size = 1000

logging = file

panic action = /usr/share/samba/panic-action %d


# Security settings

security = user

map to guest = Bad User
```

- [RoverHome]

```
comment = Share for Home Directory

path = /home/user-rover/

browseable = yes

read only = yes

guest ok = no

valid users = user-rover

force user = user-rover
```

- o [RoverShare]

```
comment = Farm Dog RTK Final Test Build  
path = /home/user-rover/Downloads/Farm-Dog-RTK-  
GNSS-System-Final-Test-Buildv3  
browseable = yes  
read only = no  
guest ok = no  
valid users = user-rover  
force user = user-rover
```

- **Step 5:** Save the configuration file by pressing **CTRL+O**, press **Enter** to confirm the file name, and then press **CTRL+X** to exit the text editor.

11.2 BASE

- **Step 1:** Update and refresh the Base Station system packages by executing the following command:

```
o sudo apt update && sudo apt upgrade -y
```

- **Step 2:** Install the required Samba NAS packages to enable network file sharing on the Base Station:

```
o sudo apt install
```

11.3 SETTING UP SAMBA NAS

- **Step 1:** To install the required Samba NAS packages and enable network file sharing, execute the following command:

```
o sudo apt install samba samba-common-bin -y
```

12 HARDWARE ASSEMBLY

Before beginning the physical assembly of the Base Station and Rover enclosures, ensure all necessary tools, fasteners, and components are organized and readily available.

12.1 REQUIRED TOOLS

- Drill
- Hammer
- Assortment of drill bits
 - 3/32 drill bit
 - Serrated drill bit
- Ruler
- Pencil
- Masking Tape

12.1.1 ADDITIONAL PARTS

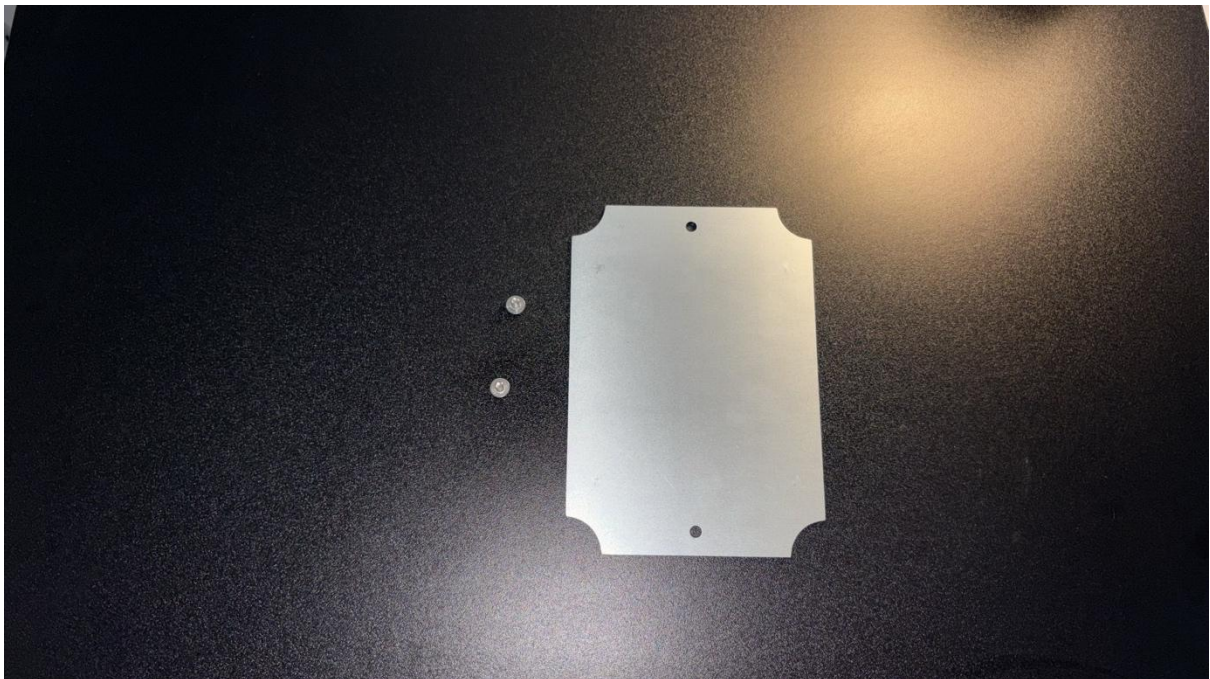
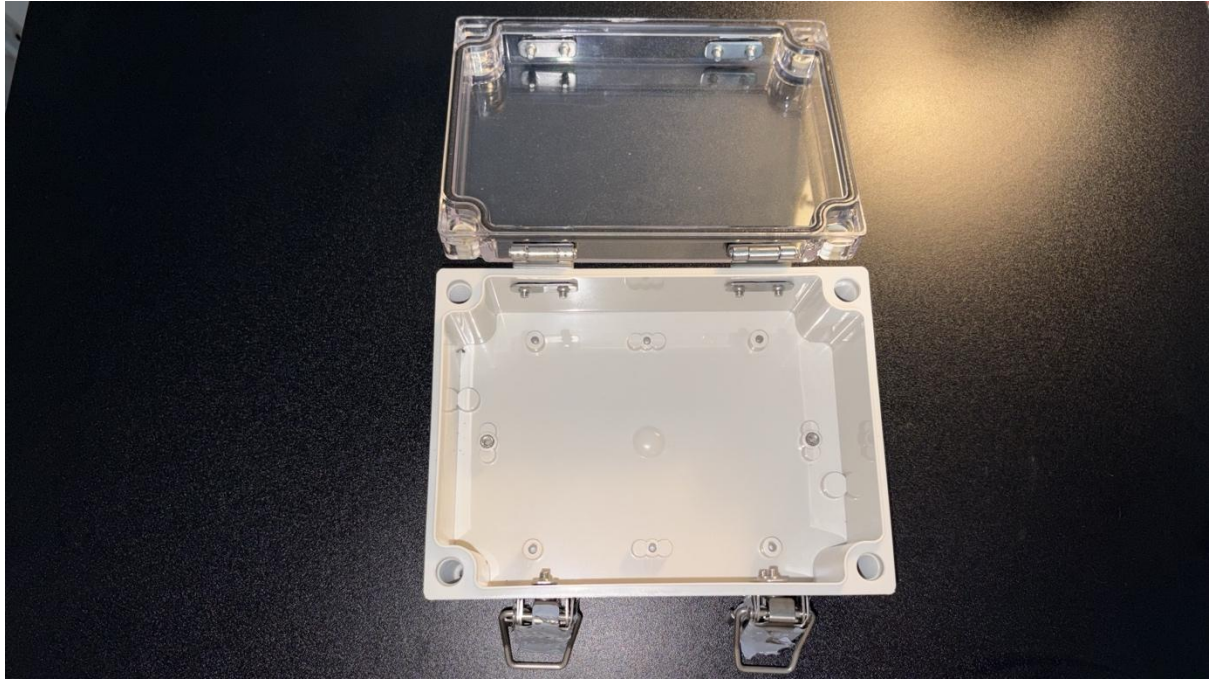
- 12x - M.5 Screws
- 12x - M.5 Nuts
- 9x - M.25*20+6mm standoffs
- 4x - M.25*11+6mm standoffs
- 2x Waterproof Cable Glands PG9 4mm – 8mm

12.2 ADDITIONAL COMPONENTS

- 1x Right Angle SMA Male to Female connector
- 2x 15cm Male to Female extension cable

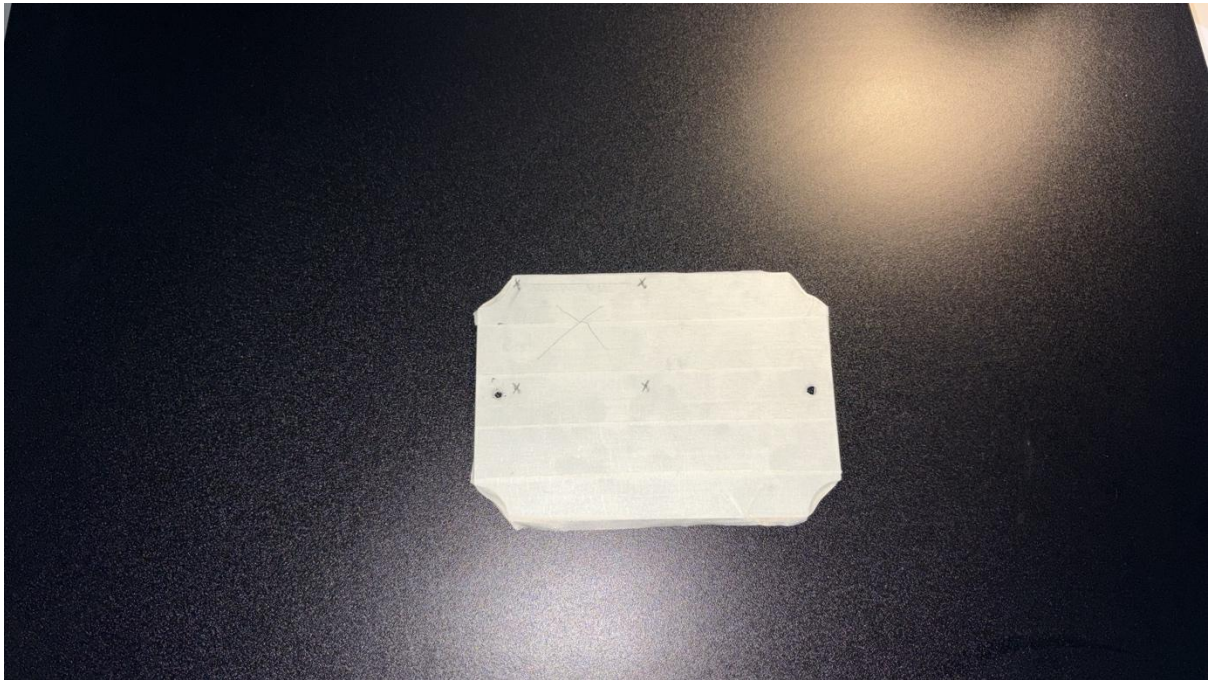
12.3 INSTALLATION OF COMPONENTS FOR THE MOBILE ROVER

Step 1: Unscrew 2x Philips screw to remove metal plate from IP66 Small ABS Box



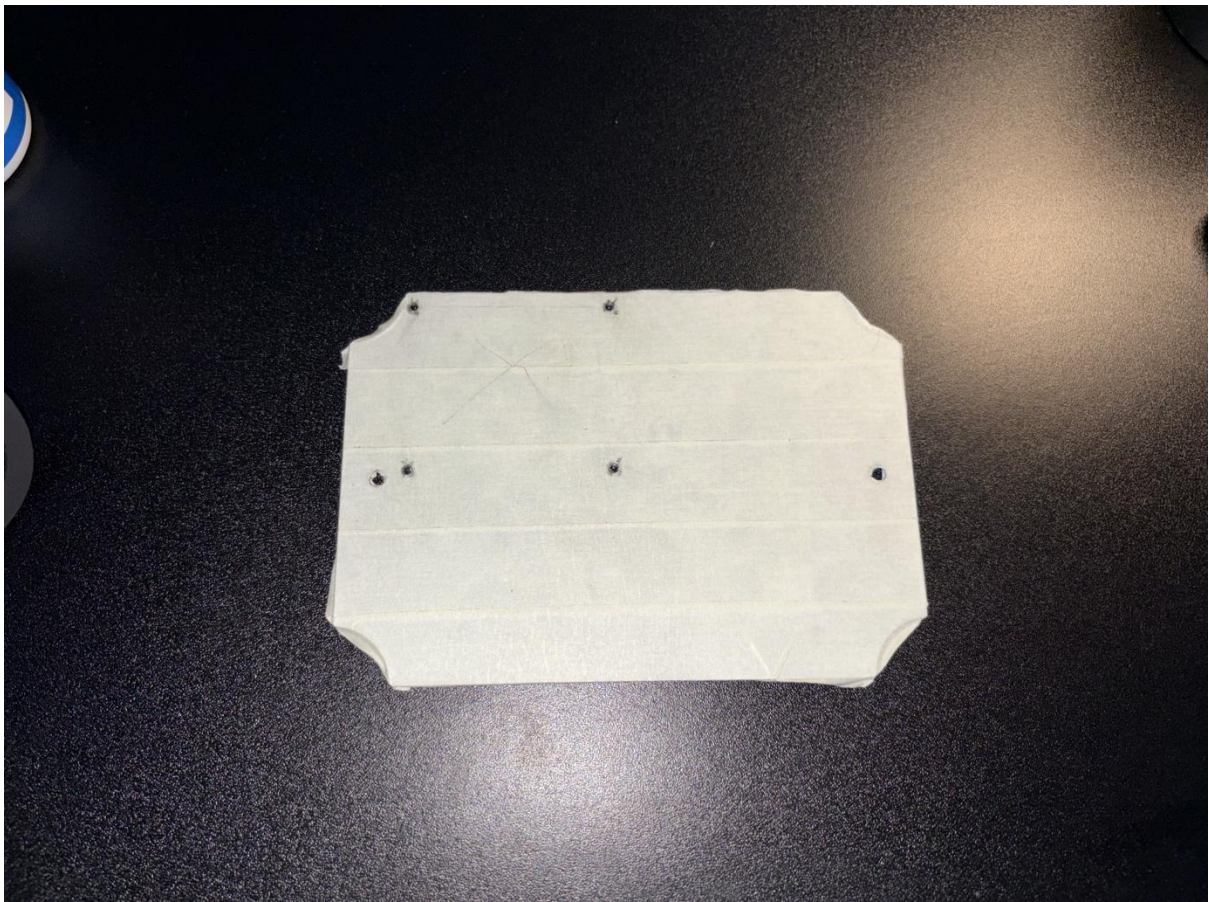
FARM DOG

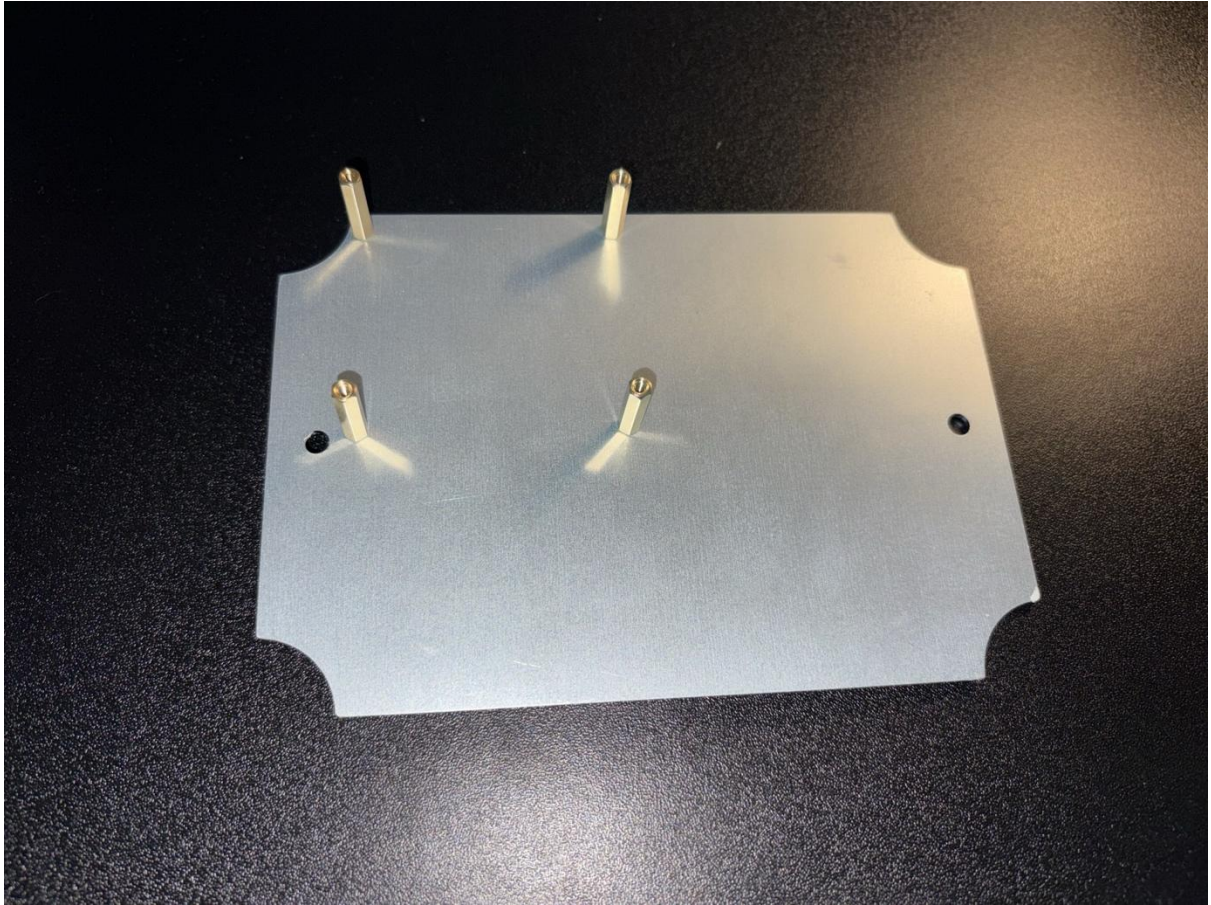
Step 2: Use masking tape to tape up the metal mounting plate, this will allow to mark the holes/dimension to easily mount up the standoff to then mount the Raspberry Pi



Step 3: Temporally place the Raspberry Pi onto the masked metal plate in the top left corner and using a pencil mark out the mounting holes on the Raspberry Pi onto masked mounting plate. (Please be mindful of the placement of the Raspberry Pi to close the edge of the mounting plate as it will not provide enough clearance when install the mounting plate back into the ABS box)

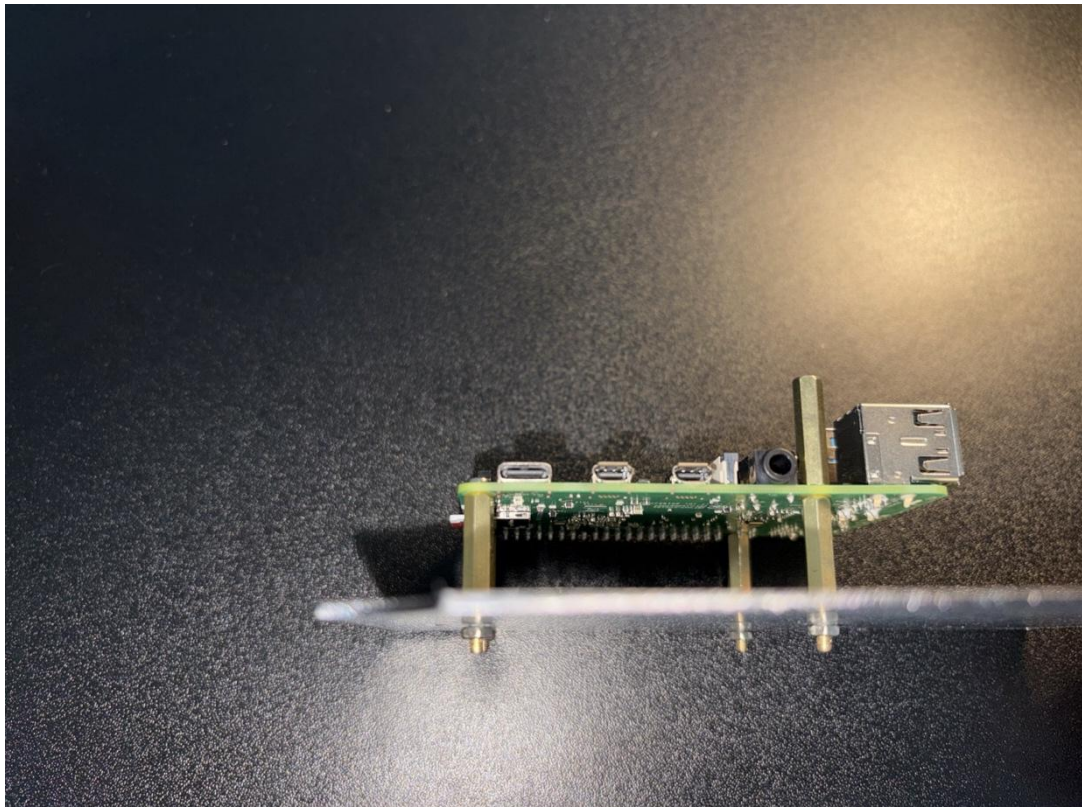
Step 4: Using a hammer and centre punch, create small indentation on the 4 marked crosses. Using a drill and 3/32 drill bit carefully drill out the 4 marked holes for mounting the metal standoffs. Afterwards, remove the masking tape and align the metal stand offs and using a 4x M.5 Nuts tighten down the 4x M.25*20+6mm stand offs to the metal base.

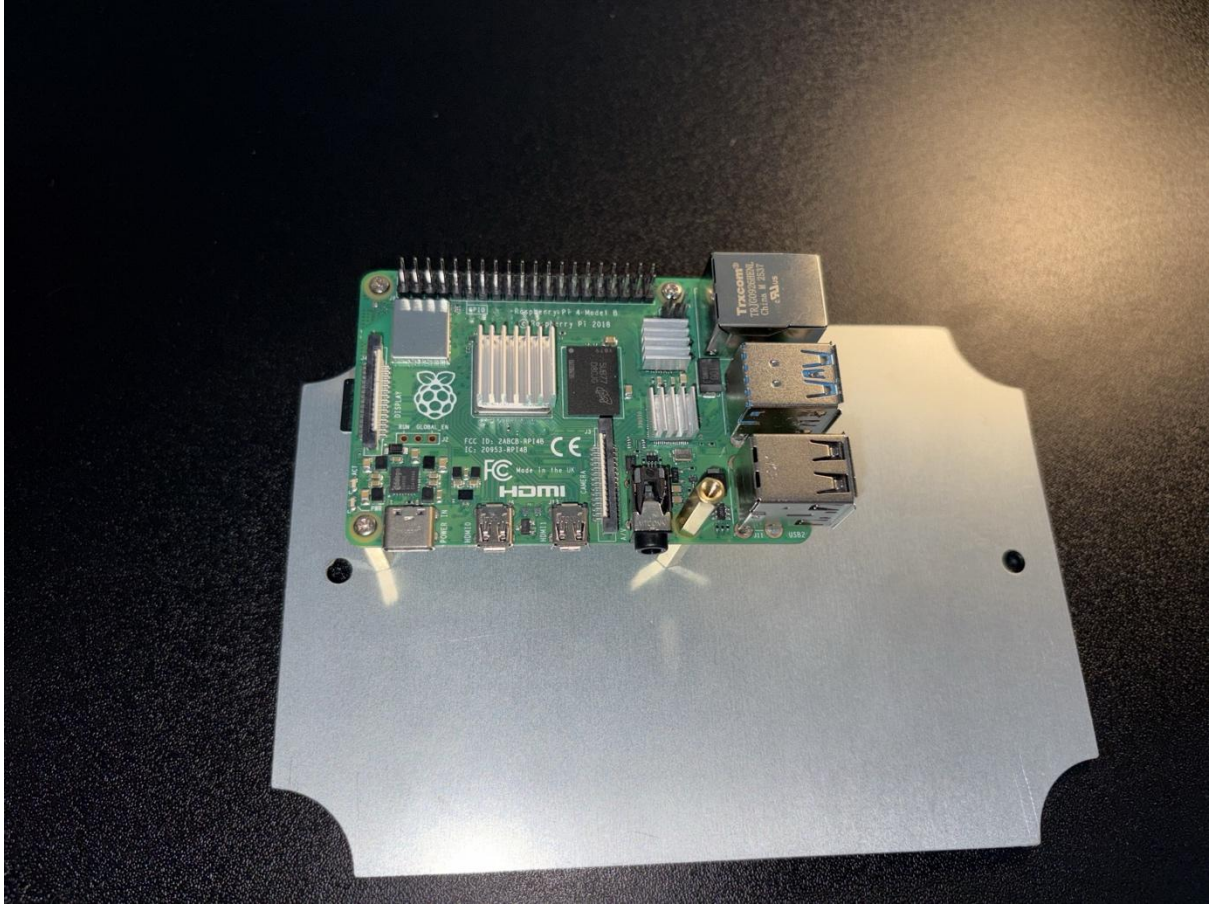




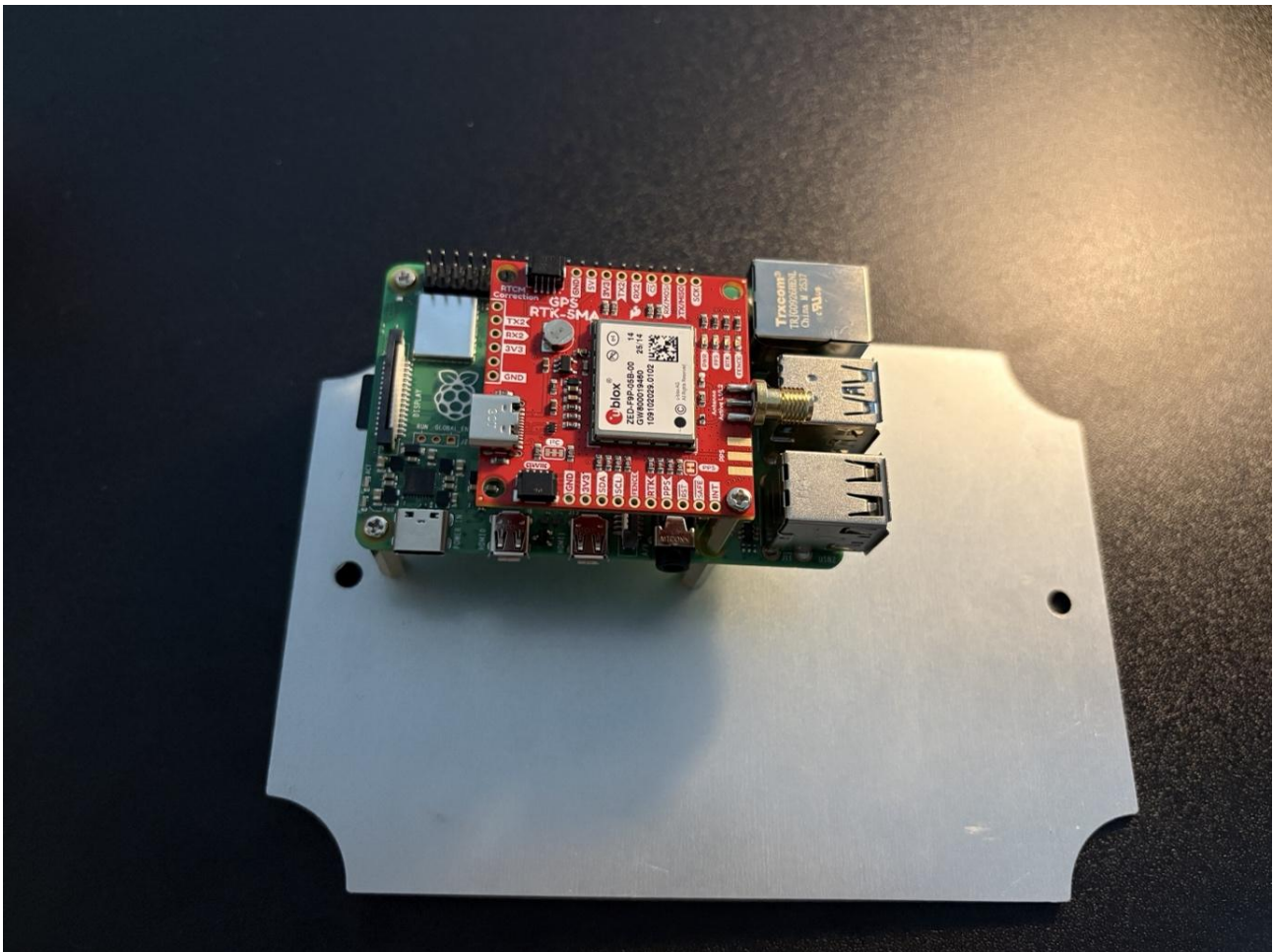
FARM DOG

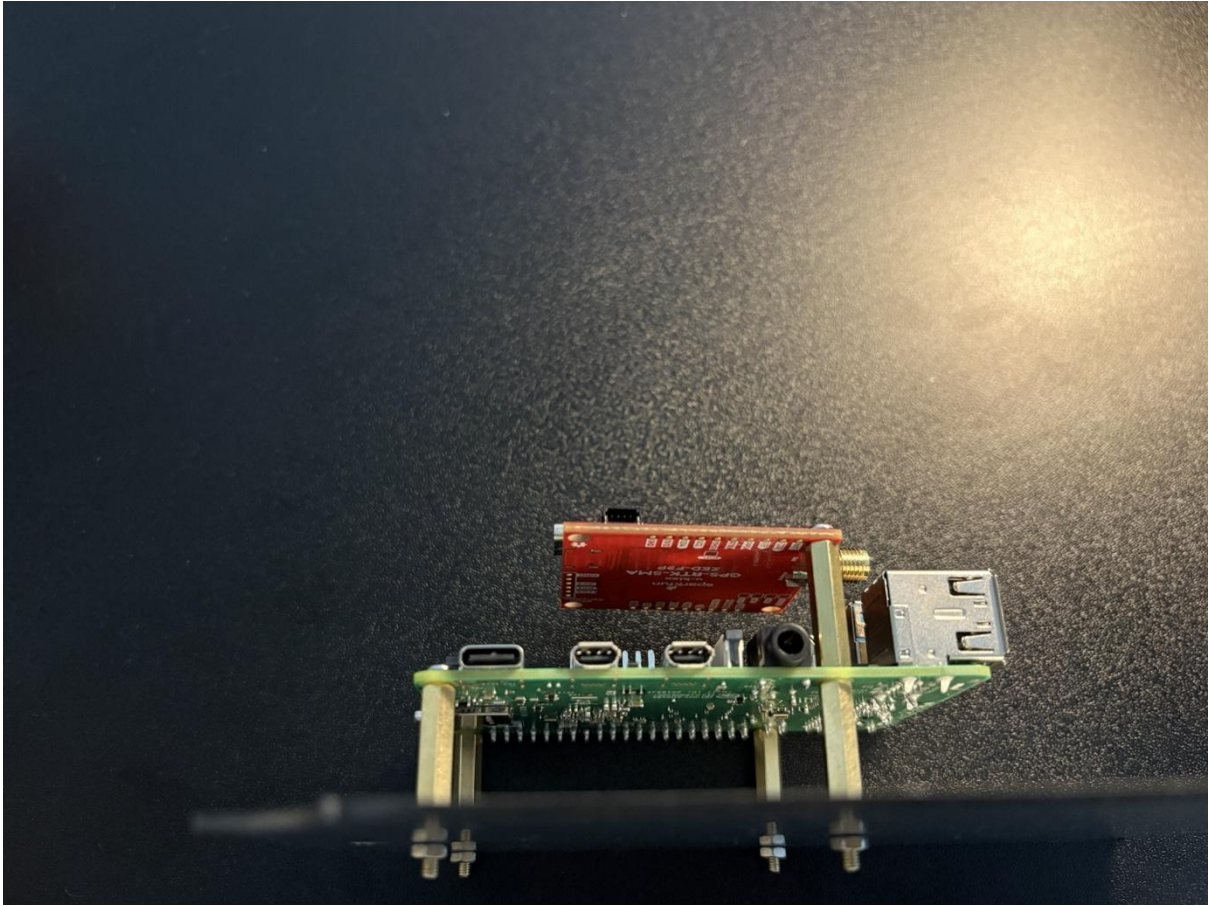
Step 5: Using 3x M.5 screws, carefully screw down the raspberry pi onto the 4x M.25*20+6mm stand offs. Using another standoff screw it into the bottom right position.





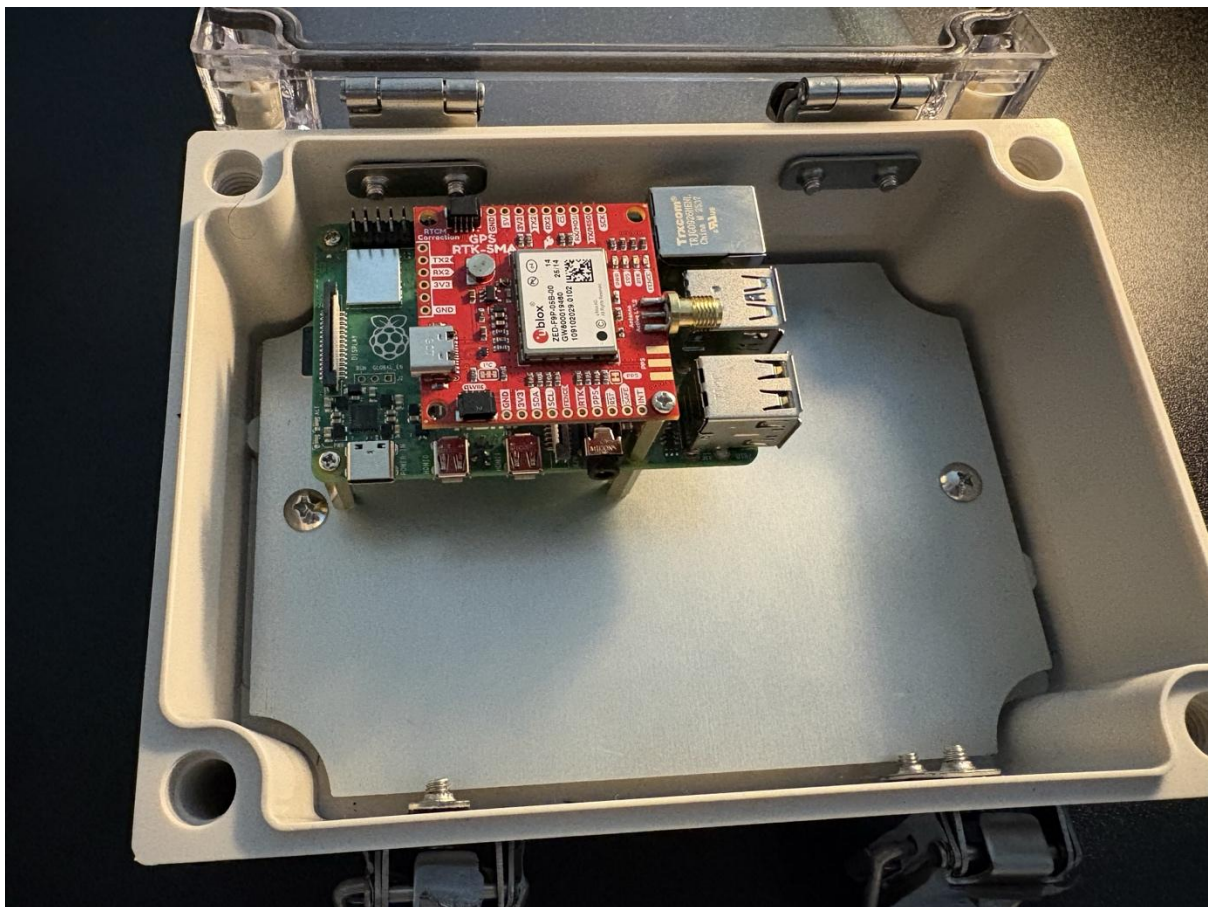
Step 6: Carefully place the GPS module on top of the mount and screw it on top of the standoff. This will give enough clearance to prevent the components from short circuiting. Plug in the external Simplecom NW621 USB Wi-Fi Module and screw in the female SMA 15cm cable to the end of the male SMA connector

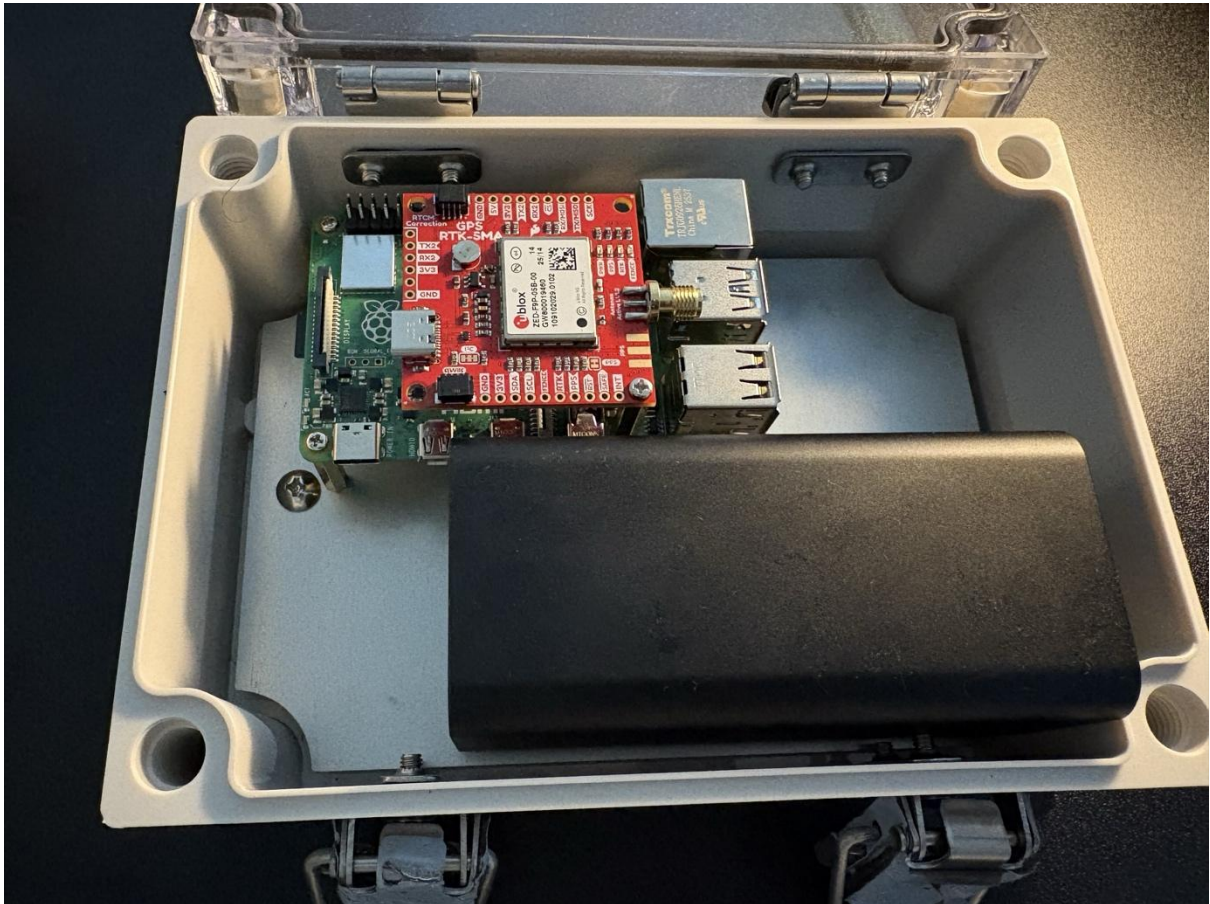




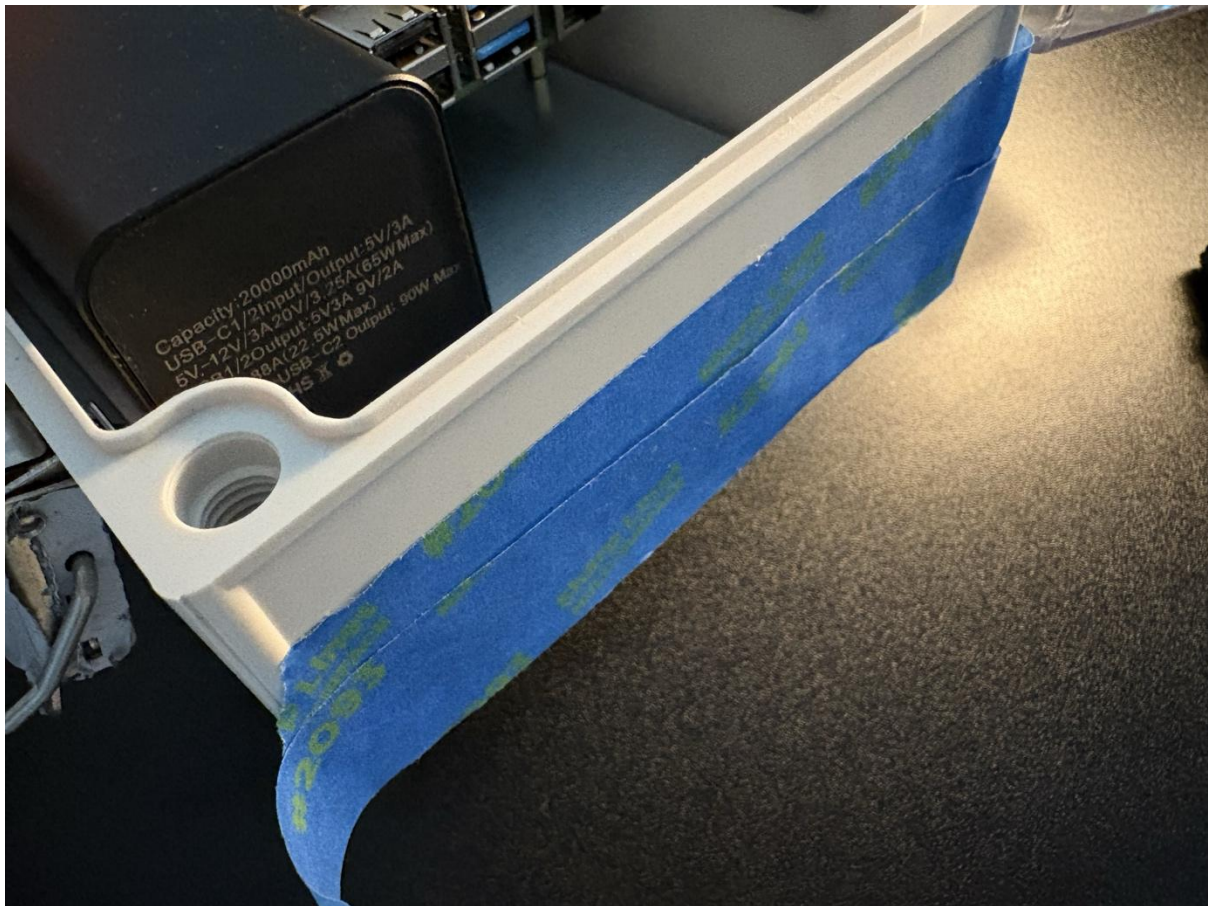
Step 7: Carefully lowering the mounted components on the metal mounting base plate into the ABS box and carefully screw it into the original screw down points.

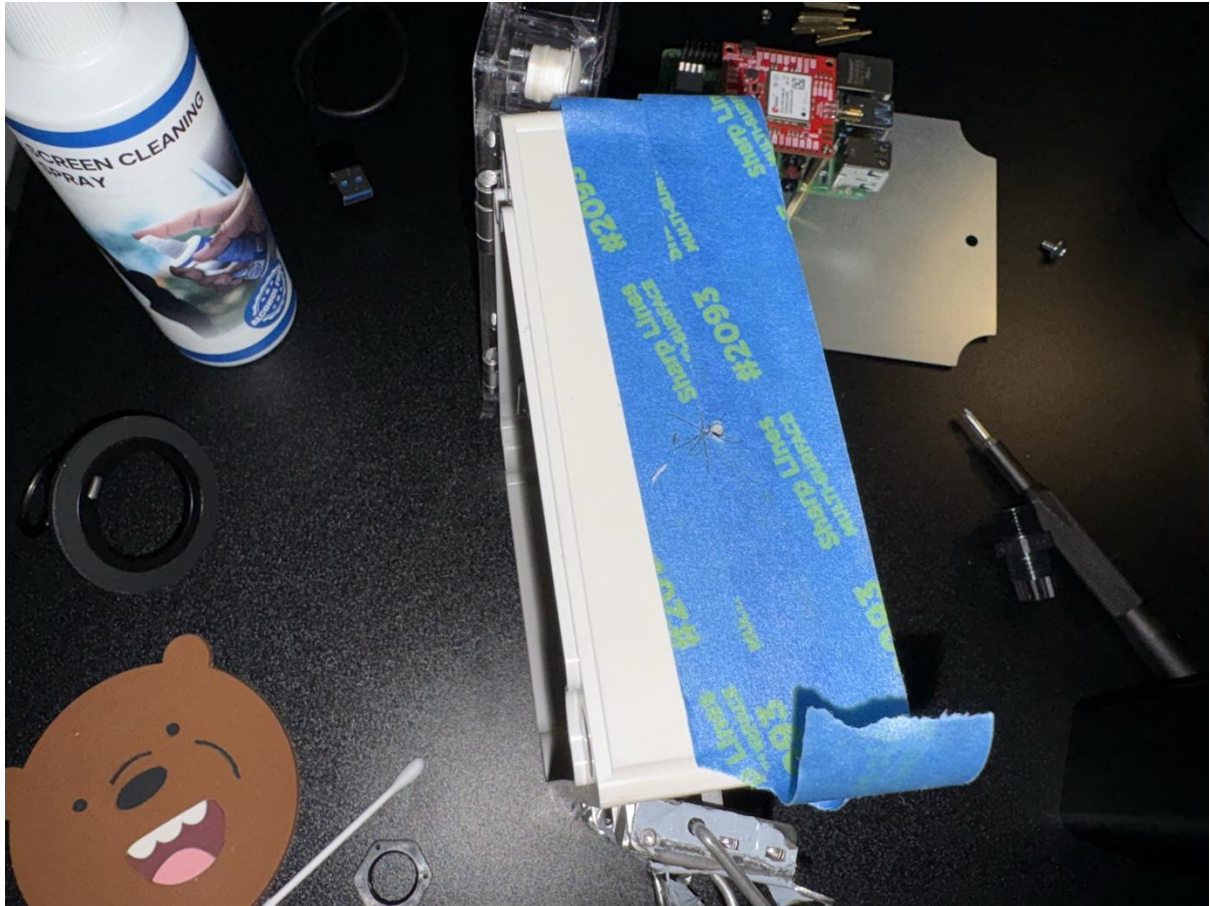
Step 8: Temporarily place the power bank into the case/box for alignment, once satisfied with the location of the battery use Velcro tape and remove the adhesive tape place one side of the Velcro into positioning. Place the other Velcro tape onto the battery, cut off any excess.



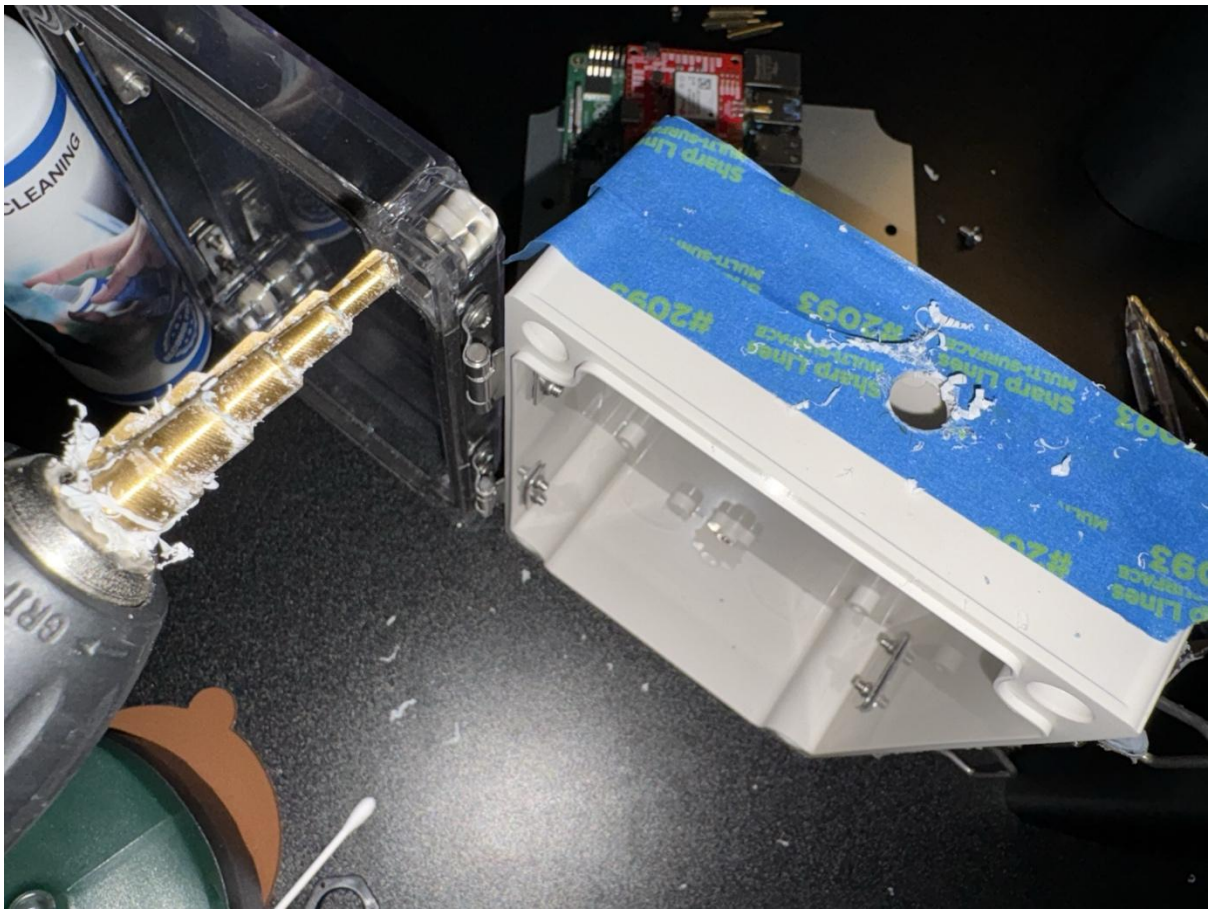


Step 9: Using masking tape, tape up the short right side of the ABS box masking tape, eyeball and using pencil to mark out the cut out for the cable glands and male SMA connector. Please be mindful of the metal plate.





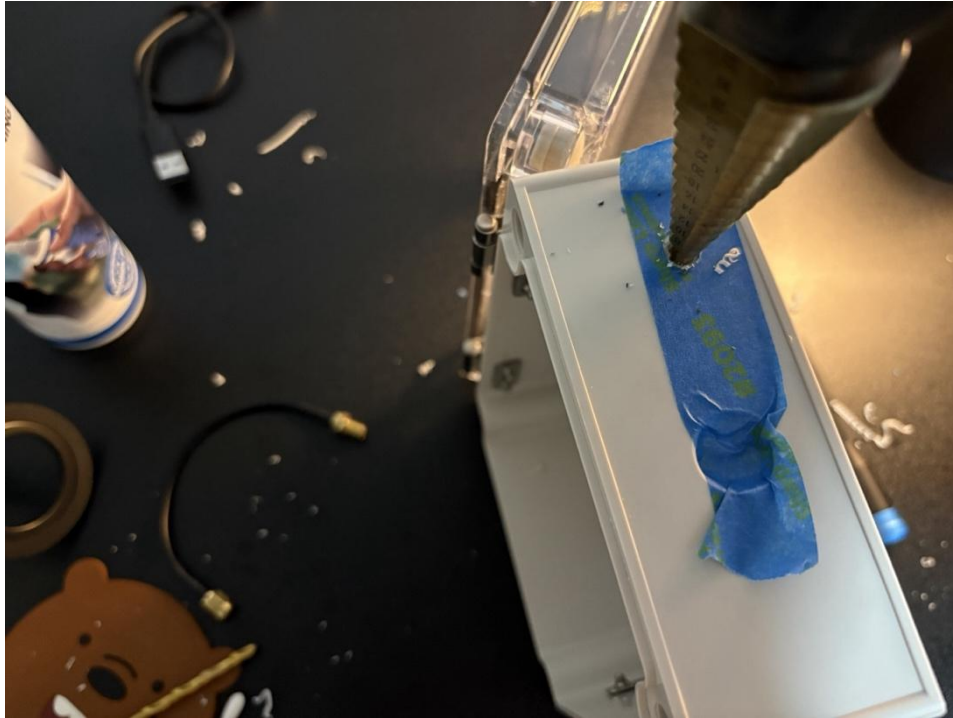
Step 10: Carefully drill out both holes for the cable gland connector and the male SMA connector. Use the appropriate drill bits in this case a 3/32 and serrated drill bit was used. After the desired size hole is drill out, remove the masking tape. Removing any debris or material.





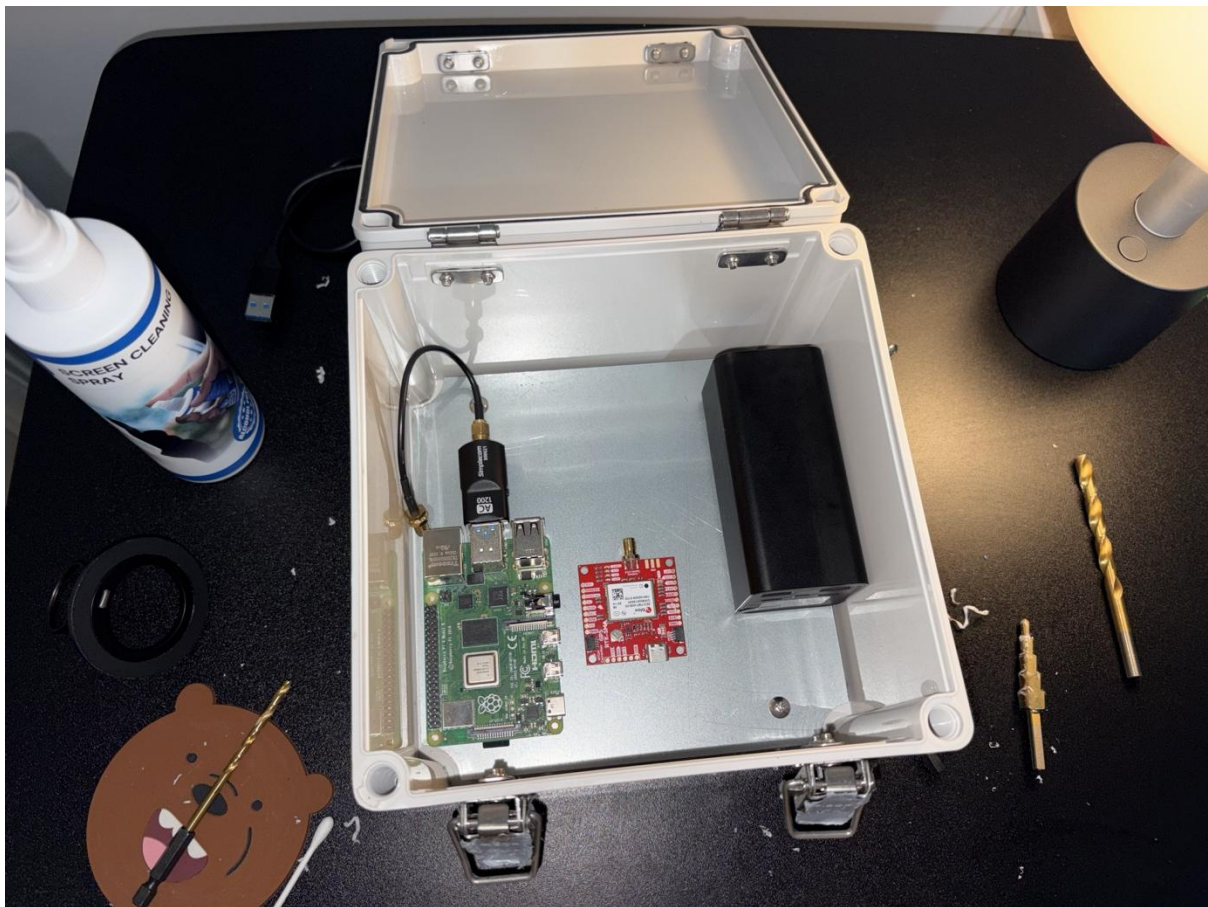
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Step 11: Clean up the excess and carefully and screw in and remount the metal mounting plate with mounted components back into the box.



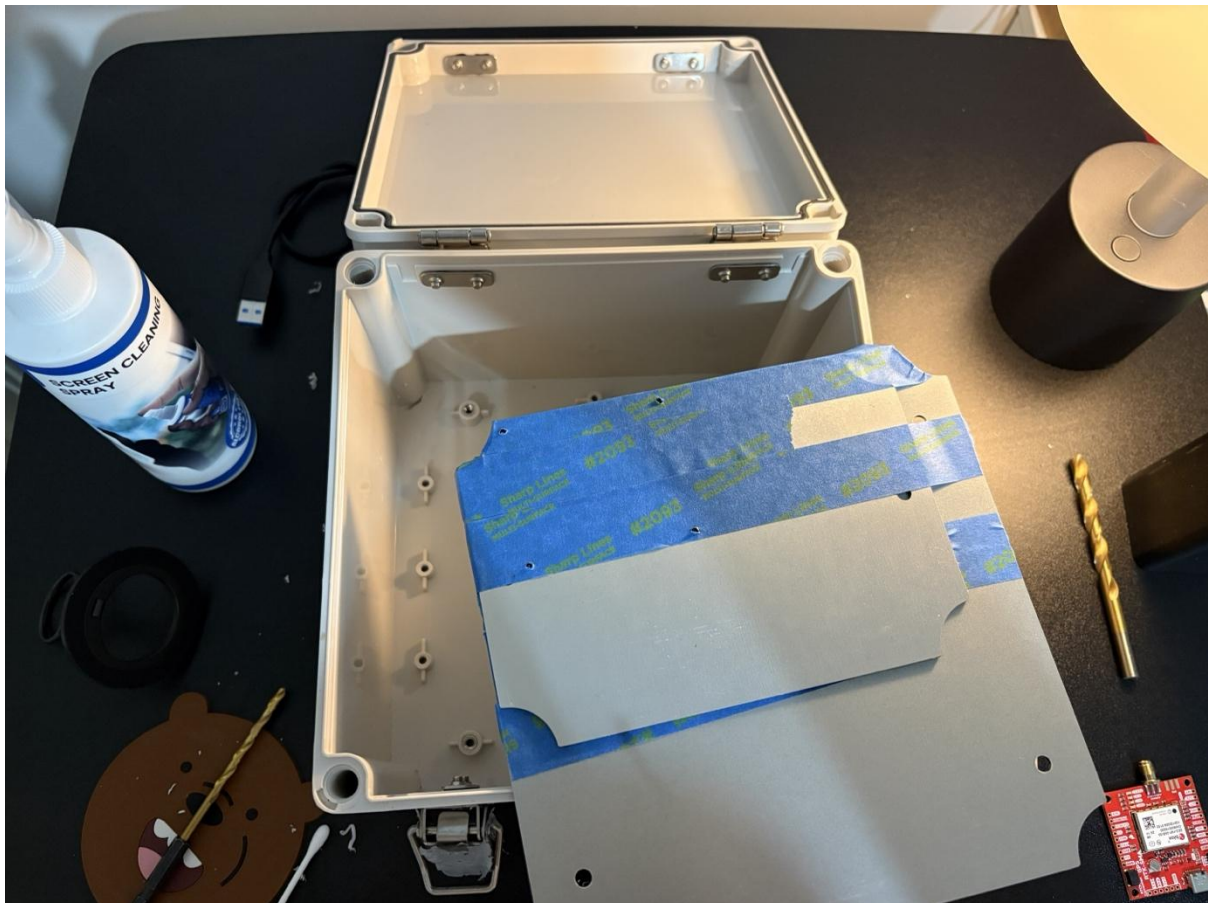
12.4 INSTALLATION OF COMPONENTS FOR THE BASE STATION

Step 1: Temporally place the components into the box to see spacing and where the components would be mounted permanently.

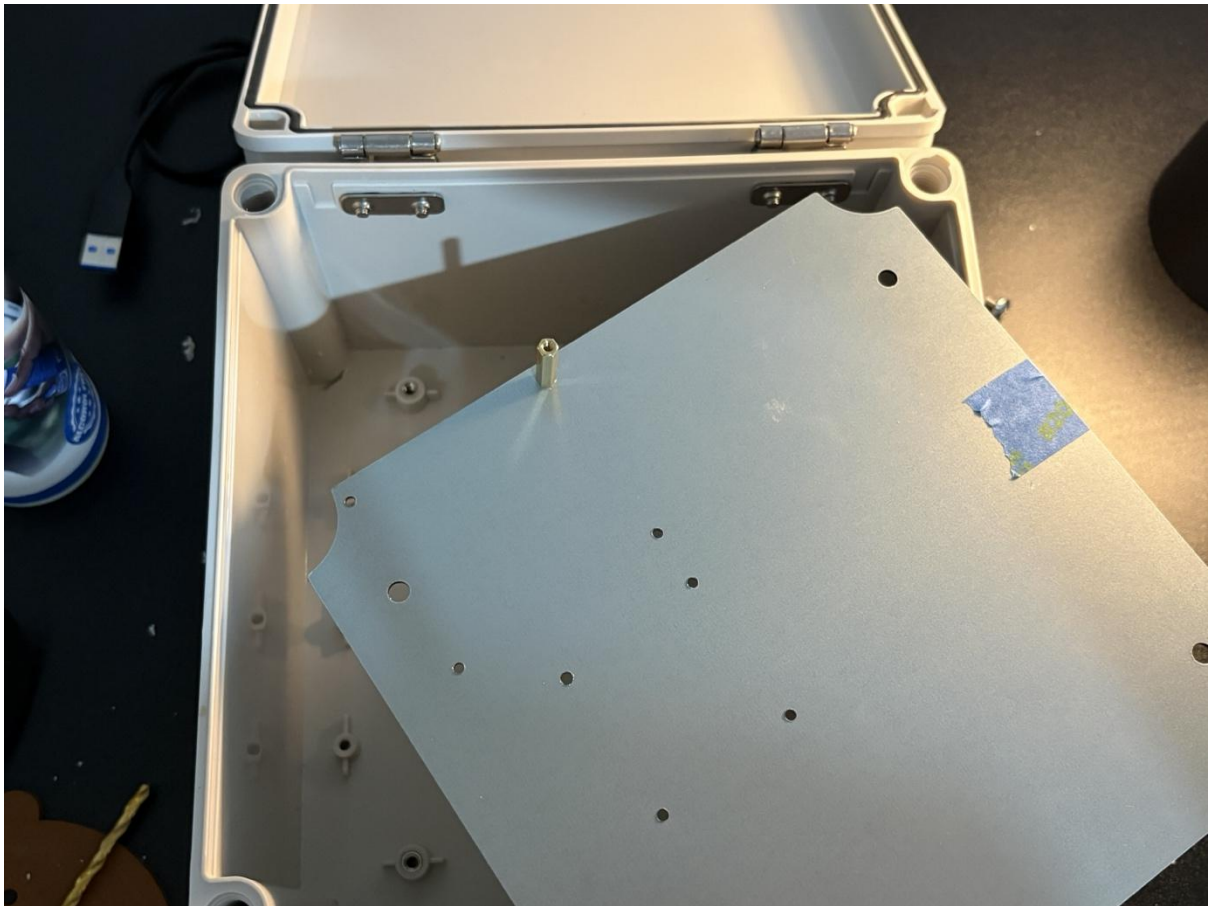


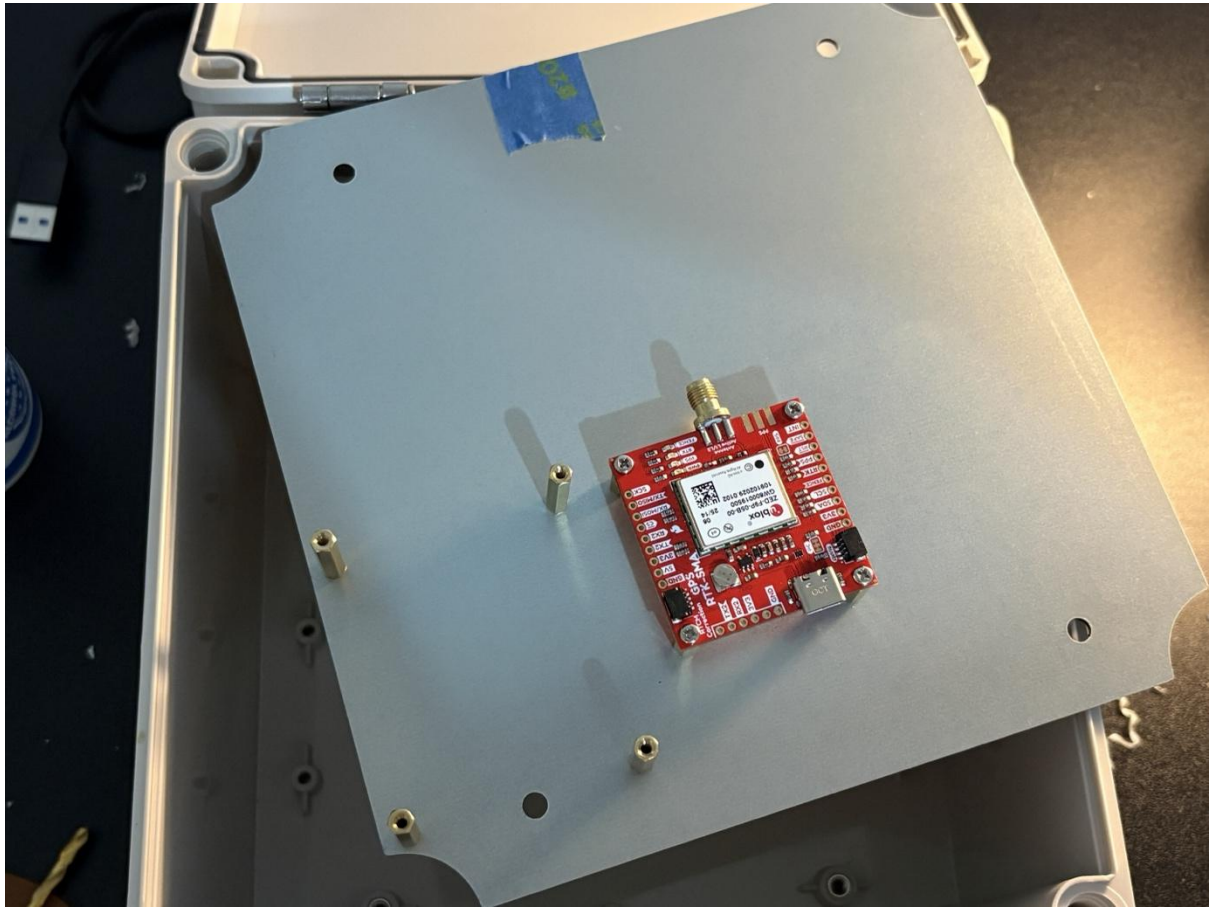
Step 2: Unscrew and use masking tape to tape up the metal mounting plate, this will allow to mark the holes/dimension to easily mount up the standoff to then mount the Raspberry Pi and SparkFun GPS-RTK-SMA ZED-F9P

Step 3: Trace out the holes of the mounting points from the Raspberry Pi and SparkFun GPS-RTK-SMA ZED-F9P, this can be achieved using the existing base mounting plate from the “Mobile Rover” to outline the holes for the Raspberry Pi. Continue trace out the mounting holes from the SparkFun GPS-RTK-SMA ZED-F9P.

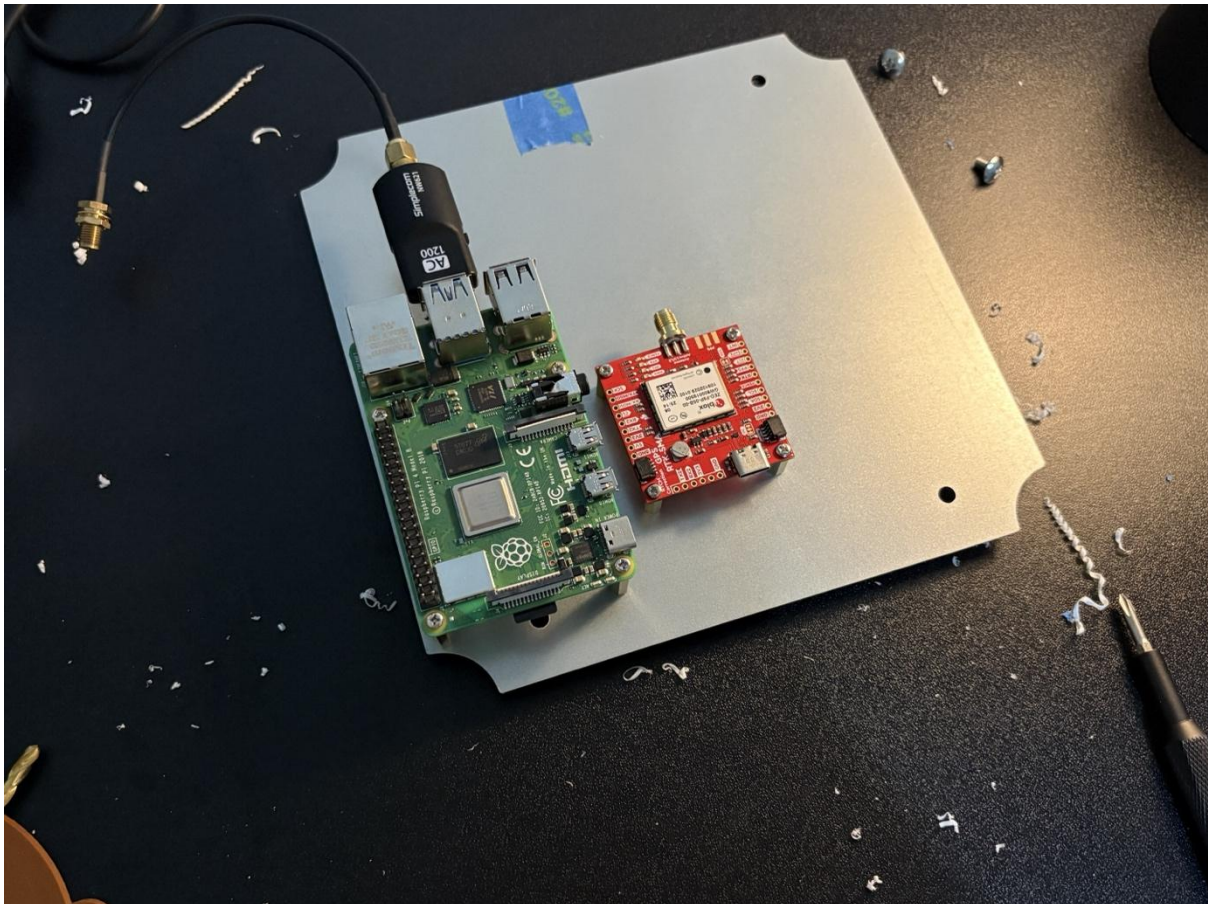


Step 4: Using a hammer and centre punch, create small indentation on the 4 marked crosses. Using a drill and 3/32 drill bit carefully drill out the 8 marked holes for mounting the metal stand offs. Afterwards, remove the masking tape and align the metal stand offs and screw down the 4x M.25*20+6mm stand offs and 4x M.25*11+6mm to the metal base. Afterwards, you can temporarily test fit the components to see if they are aligned.



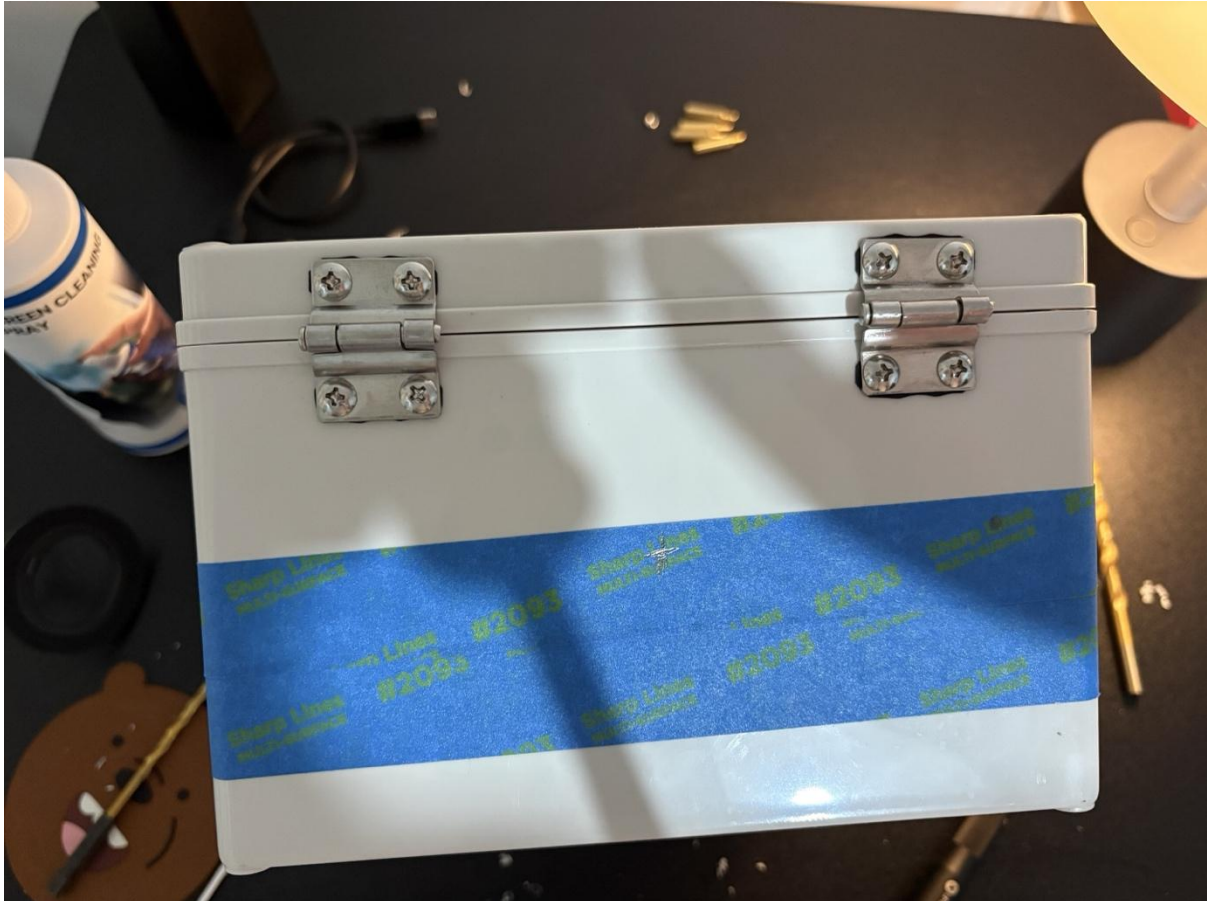


Step 5: Carefully screw in the remaining component to the corresponding standoffs.



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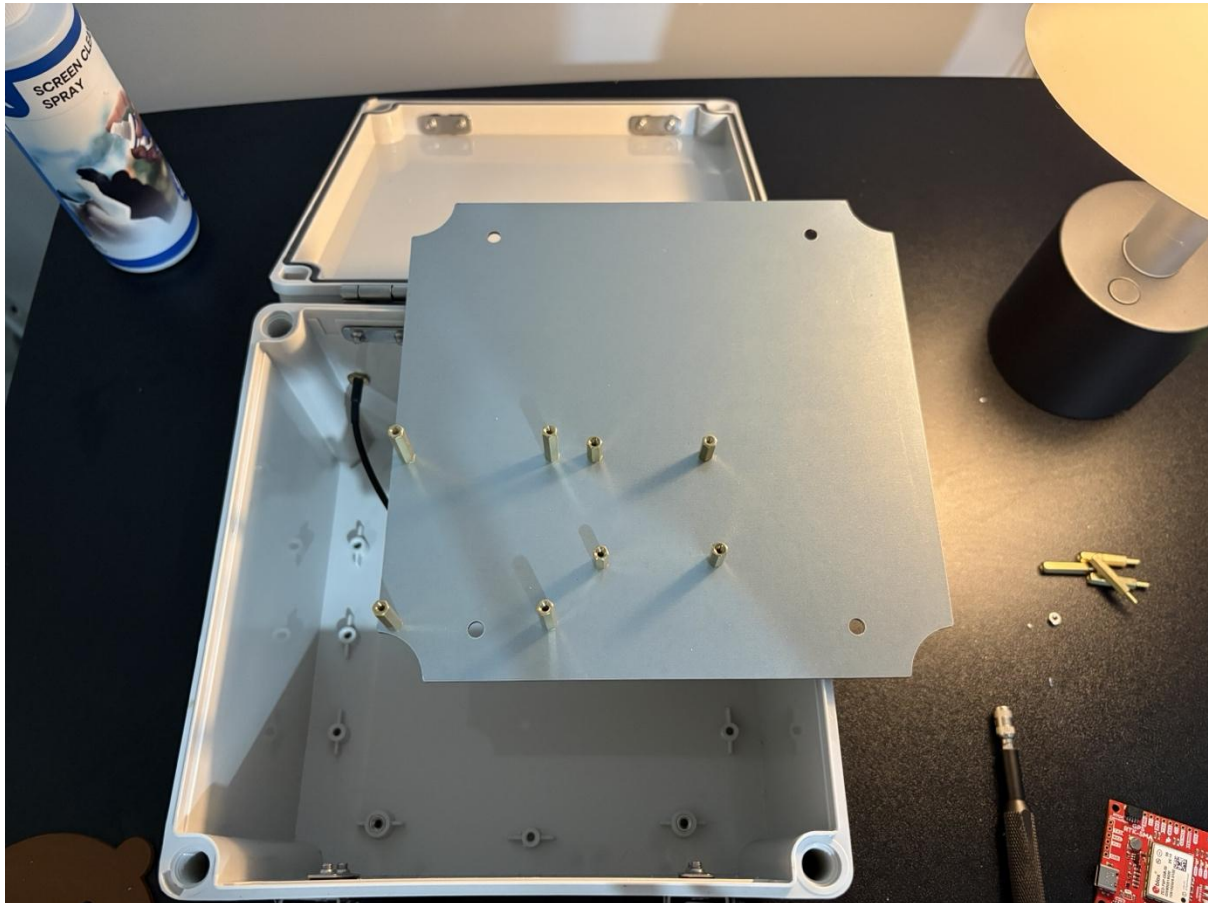
Step 6: Moving over to the exterior Large ABS IP66 Enclosure, mask up and use a ruler to centre and mark the cable gland hole and another hole for the SMA male connector hole. Use a drill and serrated drill bit to carefully drill out both holes to the corresponding size.



Step 7: Install and mount the cable gland and SMA connector into the drilled-out holes.



Step 8: Finish mounting and screw in the metal stand offs onto the mounting plate for the mounting plate, afterwards mount and screw in both Raspberry Pi and ZED F9P GPS modules.

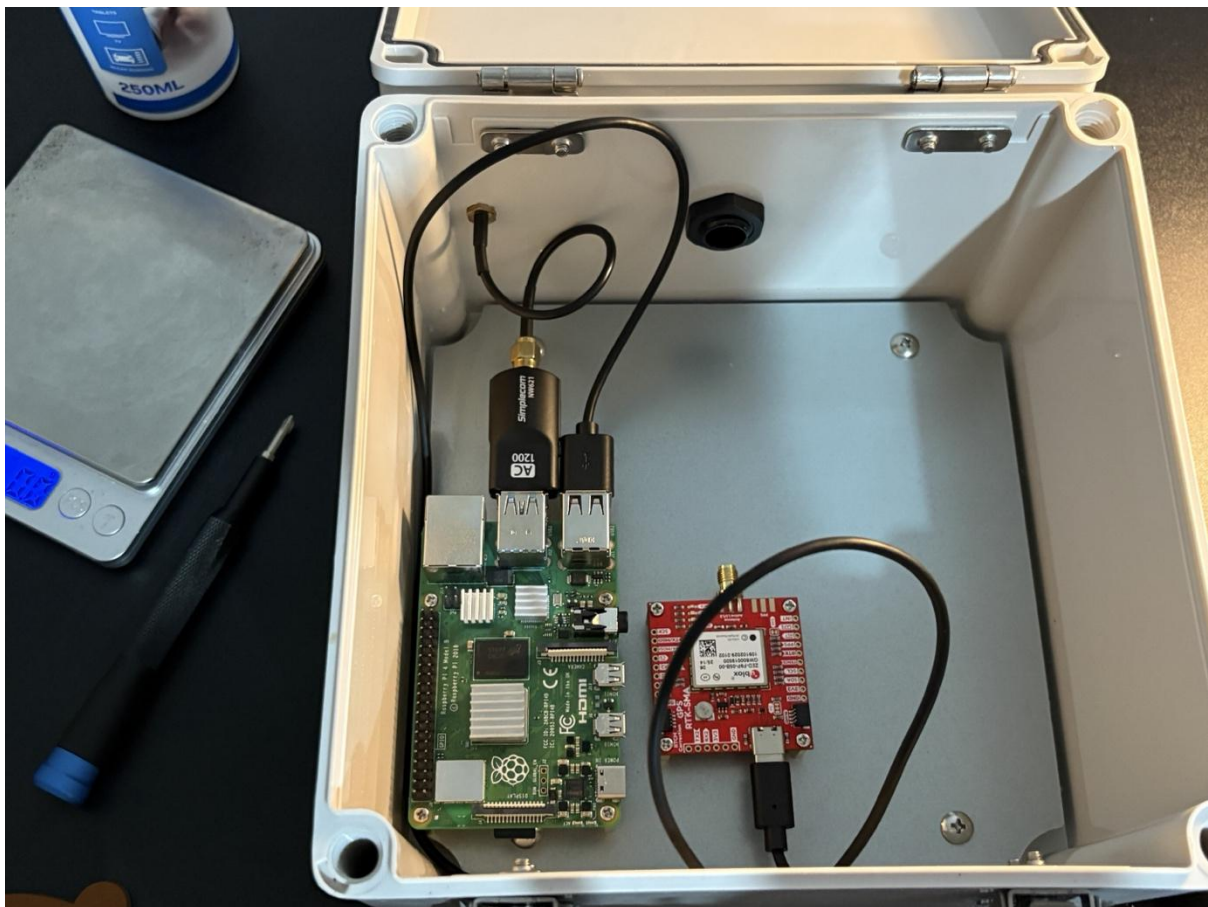


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Step 7: Cable and plug in the Simplecom NW621 USB Wi-Fi Module and screw in 15 cm SMA extension cable to the end of the USB Wi-Fi module.

Step 8: Carefully lowering the mounted components on the metal mounting base plate into the ABS box and carefully screw it into the original screw down points.

Step 9: Connected the components into the corresponding USB ports, connect the SMA male to female extension cable



Step 10: Temporarily place the power bank into the case/box for alignment, once satisfied with the location of the battery use Velcro tape and remove the adhesive tape place one side of the Velcro into positioning. Place the other Velcro tape onto the battery, cut off any excess.



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